ALTERNATING BUNDLES AND ANALYTIC DESCENT

JACK MCCARTHY

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Abstract. Regular functors on normed spaces induce functors on vector bundles, and natural transformations pass fiberwise to bundles. We study this principle for bounded alternating maps over nontrivially normed fields. Their inclusion into all multilinear maps reduces the analytic question to one problem: lifting homogeneous coefficients through a closed linear subspace with controlled norms. We give an exact descent criterion and two mechanisms for satisfying it, using retractions in the fibers or coordinates on the parameter space.

The alternating bifunctor is always C^∞ and is analytic on all hom spaces exactly when $k! \neq 0$ in the field. Nonarchimedean spherically complete targets give analyticity in every degree. Independently, bases with open K^d charts support analytic alternating bundles with arbitrary normed fibers in every characteristic; summable infinite-dimensional parameter spaces give further cases. Counterexamples show the limits of both general and scalar-valued constructions, and polynomial shears realize the failures in bundle and tangent transitions. The exterior-power representation identifies the universal action and its scalar bidual test. Finally, over complete fields, ambient analytic differential forms retain pullback, wedge products and exterior differentiation even when the natural alternating atlas is not analytic. The counterexamples and supplementary classification results are proved in the appendices. Every theorem, proposition, lemma and corollary except Proposition 6.6 has a Lean proof of a corresponding statement; Section 9 lists the main places where those statements are narrower than the text.

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The original counterexample and its initial write-up were developed with Anthropic's Claude. Subsequent research and exposition used Claude and OpenAI's Codex; the organization of the paper was prepared with Codex. AI assistants contributed to mathematical arguments and independent proof audits. The original Lean formalization used Codex and Autoform. The remaining claims and the per-claim ledger were formalized with Anthropic's Claude Code in October 2026. Proposition 6.6 was developed and written with Claude and is not formalized. Section 9 records the verification scope.

1. Introduction

Linear algebra supplies differential geometry with dual bundles, tensor bundles, and bundles of differential forms. These constructions reflect a general principle: a sufficiently regular functor on normed spaces extends fiberwise to vector bundles and their morphisms, and natural transformations extend with it. The analytic requirement concerns the action on linear maps: C^n families must give rise to C^n families of induced maps. Every C^n functor on normed spaces satisfies this condition. Thus, for a fixed C^n manifold M, fiberwise application defines a functor

$$\{C^n \text{ functors on normed spaces}\} \longrightarrow \{\text{functors on } C^n \text{ vector bundles over } M\},$$
$$\mathcal{F} \longmapsto \mathcal{F}_M.$$

Its morphisms are natural transformations: $\eta\colon \mathcal{F} \Rightarrow \mathcal{G}$ gives $\eta_M\colon \mathcal{F}_M \Rightarrow \mathcal{G}_M$. The same construction applies to several variables and to contravariant variables. It is this passage, rather than the construction of fibers alone, that makes regularity of a linear-algebraic functor a geometric question.

We study the alternating-map bifunctor

$$\mathrm{Alt}^k\colon \mathsf{Vec}_K^{\mathrm{op}} \times \mathsf{Vec}_K \longrightarrow \mathsf{Vec}_K, \qquad (E,F) \longmapsto \mathrm{Alt}^k(E;F),$$

where Vec_K denotes normed spaces and bounded linear maps over a nontrivially normed field K. The spaces need not be complete, and the degree k is any nonnegative integer. Our main question is when applying this bifunctor to analytic input bundles produces an analytic bundle, naturally in the inputs and their morphisms.

The answer is governed by one local problem. Alternating maps form a closed linear subspace of all bounded multilinear maps. The multilinear construction is always analytic. Its restriction to alternating maps is analytic precisely when the homogeneous coefficients can be represented by bounded multilinear maps with alternating values and suitable norm bounds. We call this passage *analytic descent through a closed linear inclusion*. Theorem 3.1 gives an exact criterion. It separates two ways to obtain alternating bundles: split the inclusion on the fiber side, or construct the coefficients using the coordinates of the base.

The principal conclusions. The bifunctor Alt^k is always C^∞. It is analytic on every hom space if and only if $k! \neq 0$ in K, and this classification is unchanged on the full subcategory Ban_K of complete spaces. Thus every characteristic-zero field is covered in every degree. In characteristic $p > 0$, the restriction to degrees $k < p$ is sharp for arbitrary normed targets. In positive characteristic there is also an analytic bifunctor in every degree

$$\mathrm{Alt}^k\colon \mathsf{Vec}_K^{\mathrm{op}} \times \mathsf{Vec}_K^{\circ} \longrightarrow \mathsf{Vec}_K^{\circ},$$

where Vec_K° is the full subcategory of nonarchimedean, spherically complete spaces. The last assertion includes closure of that target category under alternating-map spaces (Theorem 4.2).

For bundles the condition is weaker. An analytic base with open K^d charts tests only finite-coordinate families of operators. Every such family is preserved by Alt^k, in every characteristic and degree, with arbitrary normed fibers. Consequently Alt^k induces a bifunctor on analytic normed bundles over every such base, including their operator-valued analytic morphisms. There are also infinite-dimensional parameter classes for which descent holds, with explicit hypotheses on the final target fiber (Theorem 4.5). A single bundle theorem, Corollary 4.6, collects these cases.

The distinction between whole hom spaces and the families tested by a base is essential. If $k! = 0$ in K, analytic input bundles over a suitable Banach base can have a naturally induced alternating atlas with a nowhere analytic transition map. An affine family of invertible operators

realizes this failure, so restricting to invertible transitions does not remove the obstruction. For scalar forms there is an exact classification over complete nonarchimedean fields:

$$\mathrm{Alt}^k(-;K) \text{ is analytic on all normed sources} \quad \Longleftrightarrow \quad k! \neq 0 \text{ or } K \text{ is spherically complete.}$$

In the negative case, even the alternating cotangent atlas of an analytic Banach manifold can fail. With Banach or tangent values the corresponding atlas can fail over every field of characteristic p once $k \geq p$ (Proposition 6.6). Section 6 proves the geometric reductions from the two operator obstructions.

An analytic calculus of forms nevertheless survives. Over every complete field, alternating-valued analytic sections of the multilinear cotangent bundle admit pullback, shuffle products, and exterior differentiation, and form a differential graded algebra. They agree with analytic sections of the alternating bundle under the descent hypotheses stated in Section 7. The distinction is substantive: a polynomial pullback can preserve ambient analyticity while destroying intrinsic alternating-valued analyticity.

Method and scope. The categorical construction is proved once in Section 2. Section 3 isolates its analytic obstruction, and Section 4 proves the mechanisms that remove it. Section 5 identifies the representing exterior power and explains what scalar and dual-valued forms detect. The geometric obstructions and the resulting calculus occupy Sections 6 and 7. Section 8 proposes a Mathlib interface with one alternating-bundle instance, backed by the family-preservation theorems.

The generality in the fibers matters. The ordinary projective exterior power represents alternating maps with arbitrary normed targets; the nonarchimedean projective exterior power represents complete nonarchimedean targets. Their analyticity properties can differ. Completeness assumptions below are attached to the arguments that need them, rather than imposed on the category at the outset.

Our bundle construction follows Lang's functors of class C^n [7, Ch. III, §4] and Bourbaki's vector functors [3, n° 7.6]. Gouëzel's characteristic-zero alternating-map construction [4, 5] provides the factorial case, and closed-range reflection of smoothness is the argument formalized in [11]. Bourbaki introduces the alternating bifunctor in [3, n° 7.8], as the functor α_k of continuous alternating k-linear maps with $\alpha_k(f_0, f_1)(m) = f_0 \circ m \circ (f_1, \ldots, f_1)$, under the assumption that K has characteristic 0 or that the bundles have finite rank, and states that it is not known whether this definition can be made without any restriction. The paragraph *Bourbaki's question* in Section 6 says what our results give for this question. Gouëzel notes in [4] that the finite-dimensional case works in positive characteristic; over a complete field this is Proposition 4.1(2). The nonarchimedean extension mechanism belongs to the functional analysis of spherical completeness; we use its norm-controlled form and relate it to [6, 15, 16, 18].

The long constructions are proved in the appendices. Appendices B–E establish the general operator obstruction, and Appendix F proves the scalar obstruction. Appendix G characterizes universal analytic reflection by tensor projections; Appendix H proves that a largest full analytic domain need not exist. Further coordinate methods and a nonspherical analytic target appear in Appendix I. These results supplement the common descent framework without interrupting the bundle arguments. Section 9 distinguishes the Lean-checked statements from those with written proofs only; a proof in this article is not by itself a claim of formal verification.

Conventions. The field K is nontrivially normed throughout. We write $\mathcal{L}(E, F)$ for bounded linear maps with their operator norm, $\mathrm{Mult}^k(E; F)$ for bounded k-linear maps, and $\mathrm{Alt}^k(E; F)$ for those vanishing whenever two arguments coincide. This last condition, rather than antisymmetry alone, defines alternation in characteristic two. The degree-zero space is F. Products of

normed spaces carry the maximum norm. The letters E, D, F usually denote fibers; P denotes the model space of a base or a parameter domain. *Finite coordinates* means a bounded linear isomorphism with K^d. Over a complete field this is equivalent to finite algebraic dimension; over an incomplete field the coordinate condition is retained.

A norm is *nonarchimedean* when $\|x+y\| \le \max(\|x\|, \|y\|)$. Every normed field of characteristic $p > 0$ has this property: Frobenius gives $|x+y|^{p^m} \le 2\max(|x|,|y|)^{p^m}$, and taking roots and letting $m \to \infty$ proves it. For $n \le \infty$, C^n refers to iterated Fréchet differentiability. Power-series analyticity means local expansion in bounded multilinear coefficients with a positive norm-controlled radius. For manifolds and C^ω bundles we use the analytic calculus in which the iterated derivatives are analytic as well, as in Mathlib's C^ω convention [9]. On open sets this agrees with power-series analyticity at every point, even for incomplete ranges: the derivative series at each point converges in the completion to the Fréchet derivative, which exists in the original operator space because every nearby point is again a point of analyticity. The two notions differ only for germs at a single point.

2. From linear constructions to bundles

A vector bundle allows a linear construction to vary over a manifold. The construction must act on linear maps, so that it transports changes of coordinates; its regularity must also be strong enough to transport families of those maps. These are separate requirements. Functoriality supplies the compatibility of the new coordinates, while analysis determines whether they define a smooth or analytic bundle.

Throughout, K is a nontrivially normed field and Vec_K is the category of normed K-spaces and bounded linear maps. No completeness assumption is imposed on the spaces. Write $\mathcal{L}(E, E')$ for its hom spaces, equipped with the operator norm. For $n \in \mathbb{N} \cup \{\infty, \omega\}$, regularity of order n means continuity when $n = 0$, iterated Fréchet smoothness for finite n and $n = \infty$, and the analytic manifold regularity class for $n = \omega$. When the range is incomplete, this last convention includes the derivative regularity required by that calculus.

2.1. **The bundle construction.** Fix a C^n manifold M modeled on a normed space P. The parameter space P describes local coordinates on the base; it need not be related to the typical fibers E, F of the bundles under consideration. Write $\mathsf{VB}^n_K(M)$ for the category of C^n normed vector bundles over M, with morphisms over id_M whose local *operator-valued* expressions are C^n. Thus a morphism from a bundle with fiber E to one with fiber F is locally a C^n map

$$x \longmapsto T(x) \in \mathcal{L}(E, F).$$

This convention records the regularity needed here; merely requiring regularity of the associated map of total spaces would give a different definition. Bounded bilinearity of composition makes the convention independent of the trivializations.

Let $\epsilon = (\epsilon_1, \dots, \epsilon_r)$ specify a variance in each variable, and put

$$\mathbf{C}^\epsilon = \mathsf{C}_1 \times \cdots \times \mathsf{C}_r, \qquad \mathsf{C}_a = \mathsf{Vec}_K \text{ or } \mathsf{Vec}_K^{\mathrm{op}}.$$

A functor $\mathcal{F}\colon \mathbf{C}^\epsilon \to \mathsf{Vec}_K$ is C^n if all its maps on products of hom spaces are jointly C^n. Products carry the maximum norm. Let $\mathbf{B}^\epsilon_M$ denote the corresponding product of bundle categories, with the same variances. Natural transformations between such functors have bounded linear components, since their codomain is Vec_K.

Theorem 2.1 (lifting linear constructions to bundles)**.** *For a fixed C^n manifold M, fiberwise application defines, up to canonical natural isomorphism, a functor*

$$\mathcal{B}^n_M\colon \operatorname{Fun}^{C^n}(\mathbf{C}^\epsilon, \mathsf{Vec}_K) \longrightarrow \operatorname{Fun}(\mathbf{B}^\epsilon_M, \mathsf{VB}^n_K(M)),$$
$$\mathcal{F} \longmapsto \mathcal{F}_M, \qquad \eta \longmapsto \eta_M.$$

Here the category on the left is the full subcategory of the ordinary functor category on the functors regular on hom spaces. The resulting bundles, bundle morphisms, and natural transformations are given fiberwise by $\mathcal{F}$, its action on maps, and the components of η, respectively.

The same conclusion holds if regularity on whole hom spaces is replaced by preservation of C^n families parametrized by open subsets of the model space P.

Proof. Choose a common trivializing cover for the input bundles. For the ath bundle, let $g^a_{ij}(x)$ convert coordinates in chart j to coordinates in chart i. These are C^n operator-valued maps and satisfy

$$g^a_{ij}(x)g^a_{j\ell}(x) = g^a_{i\ell}(x), \qquad g^a_{ii}(x) = \mathrm{id}.$$

In a covariant variable use g^a_{ij}; in a contravariant variable use g^a_{ji}. Denote the resulting tuple of arrows in $\mathbf{C}^\epsilon$ by $\mathbf{g}_{ij}$. Functoriality gives

$$\theta_{ij}(x) = \mathcal{F}(\mathbf{g}_{ij}(x)), \qquad \theta_{ij}(x)\theta_{j\ell}(x) = \theta_{i\ell}(x).$$

The maps θ_{ij} are C^n by composition. They therefore glue the products with typical fiber $\mathcal{F}(E_1, \dots, E_r)$ into a C^n vector bundle: the associated coordinate changes are C^n because operator evaluation is bounded bilinear. Inverse input transitions are already present among the g^a_{ij}, so this argument needs no theorem about inversion on an operator algebra.

Apply $\mathcal{F}$ also to the local operator families of a tuple of bundle morphisms, reversing their directions in the contravariant variables. Their compatibility with transitions follows from functoriality, and their regularity follows from composition. This gives $\mathcal{F}_M$. The same argument applied to changes of trivialization identifies different choices canonically and preserves identities and composition.

For a natural transformation $\eta\colon \mathcal{F} \Rightarrow \mathcal{G}$, its component on the induced bundles is locally the constant operator $\eta_{(E_1,\dots,E_r)}$. Naturality says precisely that this operator intertwines the two transition systems:

$$\eta_{\mathbf{E}}\, \mathcal{F}(\mathbf{g}_{ij}(x)) = \mathcal{G}(\mathbf{g}_{ij}(x))\, \eta_{\mathbf{E}}.$$

It consequently defines a C^n bundle morphism. The same identity for arbitrary input morphisms proves naturality of η_M. Identities and compositions of transformations are preserved fiberwise. No extra regularity of η is needed: its local expression is constant.

For the final assertion, write every transition and local morphism family in a base chart. Its domain is an open subset of P, and preservation of such families supplies exactly the regularity used above. The argument otherwise remains unchanged. □

The familywise hypothesis is a condition on $\mathcal{F}$ for *all* families with the specified parameter space: for fixed objects and every C^n map

$$\gamma\colon U \longrightarrow \prod_{a=1}^{r} \operatorname{Hom}_{\mathsf{C}_a}(E_a, E'_a), \qquad U \subseteq P \text{ open},$$

the operator family $\mathcal{F} \circ \gamma$ must be C^n. It is weaker than regularity on whole hom spaces. In particular, preserving finite-coordinate analytic families can suffice for bundles on a finite-coordinate base even when the functor fails to be analytic on an infinite-dimensional hom space. The distinction between these two requirements is central to this paper.

The theorem also applies to restricted categories of fibers, provided the prescribed bundle morphisms take values in their hom spaces. It is the familiar transport of a linear construction to bundles, expressed so that both mixed variance and the required operator regularity are explicit; compare [5, 7] and Bourbaki's vector functors [3, n°s 7.6.1–7.6.4].

2.2. **Alternating maps inside multilinear maps.** For normed spaces E, F, let $\mathrm{Mult}^k(E;F)$ be the normed space of bounded k-linear maps $E^k \to F$, and let $\mathrm{Alt}^k(E;F)$ be its alternating subspace. Alternating means vanishing whenever two arguments agree; this definition applies unchanged in characteristic two. The inclusion

$$j_{E,F}\colon \mathrm{Alt}^k(E;F) \longrightarrow \mathrm{Mult}^k(E;F)$$

is a linear isometry with closed range. Indeed, alternation is an intersection of closed conditions under evaluation at fixed tuples. Both constructions are contravariant in E and covariant in F, and the inclusions form a natural transformation. If F is complete, both spaces are complete: operator-norm Cauchy families converge pointwise in F, and their boundedness and defining identities pass to the limit. Thus the alternating bifunctor also restricts to Banach spaces.

For $f \in \mathcal{L}(E,E')$ and $g \in \mathcal{L}(F,F')$, their joint action is

$$B(f,g)(m)(x_1,\dots,x_k) = g\big(m(fx_1,\dots,fx_k)\big).$$

On alternating maps, write

$$A = A^k_{E,E';F}\colon \mathcal{L}(E,E') \longrightarrow \mathcal{L}(\mathrm{Alt}^k(E';F), \mathrm{Alt}^k(E;F)), \qquad A(f) = B(f, \mathrm{id}_F),$$

for precomposition. The elementary bound

$$\|B(f,g)(m)\| \le \|g\|\,\|m\|\,\|f\|^k$$

shows that the actions are bounded operators. Their dependence on (f,g) requires more care.

Proposition 2.2 (ambient polynomial action)**.** *The joint action on multilinear maps*

$$B\colon \mathcal{L}(E,E') \times \mathcal{L}(F,F') \longrightarrow \mathcal{L}\big(\mathrm{Mult}^k(E';F), \mathrm{Mult}^k(E;F')\big)$$

is the diagonal of a bounded $(k+1)$-linear map, and hence is C^ω. Its restriction to alternating inputs is analytic as a map into

$$\mathcal{L}\big(\mathrm{Alt}^k(E';F), \mathrm{Mult}^k(E;F')\big),$$

and all its values belong to the closed subspace of operators taking values in $\mathrm{Alt}^k(E;F')$.

Proof. For $v_i = (f_i, g_i)$ in the product of hom spaces, define

$$H(v_0,\dots,v_k)(m)(x_1,\dots,x_k) = g_0\big(m(f_1x_1,\dots,f_kx_k)\big).$$

This is multilinear in the v_i and has norm at most 1. Its diagonal is B. A diagonal of a bounded multilinear map has finite polynomial derivatives of every order, so it is C^ω even with incomplete range. Restriction of the input m to alternating maps is bounded linear and preserves analyticity. On the diagonal, two equal x_i give two equal fx_i, so the output remains alternating. Closedness of the operator subspace follows by applying the closedness of $j_{E,F'}$ pointwise. For $k = 0$, the same formula is simply postcomposition by g_0. □

For bundles $\mathcal{E}, \mathcal{H}$ with transition families g_{ij}, h_{ij}, the induced alternating transitions are therefore

$$m \longmapsto h_{ij}(x) \circ m \circ \big(g_{ji}(x),\dots,g_{ji}(x)\big).$$

Their images in the multilinear operator space are analytic whenever the input families are analytic. The unresolved step is whether their power-series coefficients can be chosen in the alternating operator space itself. The mixed coefficient H above need not take alternating

values: different f_i do not preserve equality of arguments. Consequently, the closed inclusion alone does not settle analyticity. This is the descent problem studied in Section 3. Once it is solved for a class of fibers or parameter spaces, Theorem 2.1 supplies the corresponding bundle construction without further gluing arguments.

3. Analytic descent through closed subspaces

The ambient polynomial of Proposition 2.2 puts every alternating construction into the same situation: a map takes values in a closed subspace, and becomes analytic after inclusion into the ambient space. The obstruction is not convergence of its values. It is the existence of multilinear coefficients in the smaller space. We first make this distinction exact, and then explain why ordinary smoothness has no corresponding obstruction.

Throughout this section, K is nontrivially normed, P, W, Z are normed K-spaces, and $j\colon W \to Z$ is a linear isometry with closed range. None of these spaces is assumed complete. We identify W with its image when convenient. If $U \subseteq P$ is open and $f\colon U \to W$, an ambient expansion at $x_0 \in U$ means

$$jf(x_0+h) = \sum_{n\geq 0} b_n(h,\dots,h), \qquad \sup_n \|b_n\| R^n < \infty \tag{3.1}$$

for some $R > 0$, with bounded n-linear $b_n\colon P^n \to Z$ and convergence to the displayed value for $\|h\| < R$. Degree zero denotes a constant. Shrinking R when necessary gives the usual local definition of analyticity. For nonarchimedean normed spaces over a complete nonarchimedean field, this use of multilinear representatives agrees with the homogeneous-polynomial lifting norms of Bertram–Glöckner–Neeb [2, 7.14–7.19]; mere continuity of an algebraic polynomial is a different condition.

Theorem 3.1 (the coefficient descent criterion)**.** *Suppose* (3.1) *holds. Then every diagonal $b_n(h,\dots,h)$ belongs to $j(W)$. The map f is analytic at x_0 if and only if there are bounded n-linear maps $q_n\colon P^n \to W$ and $r > 0$ such that*

$$jq_n(h,\dots,h) = b_n(h,\dots,h) \quad (n \geq 0,\ h \in P), \qquad \sup_n \|q_n\| r^n < \infty.$$

Only the diagonals must agree; the original mixed values of b_n need not lie in $j(W)$.

Proof. Fix $h \in P$ and restrict the expansion to the scalar line $x_0 + th$. Its coefficients satisfy a geometric bound. Such a convergent one-variable series has unique coefficients over any nontrivially normed field: if N is the first nonzero coefficient of a series representing zero, division by t^N bounds its norm by a geometric tail tending to zero as $t \to 0$, $t \neq 0$. This argument uses only convergence of the given series, not completeness of its target.

Closedness also recovers each coefficient separately. Starting with $b_0 = jf(x_0)$, suppose the earlier diagonals lie in $j(W)$. For sufficiently small nonzero t,

$$t^{-n}\left(jf(x_0+th) - \sum_{m<n} t^m b_m(h,\dots,h)\right) \in j(W).$$

The geometric tail estimate makes this expression converge to $b_n(h,\dots,h)$. Hence that vector lies in the closed subspace.

If f is analytic, apply j to one of its expansions and compare on every scalar line. Uniqueness gives precisely the required diagonal identities. Conversely, shrink r below R. The proposed partial sums in W have images converging to $jf(x_0+h)$ term by term. The isometry implies their convergence to the already specified vector $f(x_0+h)$ in W, and the bound supplies a positive radius. □

This is a local power-series criterion at a point, not an assertion that `AnalyticAt` alone supplies the full C^ω derivative tower used in manifold calculus. When the target is incomplete, an expansion there need not provide expansions throughout a neighborhood; analyticity on an open set requires the criterion at every point. The theorem also explains why derivative jets do not settle the problem in positive characteristic: a nonzero Frobenius term can have all ordinary derivatives zero while remaining a nonzero homogeneous term.

Lemma 3.2 (smooth descent)**.** *For $n \in \mathbb{N} \cup \{\infty\}$, a map $f\colon U \to W$ is C^n if and only if jf is C^n.*

Proof. Continuity reflects through an isometry. If jf has Fréchet derivative T at x, then

$$Th = \lim_{t\to 0,\ t\neq 0} \frac{jf(x+th) - jf(x)}{t} \in j(W).$$

Thus $T = jS$ for a unique bounded linear $S\colon P \to W$, with the same norm. The isometry transfers the derivative remainder estimate, so $Df = S$. Postcomposition by j is itself a closed linear isometry $\mathcal{L}(P, W) \to \mathcal{L}(P, Z)$: its image consists exactly of operators whose values lie in $j(W)$, a closed condition under each evaluation. Applying the same argument to derivative maps proves every finite order by induction, and hence C^∞. □

In particular, the full alternating bifunctor is C^∞ on its operator spaces: its ambient action is a bounded polynomial, and the inclusion of alternating maps is closed and isometric. For the precomposition action $A = A^k_{E,E';F}$, the argument is the commutative triangle

$$\begin{array}{ccc} & \mathcal{L}(E, E') & \\ {\scriptstyle A}\swarrow & & \searrow{\scriptstyle u \mapsto (m \mapsto m\circ(u,\dots,u))} \\ & \text{polynomial} & \\ \mathcal{L}\big(\mathrm{Alt}^k(E';F), \mathrm{Alt}^k(E;F)\big) & \xrightarrow[j_{E,F}\circ -]{\text{closed linear isometry}} & \mathcal{L}\big(\mathrm{Alt}^k(E';F), \mathrm{Mult}^k(E;F)\big) \end{array}$$

in which the right-hand arrow is the diagonal of a bounded k-linear map (Proposition 2.2 with $g = \mathrm{id}_F$), and Lemma 3.2 is applied with the bottom arrow as j. This is the smooth descent argument used in [11]. Theorem 2.1 therefore constructs alternating bundles and their morphisms at every finite regularity and at C^∞, over arbitrary normed model spaces and fields. Analyticity is the additional coefficient problem.

We call a finite sum of diagonals of bounded multilinear maps a *continuous polynomial*. For a homogeneous ambient polynomial, the descent problem has a particularly economical form.

Proposition 3.3 (one lift, every base point)**.** *Let $k \geq 1$, and suppose $a\colon H \to W$ satisfies $ja(h) = B(h, \dots, h)$ for a bounded k-linear map $B\colon H^k \to Z$. The following are equivalent:*

1. *a is analytic at some point of H;*
2. *there is a bounded k-linear $Q\colon H^k \to W$ with $Q(h, \dots, h) = a(h)$;*
3. *a has a finite power-series expansion at every point, of infinite radius.*

Consequently each precomposition action $A^k_{E,E';F}$ is either a continuous polynomial on its whole domain or analytic nowhere.

Proof. An expansion at h_0, compared along $h_0 + th$ with the polynomial $B(h_0 + th, \dots, h_0 + th)$, has degree-$k$ diagonal $B(h, \dots, h)$. Its degree-k coefficient is the required Q. Conversely, expand $Q(h_0 + h, \dots, h_0 + h)$ by multilinearity. The coefficient of degree r is the sum over the r occupied slots, with norm at most $\binom{k}{r}\|Q\|\|h_0\|^{k-r}$. This is a finite expansion. The last assertion follows from Proposition 2.2. □

4. When alternating bundles are analytic

There are two ways to solve the coefficient problem. One can retract the ambient space onto the alternating subspace, thereby handling every parameter space at once. Alternatively, one can use coordinates on the parameter space to replace the ambient coefficients. The first mechanism concerns the fibers; the second concerns the base and the families of morphisms being tested.

4.1. **Retractions and lifts in the fibers.**

Proposition 4.1 (splitting the alternating inclusion)**.** *If the inclusion* $j_{E,F}\colon \mathrm{Alt}^k(E;F) \hookrightarrow \mathrm{Mult}^k(E;F)$ *has a bounded linear retraction* ρ*, then* $A^k_{E,E';F}$ *has a bounded* k*-linear lift of norm at most* $\|\rho\|$*, for every normed* E'*. Such a lift also exists in each of the following cases:*

1. $k! \neq 0$ *in* K*;*
2. *either* E *or* E' *has a finite algebraic basis with continuous coordinate functionals.*

No completeness hypothesis is required.

Proof. For a retraction, set

$$Q(u_1,\dots,u_k)(m) = \rho\big(m \circ (u_1,\dots,u_k)\big).$$

The ambient mixed composition has norm at most $\|m\| \prod_i \|u_i\|$, and its diagonal is already alternating.

If $k! \neq 0$, use instead

$$Q(u_1,\dots,u_k)(m)(x) = \frac{1}{k!} \sum_{\sigma \in S_k} m(u_{\sigma(1)}x_1,\dots,u_{\sigma(k)}x_k).$$

When two x-arguments agree, pairing permutations that exchange their positions proves vanishing, in every characteristic. On the diagonal all $k!$ summands coincide. The operator norm is bounded by $k!/|k!|$, where the numerator is the real number counting summands. Equivalently, normalized antisymmetrization retracts $\mathrm{Mult}^k(E;F)$ onto $\mathrm{Alt}^k(E;F)$ with this bound.

For continuous coordinates $(e_i, \varepsilon_i)_{1\le i\le d}$ on E, sum over increasing words $s_1 < \cdots < s_k$:

$$Q(u_1,\dots,u_k)(m)(x) = \sum_s \det\big(\varepsilon_{s_a}(x_b)\big)_{a,b} m(u_1 e_{s_1},\dots,u_k e_{s_k}).$$

If the coordinates are instead on E', replace the summand by $\det(\varepsilon_{s_a}(u_a x_b))_{a,b}\, m(e_{s_1},\dots,e_{s_k})$. Equal columns give alternation; the exterior-basis expansion gives the diagonal identity. Each is a finite bounded sum, with norm at most $k! \sum_s \prod_a \|\varepsilon_{s_a}\| \|e_{s_a}\|$. If $k > d$, the relevant alternating space is zero and the sum is empty. □

Over a complete field, every finite-dimensional normed space has continuous coordinates. One can prove this by induction on dimension: a codimension-one kernel is complete and hence closed; its positive distance from a fixed vector outside it bounds the corresponding coordinate functional. The coordinate map and its inverse are then bounded, giving completeness for the induction. Over an incomplete field this condition must be retained: if $a \in \widehat{K} \setminus K$, the a-coordinate on $K + Ka \subseteq \widehat{K}$ is discontinuous. Finite dimension of F alone does not provide a substitute; scalar-valued forms already exhibit the obstruction of Section 6. Over a complete field, cases (1) and (2) contain the two hypotheses under which Bourbaki defines the bundle of alternating maps: characteristic zero, and finite rank [3, n° 7.8].

A nonarchimedean normed space is *spherically complete* if every nonempty family of pairwise-intersecting closed balls has a common point. In a nonarchimedean metric, intersecting balls are nested.

Theorem 4.2 (spherically complete targets)**.** *Suppose K is nonarchimedean and F is nonarchimedean and spherically complete. Then, for every normed E, the space $\mathrm{Alt}^k(E;F)$ is nonarchimedean and spherically complete, and its inclusion into $\mathrm{Mult}^k(E;F)$ has a contractive linear retraction. Thus $A^k_{E,E';F}$ is a continuous polynomial for every normed E, E' and every k. Neither the field nor the source spaces need be complete.*

Proof. We recall the extension argument with the simultaneous bounds needed here. Let V be a nonarchimedean seminormed space and G a nonarchimedean spherically complete space. Suppose a nonempty family of bounded operators $T_\alpha\colon V \to G$ and radii $r_\alpha \geq 0$ satisfy $\|T_\alpha - T_\beta\| \leq \max(r_\alpha, r_\beta)$, and a linear map T_0 on a subspace D satisfies $\|T_0 d - T_\alpha d\| \leq r_\alpha \|d\|$. To extend over $x \notin D$, choose its value in all balls

$$\overline{B}\big(T_\alpha(d+x) - T_0 d,\ r_\alpha\|d+x\|\big), \qquad \alpha,\ d \in D.$$

These balls meet pairwise: for $r_\alpha \leq r_\beta$, the difference of their centers is bounded by $\max(r_\alpha\|d+x\|, r_\beta\|e+x\|)$, using the hypotheses and the nonarchimedean inequality. Spherical completeness supplies a common value. Zorn's lemma then extends T_0 to all of V, preserving every bound. Taking one center zero and radius one gives norm-preserving extension of contractions. This is the nonarchimedean extension mechanism underlying Ingleton's theorem [6].

Apply the simultaneous form to the free vector space on symbols $[x_1, \ldots, x_k]$, with seminorm

$$\left\| \sum_x c_x[x] \right\| = \max_x |c_x| \prod_i \|x_i\|,$$

and let D be the span of the multilinearity and repeated-input relations. Alternating maps to F correspond isometrically to bounded linear maps on this free space vanishing on D: one direction uses the nonarchimedean inequality in F, and the other evaluates on generators. A family of pairwise-meeting balls in $\mathrm{Alt}^k(E;F)$ therefore supplies centers T_α as above. Extending $T_0 = 0$ from D gives a common point of these balls. This proves spherical completeness without any separation assumption on the continuous dual of E.

The spaces $\mathrm{Mult}^k(E;F)$ and $\mathrm{Alt}^k(E;F)$ have nonarchimedean operator norms. Extend the inverse of their isometric inclusion to the entire multilinear space, now using the latter alternating space as the spherically complete target. The resulting contraction is the required retraction. Proposition 4.1 finishes the proof. In degree zero the alternating space is simply F. □

Quotienting the free-space construction by its relations and then by zero seminorm gives the separated nonarchimedean exterior power; completion gives the Banach-target representation in Section 5. Its maximum seminorm is essential here; the norm on E itself need not satisfy a nonarchimedean inequality. Equivalent changes of norms preserve all analytic conclusions, so an equivalent nonarchimedean spherically complete norm on F is sufficient.

Corollary 4.3 (discretely valued fields)**.** *Suppose K is nonarchimedean with $|K^\times| = r^{\mathbb{Z}}$ for some $0 < r < 1$. Every Banach target F admitting an equivalent nonarchimedean norm makes $A^k_{E,E';F}$ analytic in every degree and for all normed E, E'. The field itself need not be complete.*

Proof. Round the equivalent nonarchimedean norm upwards to $r^{\mathbb{Z}}$, putting the value at zero equal to zero. The new norm lies between the old one and r^{-1} times that norm; monotonicity and compatibility with multiplication by $r^{\mathbb{Z}}$ preserve the norm axioms. It is complete and its

nonzero distances lie in $r^{\mathbb{Z}}$. Such a space is spherically complete. Indeed, a nested family of nonempty closed balls either has radii tending to zero, when completeness gives a common point, or has a smallest effective radius in the discrete value set, when all balls of that radius coincide and lie in the other balls. Pairwise-intersecting balls in a nonarchimedean space are nested. Apply Theorem 4.2 and transport the resulting lift back through the equivalent norms. □

4.2. **Coordinates on the parameter space.**

Theorem 4.4 (finite-coordinate parameters)**.** *Let $j\colon W \to Z$ be a closed linear isometry. If P is boundedly linearly isomorphic to K^d and $U \subseteq P$ is open, then $f\colon U \to W$ is analytic if and only if jf is analytic. For the maximum norm on K^d, ambient coefficients b_n can be replaced by W-valued coefficients q_n with*

$$\|q_n\| \le d^n \|b_n\| \quad (n \ge 1).$$

If Z is nonarchimedean, the factor d^n can be replaced by 1.

Proof. Theorem 3.1 puts every homogeneous diagonal in W. Expand it in coordinates as $b_n(h,\dots,h) = \sum_{|\alpha|=n} c_\alpha h^\alpha$. Every c_α belongs to W: pass to the algebraic quotient Z/W and use polynomial uniqueness over the infinite field K. Only the finite-dimensional span of these coefficients is involved, so the argument does not require separation of points by a continuous dual.

For each α, choose the nondecreasing coordinate word $i_1(\alpha),\dots,i_n(\alpha)$ having these multiplicities, and define

$$q_n(h_1,\dots,h_n) = \sum_{|\alpha|=n} c_\alpha \prod_{r=1}^{n} (h_r)_{i_r(\alpha)}.$$

There are exactly d^n ordered terms before grouping. Thus the triangle inequality gives the stated bound, and two maximum estimates give the nonarchimedean refinement. The diagonals agree. Their exponential bounds preserve a positive radius in Theorem 3.1. The case $d = 0$ is immediate. □

Finite-coordinate bases therefore support alternating bundles in every characteristic and degree, regardless of fiber dimension or completeness. Finite algebraic dimension of the base is not enough over an incomplete field: for $K = \mathbb{F}_p(t)$ and $k \ge p$, the end of Appendix H gives a base of dimension $p+1$ without continuous coordinates and a nonanalytic alternating atlas. Infinite coordinates require a summability mechanism. There are two distinct versions; their differing hypotheses matter.

Theorem 4.5 (summable parameter spaces)**.** *Let $j\colon W \to Z$ be a closed linear isometry.*

1. *Suppose K and Z are nonarchimedean and W is complete. For $U \subseteq c_0(I,K)$ open, analyticity of jf implies analyticity of $f\colon U \to W$. Coefficients can be lifted without increasing their norms.*
2. *Suppose Z is complete. Let $a\colon H \to W$ have an ambient bounded homogeneous polynomial representation of fixed degree d, that is, $ja(h) = B(h,\dots,h)$ for a bounded d-linear $B\colon H^d \to Z$ on a normed space H. For every analytic $\gamma\colon U \to H$, with $U \subseteq \ell^1(I,K)$ open, the composite $a\gamma$ is analytic.*

Here c_0 carries the supremum norm and ℓ^1 the ordinary sum norm. The field need not be complete. Each analyticity assertion remains valid for a parameter space P that is a bounded linear retract of the indicated space V, with bounded linear $i\colon P \to V$, $r\colon V \to P$ and $ri = \mathrm{id}_P$; in (1) the lifted coefficients then satisfy $\|q_n\| \le (\|r\|\|i\|)^n \|b_n\|$.

Proof. For c_0, order I and group the coordinate coefficients of each ambient b_n into nondecreasing words. Finite-coordinate uniqueness puts the grouped coefficients $c_{i_1,\dots,i_n}$ in W, and the nonarchimedean inequality bounds their norms by $\|b_n\|$. The formula

$$q_n(x_1,\dots,x_n) = \sum_{i_1\le\cdots\le i_n} c_{i_1,\dots,i_n}(x_1)_{i_1}\cdots(x_n)_{i_n}$$

converges: outside finite sets of words some coordinate factor is arbitrarily small, while the others are bounded. Completeness of W supplies the sum, with norm at most $\|b_n\|\prod_r\|x_r\|$. Equality of diagonals holds on finite supports and then by density. Apply Theorem 3.1.

For ℓ^1, the same sorting argument in degree d gives a lift of norm at most $d!\|B'\|$ for every bounded d-linear $B'\colon \ell^1(J,K)^d \to Z$ whose diagonal lies in W. Each grouped coefficient has at most $d!$ terms, and the resulting series is absolutely summable, bounded by $d!\|B'\|\prod_r\|x_r\|_1$.

To apply this bound to a nonlinear γ, factor its expansion locally as Tg, where $T\colon \ell^1(J,K)\to \widehat{H}$ is bounded linear. Explicitly, write $\gamma(x_0+x)=\sum_n a_n(x,\dots,x)$ and choose $s\in K$, $0<|s|$ within its radius. Set $J=\coprod_{n\ge0} I^n$ and

$$T(e_{n,i_1,\dots,i_n}) = s^n a_n(e_{i_1},\dots,e_{i_n}), \qquad g(x)_{n,i_1,\dots,i_n} = s^{-n}x_{i_1}\cdots x_{i_n}.$$

The first formula has uniformly bounded values and extends by absolute summation into $\widehat{H}$. The degree-n multilinear coefficient of g has entries $s^{-n}(x_1)_{i_1}\cdots(x_n)_{i_n}$ in its nth block and zero elsewhere, so its norm is at most $|s|^{-n}$. Its diagonal block has sum norm $(\|x\|_1/|s|)^n$, so g has a convergent multilinear expansion for $\|x\|_1<|s|$, without requiring completeness of K, and $T(g(x))=\gamma(x_0+x)$ there. Extend the fixed outer multilinear map to $\widehat{H}$ using completeness of Z; its diagonal still lies in W by closedness. The sorted lift of its composite with T is a bounded polynomial into W. Composing with g proves analyticity at x_0, and the argument applies at each point. The construction uses one sorting bound in degree d, independent of the degrees in the expansion of γ.

Finally, if $ri=\mathrm{id}_P$ for bounded maps $i\colon P\to V$, $r\colon V\to P$, apply either result to the family composed with r on $r^{-1}(U)$, and then restrict along i. □

Since Theorems 4.4 and 4.5 give analyticity at every point of an open set, the C^ω conclusion follows (see Conventions).

The ℓ^1 assertion uses one factorial bound for the fixed outer degree. Applying $n!$ separately to every Taylor coefficient of an arbitrary ambient analytic map would not yield an exponential bound, and does not prove general analytic descent. For the alternating joint action the outer degree is $k+1$, or k for precomposition alone.

4.3. **The bundle theorem.**

Corollary 4.6. *Let M be an analytic manifold over K, with normed model space P. Fiberwise alternating maps give an analytic bifunctor on analytic normed vector bundles and their operator-valued analytic morphisms in each of the following settings:*

Base model	*Hypotheses on the typical fibers*
Arbitrary P	*$k! \neq 0$ in K; or K is nonarchimedean and the target fibers admit equivalent nonarchimedean spherically complete norms; or the source fibers have finite continuous coordinates.*
Finite continuous coordinates	*Arbitrary normed fibers.*
$c_0(I,K)$, or a bounded linear retract	*K nonarchimedean; complete nonarchimedean target fibers.*
$\ell^1(I,K)$, or a bounded linear retract	*Complete target fibers.*

The fiber hypotheses apply to both pairs of bundles at the endpoints of a morphism. Each alternative in the first row specifies a separate setting. At every finite smoothness order and at C^∞, no additional hypothesis on K, P or the fibers is needed.

Proof. The fiber results give a bounded k-linear lift Q of A; then $(u_1, \dots, u_k, g) \mapsto \big(m \mapsto g \circ Q(u_1, \dots, u_k)(m)\big)$ is a bounded $(k+1)$-linear lift of the joint action B, which is therefore analytic. The parameter results instead apply directly to their ambient polynomial along transition families and local morphism families. When the final value-space fiber is complete, the ambient multilinear and operator spaces are complete; they are also nonarchimedean when that fiber is. Theorem 2.1 supplies the bundle construction in every case. The smooth assertion follows from Lemma 3.2. □

For both $\mathbb{Q}_p$ and $\mathbb{C}_p$, every finite degree belongs to the first row: these fields have characteristic zero, although their residue characteristic is p. The possibly large bound $k!/|k!|_p$ remains finite and causes no convergence problem for a finite polynomial. Spherical completeness of $\mathbb{Q}_p$, and its failure for $\mathbb{C}_p$ [16, §1], are therefore irrelevant to this conclusion.

The regularity of the input remains part of the statement. A C^∞ base and C^∞ transitions produce a C^∞ alternating bundle; they do not automatically produce an analytic atlas. For example, on the open unit ball of either field the function $f(0) = 0$, $f(x) = p^{\nu(x)^2+1}$ for $x \neq 0$, where $\nu(x) = \lfloor -\log_p|x| \rfloor$, is locally constant off zero and satisfies $f(x) = o(|x|^m)$ for every m. Hence it is C^∞ with zero derivatives, but is not analytic at zero. For $k \geq 1$, changing a frame of a trivial rank-k bundle by $\operatorname{diag}(1+f, 1, \dots, 1)$ gives smooth nonanalytic transitions on scalar top forms. The underlying bundle is still analytically trivial; it is the chosen smooth atlas that fails to be analytic. Section 6 addresses the stronger obstruction arising from analytic input families.

5. Exterior powers and scalar detection

The closed-subspace description explains how analytic coefficients descend. The exterior-power description explains which targets detect their failure. Both are useful: the first supplies proofs, while the second distinguishes a universal alternating construction from its scalar observations.

For $k \geq 1$ give the algebraic exterior power $\Lambda_K^k E$ the seminorm

$$\|z\|_\pi = \inf_{z = \sum_j x_{j1} \wedge \cdots \wedge x_{jk}} \sum_j \prod_{a=1}^{k} \|x_{ja}\|.$$

Its quotient by the zero-seminorm subspace is denoted $\Lambda_\pi^k E$. This is an uncompleted normed space, with universal alternating map $\omega_E(x_1, \dots, x_k) = [x_1 \wedge \cdots \wedge x_k]$. Put $\Lambda_\pi^0 E = K$ and $\omega_E() = 1$.

Proposition 5.1 (representation)**.** *There is a natural linear isometry*

$$\mathrm{Alt}^k(E;F) \cong \mathcal{L}(\Lambda^k_\pi E, F)$$

for every normed F. Under it, precomposition by $u\colon E \to D$ is precomposition by $\Lambda^k_\pi u\colon \Lambda^k_\pi E \to \Lambda^k_\pi D$.

Proof. The algebraic linearization $\widetilde{m}$ of a bounded alternating map satisfies

$$\|\widetilde{m}(z)\| \le \|m\| \sum_j \prod_a \|x_{ja}\|, \qquad \|\widetilde{m}(z)\| \le \|m\|\|z\|_\pi.$$

It therefore kills the null subspace and descends contractively. Conversely, $\|\omega_E(x_1,\dots,x_k)\| \le \prod_a \|x_a\|$, so composing a bounded linear map with ω_E is contractive. The two operations are inverse and hence isometric. Applying u to each wedge factor proves the intertwining identity and the bound $\|\Lambda^k_\pi u\| \le \|u\|^k$ for $k \ge 1$. In degree zero the action is the identity on K. □

The quotient removes the vectors invisible to bounded linearizations and makes the representing object normed. If F is complete, the same correspondence holds with the completion $\widehat{\Lambda^k_\pi E}$. The uncompleted version is needed for arbitrary normed targets.

Write $Y^* = \mathcal{L}(Y,K)$ and $J_Y(y)(\lambda) = \lambda(y)$ for the canonical map into the continuous bidual. The map J_Y is contractive; no point-separation or Hahn–Banach assumption is implicit.

Theorem 5.2 (universal and scalar tests)**.** *Let U be an open parameter domain and $W\colon U \to \mathcal{L}(X,Y)$ a family. Define*

$$R_F(t)(T) = T \circ W(t), \qquad R_F\colon U \to \mathcal{L}(\mathcal{L}(Y,F),\mathcal{L}(X,F)).$$

For power-series analyticity, or for any fixed class C^n, the following statements hold.

1. *W is regular if and only if R_F is regular for every normed F; the single target Y suffices.*
2. *R_K is regular if and only if $J_Y W$ is regular in $\mathcal{L}(X,Y^{**})$, and if and only if R_{G^*} is regular for every normed G. Any one nonzero continuous dual G^* suffices for the last test.*
3. *If J_Y has a bounded linear left inverse, regularity of R_K implies regularity of W. If J_F has such a left inverse, regularity of R_K implies regularity of R_F.*

Proof. Precomposition depends bounded-linearly on W, and $W(t) = R_Y(t)(\mathrm{id}_Y)$, proving the first assertion. Exchanging the arguments defines a linear isometry

$$\Theta\colon \mathcal{L}(Y^*,X^*) \longrightarrow \mathcal{L}(X,Y^{**}), \qquad (\Theta(B)x)(\lambda) = (B\lambda)(x).$$

Both operator norms are the least C bounding the displayed scalar by $C\|x\|\|\lambda\|$. The inverse exchanges arguments again, and $\Theta(R_K(t)) = J_Y W(t)$. Similarly, the isometry

$$\mathcal{L}(X,G^*) \cong \mathcal{L}(G,X^*), \qquad T \longmapsto [g \mapsto (x \mapsto T(x)(g))]$$

identifies $R_{G^*}(t)$ with $S \mapsto R_K(t) \circ S$. This proves regularity for every dual target. Conversely, choose $v \in G^*$ and $g \in G$ with $v(g) = 1$. Injection $\alpha \mapsto v\alpha$ and evaluation at g compress R_{G^*} to R_K by fixed bounded linear maps. Finally, if $P_Y J_Y = \mathrm{id}_Y$, then $W = P_Y \Theta(R_K)$. If $P_F J_F = \mathrm{id}_F$, use

$$R_F(t)(T) = P_F \circ R_{F^{**}}(t)(J_F \circ T).$$

All identities use bounded linear operations, which preserve each stated regularity class. □

Apply the theorem with $X = \Lambda^k_\pi E$, $Y = \Lambda^k_\pi D$ and $W(u) = \Lambda^k_\pi u$. Under the isometry of Proposition 5.1, R_F is the precomposition action $A^k_{E,D;F}$, which is C^∞ by Proposition 2.2 and Lemma 3.2. It follows that the exterior-power functor is C^∞, and is analytic on all hom spaces

exactly when Alt^k is. Scalar alternating maps detect its operator-norm action after composition with J_Y; this is norm analyticity in a new target, not a weak-star regularity condition. In particular every nonzero continuous-dual target has exactly the same analytic action as the scalar target, even over an incomplete field. The retraction criterion gives a sufficient route from this bidual information back to the original target; its necessity is not asserted.

There is a useful way to see the remaining difficulty. For $k \geq 1$ and a fixed $z = \sum_j \omega_E(x_{j1}, \dots, x_{jk})$, the map $u \mapsto (\Lambda^k_\pi u)z$ has the lift

$$(u_1, \dots, u_k) \longmapsto \sum_j \omega_D(u_1 x_{j1}, \dots, u_k x_{jk}).$$

For $z \neq 0$ its norm can be made at most $2\|z\|$ by the definition of the projective norm. However, the chosen mixed coefficients need not depend linearly on z. Operator-valued analyticity requires precisely such a bounded linear dependence. Thus even uniformly bounded choices of individual orbit lifts do not finish the construction. Degree zero has constant action.

Changing the target category. Over a nonarchimedean field replace the sum in the projective seminorm by a maximum, then separate and complete. Write the result as $\widehat{\Lambda}^k_{\pi,\mathrm{na}}E$, with $\widehat{\Lambda}^0_{\pi,\mathrm{na}}E = \widehat{K}$. The preceding universal-property proof, using the nonarchimedean inequality, gives

$$\mathrm{Alt}^k(E; F) \cong \mathcal{L}(\widehat{\Lambda}^k_{\pi,\mathrm{na}}E, F) \quad \text{for complete nonarchimedean } F.$$

This is the exterior version of Serre's nonarchimedean projective tensor construction [17, §4]. The sum and maximum constructions represent different target categories. Over a discretely valued field, the completed nonarchimedean exterior-power action is analytic in every degree: use its own target $F = \widehat{\Lambda}^k_{\pi,\mathrm{na}}D$ in Corollary 4.3, and evaluate the action at the universal alternating map $D^k \to \widehat{\Lambda}^k_{\pi,\mathrm{na}}D$. The general-target obstruction of Section 6 therefore concerns the sum construction; its target admits no equivalent nonarchimedean norm. This distinction is necessary when interpreting the result in the usual nonarchimedean Banach category.

6. Obstructions and their geometric realization

Analytic descent can fail even for the homogeneous polynomial underlying precomposition. The following two constructions establish sharpness. Their full proofs occupy Appendices B–F; we first state their conclusions and then show how the operator obstructions become failures of analytic bundle transitions. The scalar construction uses van der Put's description of the continuous dual of a bounded sequence space [12]; its multilinear extension and the counterexample itself are proved in Appendix F.

Theorem 6.1 (operator obstructions)**.** *Let K be a nontrivially normed field of characteristic $p > 0$, and let $k \geq p$.*

(1) *There are Banach K-spaces E, F for which*

$$A^k_{E,E;F} \colon \mathcal{L}(E, E) \longrightarrow \mathcal{L}\big(\mathrm{Alt}^k(E; F), \mathrm{Alt}^k(E; F)\big)$$

is analytic at no point. The target F admits no equivalent nonarchimedean norm.

(2) *If K is complete and not spherically complete, there are nonarchimedean Banach spaces E, D for which $A^k_{E,D;K}$ is analytic at no point. They can be chosen as closed subspaces of*

$$\ell^\infty(\Lambda, K^{k+1}) \quad \text{and} \quad \ell^\infty(\Lambda, K^k),$$

respectively, containing the corresponding c_0 spaces, with Λ countable.

For part (1), begin over a Laurent series field with a bounded-sequence source and a completed projective exterior target. The finite-field multiplier theorem of Appendix B rules out the required homogeneous lift, as shown in Appendix C. Projective extension of scalars and passage through the completion transfer the obstruction to every K in the statement (Appendix D); Appendix E assembles the proof and establishes the claim about the target norm. Over the Laurent field itself, the same construction gives the sharper c_0-base obstruction in Corollary C.7.

For part (2), the source spaces are cut out by limit conditions along chains in a countable tree. A finite-dimensional incompatibility and a multilinear tail theorem exclude a bounded lift; Appendix F gives the entire construction. In both arguments, Proposition 3.3 turns the absence of a degree-k lift into nonanalyticity at every point. Proposition I.1 shows that the scalar witnesses must both fail to be of countable type.

These inputs concern different categories. Part (1) proves that the factorial hypothesis cannot be removed for arbitrary normed targets, including over spherically complete fields. Part (2) places the entire obstruction inside nonarchimedean spaces, with the smallest nonzero target. Completeness of the witnesses is useful for sharpness; it imposes no additional restriction on the positive results for general normed fibers.

Corollary 6.2 (regularity on whole hom spaces)**.** *For every nontrivially normed field K, the bifunctor $\mathrm{Alt}^k\colon \mathsf{Vec}_K^{\mathrm{op}} \times \mathsf{Vec}_K \to \mathsf{Vec}_K$ is C^∞, and is analytic if and only if $k! \neq 0$ in K. Both assertions remain true on the full subcategory of Banach spaces.*

Proof. Smooth descent applies to the ambient polynomial action. An invertible factorial gives a finite multilinear lift, and bounded bilinear postcomposition supplies the joint bifunctor action. If the factorial vanishes, Theorem 6.1(1) excludes analyticity even on Banach spaces, after fixing the output morphism to be the identity. □

Theorem 6.3 (universal scalar classification)**.** *Let K be a complete nonarchimedean, nontrivially normed field. For any $k \geq 0$, the contravariant functor*

$$\mathrm{Alt}^k(-;K)\colon \mathsf{Vec}_K^{\mathrm{op}} \longrightarrow \mathsf{Vec}_K$$

is analytic on every hom space if and only if $k! \neq 0$ in K or K is spherically complete. The same equivalence holds when the source category is restricted to Banach spaces, or to nonarchimedean Banach spaces.

Proof. An invertible factorial gives a multilinear lift by normalized antisymmetrization. If K is spherically complete, Theorem 4.2 applies with target K. Both arguments permit arbitrary normed sources. If neither condition holds, then K has characteristic $p > 0$ with $k \geq p$, and Theorem 6.1(2) supplies nonarchimedean Banach witnesses against all three assertions. □

For a nonzero finite-dimensional target F, choose a basis v_i with bounded coordinate functionals λ_i, using completeness of K. Then

$$A_F(u)(m) = \sum_i v_i A_K(u)(\lambda_i \circ m).$$

Conversely, choosing $\lambda(v) = 1$ compresses A_F to A_K by injection along v and evaluation by λ. Thus the same criterion holds for every nonzero finite-dimensional target. Theorem 5.2 extends this equivalence to every nonzero continuous-dual target.

In particular, the scalar obstruction has no characteristic-zero analogue. It does not arise over either $\mathbb{Q}_p$ or $\mathbb{C}_p$, even though the latter is not spherically complete. The relevant characteristic is that of the field, not its residue field.

6.1. **From a bad operator family to changes of frame.** Transition functions take values in invertible operators. The following compression identity shows that this restriction cannot repair a failure on an unrestricted operator space.

Proposition 6.4 (shear realization)**.** *Suppose $A = A^k_{E,D;F}$ is not analytic at $u_0 \in \mathcal{L}(E,D)$. Set $H = D \oplus E$, with the maximum norm, and*

$$g(u)(d,e) = (d + ue, e).$$

Both g and its inverse family are affine analytic. Nevertheless, the family of pullback operators $g(u)^*$ on $\mathrm{Alt}^k(H;F)$ is not analytic at u_0.

Proof. The inverse family is $g(-u)$. Define bounded linear maps

$$J\colon \mathrm{Alt}^k(D;F) \longrightarrow \mathrm{Alt}^k(H;F), \qquad Jm = m \circ \pi_D^{\times k},$$

and $R\colon \mathrm{Alt}^k(H;F) \to \mathrm{Alt}^k(E;F)$ by restriction along $i_E(e) = (0,e)$. Since $\pi_D g(u) i_E = u$,

$$R \circ g(u)^* \circ J = A(u). \tag{6.1}$$

Bounded linear compression preserves analyticity, so analyticity of $g(u)^*$ at u_0 would contradict the hypothesis. □

Take the base $M = \mathcal{L}(E,D)$ and the trivial bundle $M \times H$. Its product trivialization and the trivialization $(u,z) \mapsto (u, g(u)z)$ form an analytic bundle atlas. The associated alternating transitions are pullback by $g(u)$ and $g(-u)$. Equation (6.1) proves that this induced atlas fails to be analytic whenever A does. Equivalently, the analytic bundle automorphism $(u,z) \mapsto (u, g(u)z)$ does not induce an analytic morphism on the alternating product bundle.

Thus every failure of the hom-space criterion has a bundle realization. Theorem 6.1 supplies Banach bases on which this failure occurs at every point, including the identity operator $g(0)$. This is a statement about the natural construction from an analytic atlas, and about its action on morphisms. The underlying pointwise bundle is trivial and still has an analytic product structure.

6.2. **Tangent bundles and individual forms.** For scalar values the obstruction can be realized by derivatives of analytic coordinate changes, rather than by an independently chosen vector bundle.

Proposition 6.5 (tangent and pullback obstructions)**.** *Suppose K is complete, has characteristic $p > 0$, is not spherically complete, and $k \geq p$.*

1. *There is an analytic Banach manifold with two global analytic charts whose induced transition on scalar alternating k-forms is analytic at no point.*
2. *There are Banach spaces N, Y, a polynomial analytic map $h\colon N \to Y$, and an analytic map $\omega_0\colon Y \to \mathrm{Alt}^k(Y;K)$ such that $h^*\omega_0$, expressed in the product coordinates on N, is analytic at no point as an $\mathrm{Alt}^k(N;K)$-valued map. After inclusion in $\mathrm{Mult}^k(N;K)$ it is analytic everywhere.*

Proof. Choose E, D from Theorem 6.1(2), and write $A = A^k_{E,D;K}$. For (1), put $X = \mathcal{L}(E,D) \times D \times E$ and use the charts id_X and

$$\psi(u,d,e) = (u, d + ue, e).$$

Operator evaluation is bounded bilinear, so ψ and its inverse are polynomial analytic. Its derivative is

$$D\psi(u,d,e)(v,a,b) = (v, a + ve + ub, b).$$

Extend forms from D along the projection $\pi_D\colon X \to D$, and restrict forms on X along $i_E(b) = (0, 0, b)$. These fixed bounded maps, again denoted J, R, give

$$R \circ A^k_{X,X;K}(D\psi(u, d, e)) \circ J = A(u).$$

Analyticity of the transition at (u_0, d_0, e_0) would make A analytic at u_0 after restriction to the affine slice (u, d_0, e_0). The reverse transition gives the same argument with $-u$.

For (2), let $B = \mathrm{Alt}^k(D; K)$, $Y = B \times D$, and $N = \mathcal{L}(E, D) \times B \times E$. Define

$$\omega_0(m, d) = m \circ \pi_D^{\times k}, \qquad h(u, m, e) = (m, ue).$$

The first coefficient map is bounded linear and the second map is polynomial. Since

$$Dh(u, m, e)(v, a, b) = (a, ve + ub),$$

restriction of the pulled-back form to the fixed E-directions gives $\Gamma(u, m) = A(u)(m)$. If $h^*\omega_0$ were alternating-valued analytic at a point, restriction of its values and of its parameter e would make Γ analytic there. Its partial derivative in m is the operator $A(u)$. A power series into a Banach target differentiates on a smaller ball, giving an analytic operator-valued derivative there. Fixing m would therefore make A analytic, a contradiction. In the ambient multilinear space, the pullback formula is a bounded multilinear expression in Dh and $\omega_0 \circ h$, and hence is analytic. □

The second assertion distinguishes a failure of intrinsic analyticity for individual forms from a failure of operator-valued bundle transitions. It also suggests the appropriate calculus when analytic descent is unavailable.

The chart in Proposition 6.5(1) realizes any nowhere analytic action $A^k_{E,D;F}$. With Theorem 6.1(1) it works over every field of characteristic p, and it extends to tangent-valued forms.

Proposition 6.6 (tangent obstructions with other values)**.** *Let K be a nontrivially normed field of characteristic $p > 0$, complete or not, and let $k \geq p$.*

1. *There are a Banach space F and an analytic Banach manifold with two global analytic charts whose induced transition on F-valued alternating k-forms is analytic at no point.*
2. *There is an analytic Banach manifold with two global analytic charts whose induced transition on tangent-valued alternating k-forms is analytic at no point.*
3. *If K is complete and not spherically complete, a manifold as in (2) exists with a nonarchimedean Banach model.*

The target F constructed for (1) admits no equivalent nonarchimedean norm; for $K = \mathbb{F}_q((u))$ as in Corollary C.8, the model space can be chosen nonarchimedean. The model space constructed for (2) admits no equivalent nonarchimedean norm.

Proof. For (1), take E, F from Theorem 6.1(1). Repeat the proof of Proposition 6.5(1) with $D = E$ and values in F. On $X = \mathcal{L}(E, E) \times E \times E$ the same maps J, R give

$$R \circ A^k_{X,X;F}(D\psi(w, d, e)) \circ J = A^k_{E,E;F}(w),$$

which is analytic at no point. The argument uses only bounded linear compression and affine slices, so completeness of K is not needed. For $K = \mathbb{F}_q((u))$ with its u-adic absolute value, Remark E.2 allows $E = \ell^\infty(\mathbb{N}, K)$ and $F = B$, the completed projective exterior power of Theorem C.1, and B has no equivalent nonarchimedean norm by Proposition E.1. Then X is nonarchimedean, since operator norms into a nonarchimedean space and maximum norms of products are nonarchimedean.

For (2) and (3), suppose that $A^k_{E,D;F}$ is analytic at no point. Put $D' = D \times F$ and $X = \mathcal{L}(E, D') \times D' \times E$, with the charts id_X and $\psi(u, d, e) = (u, d + ue, e)$ as before. Write $g = D\psi(u, d, e)$, so that $g^{-1}(v, a, b) = (v, a - ve - ub, b)$. As functions of the point (u, d, e), the induced transitions on tangent-valued forms are $m \mapsto g \circ m \circ (g^{-1}, \dots, g^{-1})$ and $m \mapsto g^{-1} \circ m \circ (g, \dots, g)$. Let $\pi_D \colon X \to D$ and $\pi_F \colon X \to F$ be the components of the middle factor, and put $i_F(y) = (0, (0, y), 0)$ and $i_E(b) = (0, 0, b)$. The bounded linear map

$$\Phi(T)(m) = \pi_F \circ T\big(i_F \circ m \circ \pi_D^{\times k}\big) \circ i_E^{\times k}$$

sends operators on $\mathrm{Alt}^k(X; X)$ to operators $\mathrm{Alt}^k(D; F) \to \mathrm{Alt}^k(E; F)$. For $f, h \in \mathcal{L}(X, X)$ it sends $m \mapsto h \circ m \circ (f, \dots, f)$ to $m \mapsto (\pi_F h i_F) \circ m \circ (\pi_D f i_E, \dots, \pi_D f i_E)$. Here $\pi_F g^{\pm 1} i_F = \mathrm{id}_F$ and $\pi_D g^{\pm 1} i_E = \pm u_D$, where $u_D \in \mathcal{L}(E, D)$ is the D-component of u. So Φ sends the two transitions to $A^k_{E,D;F}(\mp u_D) = (\mp 1)^k A^k_{E,D;F}(u_D)$. If either transition were analytic at (u_0, d_0, e_0), restriction to the affine slice $w \mapsto (u_0 + (w, 0), d_0, e_0)$, $w \in \mathcal{L}(E, D)$, would make $A^k_{E,D;F}$ analytic at the D-component of u_0.

For (2), take E, F from Theorem 6.1(1) and $D = E$. The model X contains the isometric copy $i_F(F)$ of F, so it admits no equivalent nonarchimedean norm. For (3), take E, D from Theorem 6.1(2) and $F = K$. Then X is a nonarchimedean Banach space, as in (1). □

Tangent bundles. For $\mathrm{Alt}^k(TM; F)$ the source fiber is the model space P. Finite coordinates on the base are therefore finite coordinates on the source fiber, and Proposition 4.1(2) already makes the action analytic on whole hom spaces. This is the last case of the first row of Corollary 4.6. Over a finite-coordinate base, Theorem 4.4 and the second row add cases only when the source fibers lack finite coordinates. Examples are source bundles of infinite rank, and pulled-back tangent bundles f^*TN along analytic maps $f \colon M \to N$ from a finite-coordinate manifold M into an infinite-dimensional manifold N.

Over a discretely valued field, Corollary 4.3 makes $\mathrm{Alt}^k(TM; F)$ analytic on every analytic manifold for every Banach F with an equivalent nonarchimedean norm, as noted after Corollary C.8. It also makes tangent-valued forms analytic when the model is such a space. So over $\mathbb{F}_q((u))$ the target in Proposition 6.6(1) cannot have an equivalent nonarchimedean norm, while the manifold can be modeled on a nonarchimedean Banach space. This is consistent with (3): a complete discretely valued field is spherically complete (Lemma D.1).

Bourbaki's question. In the terms of [3], Alt^k on Banach spaces is the functor α_k of [3, n° 7.8.1] for $k \geq 1$, and $A^k_{E,D;F}(u) = \alpha_k(\mathrm{id}_F, u)$. There the field is $\mathbb{R}$, $\mathbb{C}$ or a complete nonarchimedean field, the spaces are Banach spaces, norms over a nonarchimedean field are required to be nonarchimedean, and class C^r means analytic when $K \neq \mathbb{R}$, in a sense that agrees with ours for maps between Banach spaces. In that setting α_k is a vector functor of class C^r if $k! \neq 0$ in K (Corollary 6.2), and for every k if K is discretely valued (Corollary 4.3 and the proof of Corollary 4.6). It is not one if $k! = 0$ in K and K is not spherically complete: Theorem 6.1(2) gives nonarchimedean Banach spaces E, D for which $A^k_{E,D;K}$ is analytic at no point. So the definition in [3, n° 7.8.1] cannot be made without any restriction. Propositions 6.4 and 6.5 show that over such a field the restriction is needed for the bundles themselves: the alternating atlas induced from an analytic bundle, or from the tangent bundle of an analytic manifold, need not be analytic. Our results do not decide whether α_k is a vector functor of class C^r over a densely valued, spherically complete field with $k! = 0$; compare Section 9.

A derivative adapted to positive characteristic. Yaël Dillies asked whether $A = A^k_{E,E';F}$ becomes analytic in a differential calculus that keeps Taylor coefficients in every characteristic. Such a calculus exists. Let f be a map from an open subset of a Hausdorff topological vector space to another, over a Hausdorff nondiscrete topological field. Bertram, Glöckner and Neeb [2] call f

of class C^1 if the difference quotient $t^{-1}\big(f(x+th)-f(x)\big)$ extends continuously to $t=0$ as a function of (x,h,t), and of class C^n, $n\geq 2$, if the extended quotient is of class C^{n-1}; the characteristic is arbitrary. A map of class C^k has a unique expansion [2, Theorem 5.1]

$$f(x+th)=\sum_{n=0}^{k} t^n a_n(x,h)+t^k\vartheta(x,h,t)$$

with ϑ continuous and $\vartheta(x,h,0)=0$. Here $a_n(x,\cdot)$ is homogeneous of degree n, and the nth differential $d^nf(x)$, an iterated directional derivative, satisfies [2, Definition 4.7 and Theorem 5.4]

$$n!\,a_n(x,h)=d^nf(x)(h,\dots,h). \tag{6.2}$$

The usual coefficient $d^nf(x)(h,\dots,h)/n!$ requires $2,\dots,k$ to be invertible in the field; see also [1, Chapter 1].

The map A is smooth in this calculus, over every K. The difference quotient of a finite sum of diagonals of continuous multilinear maps is again such a sum, so the ambient map $u\mapsto\big(m\mapsto m\circ(u,\dots,u)\big)$ is smooth [2, 3.4 and Remark 4.2], and so is A by [2, Lemma 10.1], the counterpart of Lemma 3.2. The expansion of A at u_0 has no remainder:

$$A(u_0+tu)=\sum_{n=0}^{k} t^n a_n(u_0,u),\qquad a_n(u_0,u)(m)=\sum_{\substack{S\subseteq\{1,\dots,k\}\\ |S|=n}} m\circ(u_1^S,\dots,u_k^S),$$

where $u_i^S=u$ for $i\in S$ and $u_i^S=u_0$ for $i\notin S$. By Theorem 3.1, each $a_n(u_0,u)$ maps $\mathrm{Alt}^k(E';F)$ into $\mathrm{Alt}^k(E;F)$, and $a_k(u_0,u)=A(u)$. By (6.2), $d^kA(u_0)(u,\dots,u)=k!\,A(u)$, which is zero when $k!=0$ in K. The coefficient a_k keeps the map that the diagonal of the kth differential loses.

Analyticity is not restored, because it is not a condition on derivatives. In this paper, and for the analytic maps of [2, 7.14–7.18], each homogeneous term of the series must be the diagonal of a continuous multilinear map, and convergence is measured by the norms of such maps. Bertram, Glöckner and Neeb note that in positive characteristic, where no polarization formula is available, this requirement "might very well be stronger than the mere continuity" of the term [2, footnote to 7.14]. For the term A it can be stronger. By Proposition 3.3, A is analytic at a point if and only if it is the diagonal of a bounded k-linear map into $\mathcal{L}\big(\mathrm{Alt}^k(E';F),\mathrm{Alt}^k(E;F)\big)$, and when $k!=0$ in K, Theorem 6.1 gives spaces for which no such map exists. In part (2) of that theorem the field is complete and nonarchimedean, and the domain and the target of A are nonarchimedean normed spaces, as in [2, 7.13]. There A is continuous, and it is the diagonal of a k-linear map into the alternating operator space (compose the ambient k-linear map with a linear projection onto that space), but of no continuous one.

If the terms of a series were required only to be continuous algebraic homogeneous polynomials, as in Proposition I.2, A would be a polynomial in every characteristic. We do not know whether maps given by such series are closed under composition. In the generality of [1, 2] there are no bundles of forms: dual bundles are not constructed, and a differential form is a map on a fiber product of the tangent bundle, alternating and multilinear on each fiber, rather than a section of a bundle [2, §8.3], [1, 4.5, 22.1 and Appendix L.1]. The ambient forms of Section 7 avoid only the bundle of alternating maps: they are sections of the bundle of multilinear maps.

7. Analytic differential forms

An alternating form can be analytic as a multilinear-valued function even when it has no analytic expansion with alternating-valued coefficients (Proposition 6.5(2)). The ambient viewpoint retains exactly the regularity needed for the usual operations on differential forms.

Throughout this section K is complete and nontrivially normed. Manifold models may be arbitrary normed spaces. We use C^ω analytic charts and maps, so their derivatives are analytic also for incomplete models. Coefficient spaces of scalar multilinear forms are Banach, where this agrees with power-series analyticity. No characteristic or degree restriction is imposed.

Definition 7.1. An *ambient analytic scalar k-form* on M is a family $\omega_x \in \mathrm{Alt}^k(T_xM;K)$ whose representative in each chart is analytic after the inclusion

$$j\colon \mathrm{Alt}^k(P;K) \hookrightarrow \mathrm{Mult}^k(P;K).$$

Write $\Omega^k_{\mathrm{amb}}(M)$ for the space of such forms. In degree zero this is the space of analytic scalar functions.

This is a bundle definition: ω is an analytic section of the multilinear bundle with alternating values. The multilinear bundle is always analytic, since its operator-valued transitions have bounded multilinear lifts. Concretely, under a coordinate change ψ the representatives satisfy

$$j\omega_\phi(y) = (j\omega_{\phi'}(\psi(y))) \circ (D\psi(y), \dots, D\psi(y)).$$

The right-hand side is analytic by bounded multilinear composition. It therefore suffices to check the definition on any analytic atlas, and adjoining further analytic charts does not change the space of forms.

Theorem 7.2 (ambient differential calculus)**.** *Ambient analytic forms define a sheaf of graded algebras on every analytic manifold. Analytic maps induce pullbacks, and there is an exterior derivative*

$$d\colon \Omega^k_{\mathrm{amb}}(M) \longrightarrow \Omega^{k+1}_{\mathrm{amb}}(M).$$

The product is associative and graded commutative, $d^2 = 0$, pullback commutes with both product and exterior derivative, and

$$d(\eta \wedge \zeta) = d\eta \wedge \zeta + (-1)^k \eta \wedge d\zeta \qquad (\eta \in \Omega^k_{\mathrm{amb}}(M)).$$

Thus $\Omega^\bullet_{\mathrm{amb}}$ is a contravariant differential graded algebra construction in every characteristic.

Proof. Locality of analytic coefficient maps gives the sheaf property. Addition and multiplication by an analytic function are bounded multilinear operations on the coefficient spaces. For an analytic map h, the formula

$$h^*\eta = (\eta \circ h) \circ (Dh, \dots, Dh)$$

is analytic in the ambient space and has alternating values. The chain rule gives identity and composition laws for pullback.

To define the product, for $S \subseteq \{1, \dots, k+l\}$ let v_S list its entries in increasing order, and set

$$\epsilon(S) = (-1)^{\#\{(s,t):s\in S,\ t\notin S,\ s>t\}}.$$

Use the shuffle formula

$$(\mu \wedge \nu)(v_1, \dots, v_{k+l}) = \sum_{|S|=k} \epsilon(S)\mu(v_S)\nu(v_{S^c}).$$

It defines a bounded bilinear operation on multilinear coefficient spaces, of norm at most $\binom{k+l}{k}$. For alternating inputs it is alternating: terms with two repeated arguments in one block vanish,

and the remaining terms cancel in pairs by exchanging those arguments between blocks. This argument includes characteristic two. The formula commutes with simultaneous precomposition and hence glues in charts. Complementing S proves graded commutativity. Expanding either bracketing of a triple product yields the same signed sum over ordered three-block partitions, proving associativity.

On an open subset of a model space P, write η for an ambient coefficient map. Define

$$(d\eta)_y(v_0,\dots,v_k)=\sum_{i=0}^{k}(-1)^i D\eta(y)(v_i)(v_0,\dots,\widehat{v_i},\dots,v_k). \tag{7.1}$$

The operation applied to $D\eta$ is bounded linear with norm at most $k+1$, so the result is ambient analytic. Each $D\eta(y)(v_i)$ is alternating: difference quotients have alternating values, and that subspace is closed. If $v_a=v_b$, the summands other than $i=a,b$ vanish; alternation moves the remaining repeated argument across $b-a-1$ positions, making those two terms cancel. Hence $d\eta$ is a form.

The second derivative of an analytic map is symmetric in every characteristic. At the center of a power series, its quadratic contribution is $p_2(v,w)+p_2(w,v)$. In the expansion of $d(d\eta)$, pair the terms differentiated in directions (v_a,v_b) and (v_b,v_a) for $a<b$. Their values agree and their signs are $(-1)^{a+b-1}$ and $(-1)^{a+b}$, so $d^2\eta=0$.

Differentiate the pullback formula. Terms differentiating η give $h^*(d\eta)$ after the signed sum (7.1). The other terms insert $D^2h(v_a,v_b)$ into one argument of η. Symmetry of D^2h and alternation of η cancel the terms in pairs. Therefore $d(h^*\eta)=h^*(d\eta)$; applying this identity to chart changes proves that the local derivatives glue. Finally, differentiating the shuffle formula gives the Leibniz rule: placing the derivative index in the second block moves it past the first k arguments and contributes $(-1)^k$. Every identity uses finite sums and cancellation, with no division by a factorial. □

The same proof gives forms with values in a commutative Banach algebra; for a general bounded bilinear product β all properties except associativity and graded commutativity carry over. For such a product the shuffle bound is $\binom{k+l}{k}\|\beta\|$. The scalar case already supplies the differential graded algebra relevant to the cotangent construction.

Proposition 7.3 (comparison with alternating-bundle sections)**.** *Every intrinsically analytic alternating-valued coefficient map is ambient analytic. If, for the model space P, the inclusion j has a bounded linear retraction, then the natural alternating cotangent atlas is analytic and its analytic sections are precisely the ambient analytic forms. The same conclusion holds when P has finite continuous coordinates.*

Proof. The first statement follows by composition with j. For a bounded retraction R, the identity $\eta=R(j\eta)$ gives the converse. Moreover,

$$(u_1,\dots,u_k)\longmapsto\big[m\longmapsto R\big(m\circ(u_1,\dots,u_k)\big)\big]$$

is a bounded multilinear lift of the precomposition action. Thus the induced alternating cotangent transitions are analytic. For finite continuous coordinates, analytic descent through closed subspaces applies to the coefficient map itself, and Corollary 4.6 supplies the analytic bundle atlas. □

The retraction hypothesis holds for invertible $k!$, by normalized antisymmetrization, and for spherically complete nonarchimedean K, by Theorem 4.2. These are sufficient conditions, not a necessity claim. An arbitrarily chosen analytic structure on a pointwise alternating bundle is weaker: the product bundles in Proposition 6.5(2) have such structures, yet pullback fails to

preserve their intrinsically analytic sections. The ambient definition avoids that dependence on an unjustified analytic descent, while preserving the geometric operations that motivate differential forms.

8. A proposed Mathlib interface

The preceding results suggest a single interface for constructing alternating bundles at every regularity. Gouëzel's proposed construction [5] uses characteristic zero to prove regularity of the induced transition operators. Its assembly through `VectorPrebundle.IsContMDiff` can instead use the family-preservation premise below. The general constructor and its finite-coordinate application are proved in [10]. The proof class and its instance are a proposal: they are not Mathlib declarations and are not in [10].

8.1. **The family-preservation premise.** Fix a base M, its model with corners I, and a regularity $n \in \mathbb{N} \cup \{\infty, \omega\}$. Let E, E', F, F' be normed model fibers and put

$$\mathcal{O} = \mathcal{L}(E', E) \times \mathcal{L}(F, F'), \qquad \mathcal{T} = \mathcal{L}\big(\mathrm{Alt}^k(E; F), \mathrm{Alt}^k(E'; F')\big).$$

Write $B\colon \mathcal{O} \to \mathcal{T}$ for the joint alternating action

$$B(u, v)(a) = v \circ a \circ (u, \dots, u).$$

The proposed premise is

$$U \subseteq M \text{ open}, \quad \gamma\colon M \to \mathcal{O}, \quad \gamma|_U \text{ is } C^n \text{ relative to } I \quad \Longrightarrow \quad (B \circ \gamma)|_U \text{ is } C^n. \tag{8.1}$$

The conclusion is regularity in the operator space $\mathcal{T}$. Both the base M and its model I are parameters of the premise: I alone does not determine M. The four model fibers allow the same interface to handle morphisms between different alternating bundles. For the bundle with source model F_1 and value model F_2, the required specialization is $E = E' = F_1$ and $F = F' = F_2$.

The following complete definition uses the joint action `alternatingMapAction` from our development. All parameters and typeclass hypotheses are explicit; the local names `Op` and `Out` abbreviate $\mathcal{O}$ and $\mathcal{T}$. The definition typechecks against [10]. Stating this predicate requires a charted base, but no completeness or manifold-regularity assumption.

```
import AlternatingAnalytic.Geometry.FiniteCoordinateManifoldFamilies

open scoped Manifold ContDiff
open AlternatingAnalytic

def PreservesAlternatingFamilies
    {K P H : Type*} [NontriviallyNormedField K]
    [NormedAddCommGroup P] [NormedSpace K P]
    [TopologicalSpace H] (I : ModelWithCorners K P H)
    (n : ℕ∞ω) (M : Type*)
    [TopologicalSpace M] [ChartedSpace H M]
    (k : ℕ) (E E' F F' : Type*)
    [NormedAddCommGroup E] [NormedSpace K E]
    [NormedAddCommGroup E'] [NormedSpace K E']
    [NormedAddCommGroup F] [NormedSpace K F]
    [NormedAddCommGroup F'] [NormedSpace K F'] : Prop :=
  let Op := (E' →L[K] E) × (F →L[K] F')
  let Out := (E [⋀^Fin k]→L[K] F) →L[K]
    (E' [⋀^Fin k]→L[K] F')
  ∀ {U : Set M} {γ : M → Op}, IsOpen U →
    ContMDiffOn I 𝓘(K, Op) n γ U →
    ContMDiffOn I 𝓘(K, Out) n
      (fun x ↦ alternatingMapAction k (γ x)) U
```

With the same ambient hypotheses, the premise can be packaged as a proof class. The class and bundle-interface sketches below suppress the repeated ambient binders; `Alt k X Y` abbreviates the continuous alternating-map type indexed by `Fin k`. These sketches specify the proposed organization rather than standalone Lean programs.

```
class AlternatingFamilyRegularity
    (I) (n) (M) (k) (E E' F F') : Prop where
  preserves : PreservesAlternatingFamilies
    I n M k E E' F F'
```

This is a proof class. It contains no choice of norm, retraction, coordinates, bundle topology, or multilinear lift. Such data may be used to prove its field, but the interface records only (8.1). In particular, it does not require B to be C^n on its whole operator domain: that stronger condition would fail to capture the finite-base analytic theorem in its full generality.

8.2. **The constructor and the single instance.** The theorem `contMDiffVectorBundle_alternating_of_family` takes the family-preservation proof as an ordinary argument. It does not perform typeclass search for a reason that the premise holds. Given input bundle transitions, it pairs the reversed source transition with the forward target transition, applies the premise on their open overlap, and identifies the resulting operator with the induced alternating transition. This proves `VectorPrebundle.IsContMDiff`; Mathlib's theorem `VectorPrebundle.contMDiffVectorBundle` then supplies the desired regularity on the existing bundle.

With the same abbreviations, the proposed public assembly has the following form:

```
theorem contMDiffVectorBundle_alternating_of_family
    (k : ℕ)
    (hfamily : PreservesAlternatingFamilies
      I n M k F₁ F₁ F₂ F₂)
    [ContMDiffVectorBundle n F₁ E₁ I]
    [ContMDiffVectorBundle n F₂ E₂ I] :
    ContMDiffVectorBundle n (Alt k F₁ F₂)
      (fun x ↦ Alt k (E₁ x) (E₂ x)) I := ...

instance ContMDiffVectorBundle.continuousAlternatingMap
    [ContMDiffVectorBundle n F₁ E₁ I]
    [ContMDiffVectorBundle n F₂ E₂ I]
    [h : AlternatingFamilyRegularity
      I n M k F₁ F₁ F₂ F₂] :
    ContMDiffVectorBundle n (Alt k F₁ F₂)
      (fun x ↦ Alt k (E₁ x) (E₂ x)) I :=
  contMDiffVectorBundle_alternating_of_family k h.preserves
```

Here E_1, E_2 are the actual fiber families over M, whereas F_1, F_2 are their normed model fibers. The regularity premise mentions only the latter and the base. The constructor reuses the canonical alternating `FiberBundle` and `VectorBundle` structures; it does not introduce another topology or another atlas construction. Its preliminary prebundle proof can remain a named theorem or a local instance inside the proof. Only the final assembly above needs to be registered as the alternation-specific bundle instance.

An application with a specialized family argument may call the constructor directly. Alternatively, it may install a local proof of `AlternatingFamilyRegularity` and let the single instance perform the assembly. Neither usage requires a new global bundle instance.

8.3. **Supplying the regularity proofs.** The sufficient conditions proved in this paper belong at the family layer. They supply the following proof constructors, with commonly used ones eligible for automatic instances:

(1) For $n \le \infty$, Lemma 3.2 and the ambient polynomial action prove preservation over every nontrivially normed field, for arbitrary normed model fibers and every model with corners.

(2) If $k! \neq 0$ in K, Proposition 4.1 proves preservation at every regularity, including ω. Characteristic zero supplies this condition in every degree. Part (2) of the same proposition proves the same when a source model fiber E or E' has finite continuous coordinates; this covers $\mathrm{Alt}^k(TM; F)$ over a finite-coordinate base.

(3) For ordinary open-chart models $I = \mathcal{I}(K, P)$, continuous coordinates $P \simeq_L K^d$ give analytic family preservation with arbitrary normed fibers (Theorem 4.4). The manifold application assumes the corresponding analytic manifold structure. A convenience

theorem can obtain the coordinates from completeness of K and finite dimensionality of P; the primitive coordinate theorem needs neither completeness of the field nor completeness of the fibers.

(4) Over a nonarchimedean field, a nonarchimedean spherically complete final target F' gives preservation at every regularity by Theorem 4.2. The summable-parameter results of Theorem 4.5 give further model-specific proof constructors under their stated target hypotheses.

No blanket Banach assumption belongs in the interface. Nor should the finite-coordinate theorem be silently extended to arbitrary models with corners or arbitrary-set `ContMDiffWithinAt` statements: the argument supplied here uses open parameter domains.

The distinction between a property of an action and a property of its input families also matters for regularity reduction. Preservation of analytic families does not, merely by lowering the regularity of the output, prove preservation of arbitrary C^n families: the latter need not be analytic. Thus one should not copy the regularity-lowering instances for `ContMDiffVectorBundle` into the new family class without a separate theorem. The smooth family proof is available independently from closed-range reflection. By contrast, once a particular bundle has been proved analytic, its lower-order bundle regularity follows through Mathlib's existing instances.

8.4. **Coherence, morphisms, and verification scope.** Mathlib's `ContMDiffVectorBundle` is itself a `Prop` class: it asserts regularity of the transition operators and carries no new bundle data. Proof irrelevance identifies different proofs of the same proposition [8]. Consequently overlapping proofs of the proposed family class cannot create incompatible bundle topologies. They can still introduce alternative typeclass-search paths. The design therefore gives one alternation-specific assembly instance, not a claim that instance search has a unique path: ordinary regularity-lowering instances and the general order-zero instance already provide other paths.

The instance graph should remain directed from analytic hypotheses to family regularity and then to bundle regularity. Its family proofs should not ask for the output alternating-bundle instance. Converse results should remain theorems, rather than reverse search rules. Specialized cases can remain named proof constructors when automatic search would add unnecessary branching; priorities and elaboration behavior of the common cases must be checked when implementing the proposal.

The four-endpoint premise also supports the morphism construction of Section 2. Given regular input morphisms and regular alternating bundles at both endpoints, apply it to their local operator coordinates to obtain a regular morphism between the alternating bundles. The self-endpoint premise used for a bundle instance alone does not establish this cross-endpoint condition. The development [10] implements this distinction in `alternatingBundleHom_of_family`.

For an individual chosen atlas the family premise is sufficient and may be stronger than necessary; only the transition families occurring in that atlas must be regular. Universally over all normed fibers and all compatible input frames, its necessity is detected by the shear construction of Proposition 6.4. An arbitrary family $u(x)\colon E \to D$ factors as

$$u(x) = \pi_D S_x i_E, \qquad S_x(d,e) = (d + u(x)e, e), \qquad S_x^{-1}(d,e) = (d - u(x)e, e).$$

The two frames of the trivial $(D \oplus E)$-bundle defined by the identity and S_x test the induced action on S_x; fixed bounded compression then recovers the action on $u(x)$. Together with bounded bilinear postcomposition this recovers the joint alternating family. This converse requires the universal quantifiers and the enlarged source fiber, and is not a converse instance for one fixed bundle or one fixed pair of models.

For the ordinary model $I = \mathcal{I}(K, P)$ and a nonempty C^n manifold M, whether (8.1) holds depends only on P, n, k and the model fibers, not on M. Regularity on M is checked in charts. Conversely, a family on an open subset of P near x_0 can be composed with $y \mapsto x_0 + \lambda(y - y_0)$, for $\lambda \in K^\times$ small, and with a chart of M whose image contains a ball around y_0; the premise on M then gives regularity near x_0. With the shear converse, if the alternating construction preserves C^n bundles with all normed fibers over one nonempty manifold modeled on P, it does so over every such manifold.

The theorem kernel in [10] consists of the first three declarations below; the fourth is the supplied-coordinate application.

```
contMDiffOn_continuousAlternatingMapCoordChange_of_family
alternatingVectorPrebundle_isContMDiff_of_family
contMDiffVectorBundle_alternating_of_family
contMDiffVectorBundle_alternating_of_finiteCoordinates
```

These declarations use `Fin k`. An upstream presentation should retain Mathlib's arbitrary finite index type, transporting the coordinate-based proofs along a bijection with `Fin k`, where k is its cardinality. The new proof class, its automatic instances, this index-type adaptation, and the proposed search organization have not been implemented or checked as an upstream patch. They are separate from the verification claims in Section 9.

9. Further questions and verification

The descent formulation leaves three closely related questions in the obstructed range $\operatorname{char} K = p > 0$, $k \geq p$.

Which fibers remove the obstruction? Call F a universal target if $\mathrm{Alt}^k(-; F)$ is analytic on every hom space. A nonarchimedean, spherically complete norm is sufficient, as is any equivalent norm of that kind. Over a complete field that is not spherically complete, however, every universal target must have zero continuous dual. Indeed, if $\lambda(v) = 1$ for $\lambda \in F^*$ and $v \in F$, then

$$A^k_{E,D;K}(u)(a) = \lambda \circ A^k_{E,D;F}(u)(va).$$

The scalar counterexample excludes analyticity of the right-hand action. An intrinsic characterization of universal targets remains open; zero dual alone has not been shown sufficient. Remark I.3 shows separately that spherical completeness of the given norm is not necessary: an equivalent spherically complete norm can suffice. Over a densely valued, spherically complete field, it is also open whether every nonarchimedean Banach target is universal. The general-target obstruction has no equivalent nonarchimedean norm and does not decide this question.

Which bases permit descent? The finite-coordinate, c_0 and ℓ^1 arguments impose different conditions on the final target. The remaining problem is to characterize model spaces P for which every analytic family of operators parametrized by an open subset of P is preserved by Alt^k, separately for arbitrary normed, complete, and complete nonarchimedean targets. Over $\mathbb{F}_q((u))$ with a u-adic absolute value, Corollary C.8 answers this for arbitrary normed targets and Banach models with an equivalent nonarchimedean norm: exactly the finite-dimensional models qualify. Over a complete field, universal Banach-valued analytic reflection through closed linear inclusions is sufficient for complete targets. Over a complete nonarchimedean field it is stronger than necessary for this fixed-degree action, by Proposition G.4 and Theorem 4.5(2). Over $\mathbb{R}$ and $\mathbb{C}$ every Banach space has it, by symmetrization (Appendix G). For Banach parameter spaces, Theorem G.3 characterizes this reflection property by exponentially bounded projections onto the closed spans of pure powers in projective tensor powers. The proof and

the sharp factorial projection bounds of Proposition G.4 appear in Appendix G; they explain why the fixed-degree ℓ^1 argument goes beyond universal analytic reflection.

When do the two notions of analytic form coincide? For a fixed model P, a bounded retraction onto alternating forms gives both an analytic cotangent atlas and equality of ambient and intrinsic analytic forms. It is not known whether compatibility with every analytic coordinate change on P already forces that equality. A chosen analytic product structure is insufficient, by Proposition 6.5. The question concerns the natural atlas and its full class of coordinate changes.

These questions concern fibers, bases and sections within the same descent problem. Restricting the objects to a largest full analytic domain is a different extremal problem. Theorem H.4, proved in Appendix H, shows that such a largest domain need not exist over the incomplete field $\mathbb{F}_p(t)$: two objects may each have an analytic endomorphism action while their cross-action is nonanalytic. The construction uses finite algebraic dimension without continuous coordinates. Whether an analogous failure occurs over complete fields or among Banach pairs remains open.

Formal verification. The Lean proofs are in [10]. Its file `challenge.md` is a ledger of 76 claims of this paper. Each claim has a challenge file, stating it in Lean with `sorry` in place of the proof, and a solution file proving the identical statement. The command `python3 scripts/check_claim.py ID`, with `ID` such as `Thm4_4`, builds both files, checks that the proof uses only the axioms `propext`, `Classical.choice` and `Quot.sound`, and checks that the two statements agree when printed with all implicit arguments shown (`pp.all`). All 76 claims pass. The check uses Lean `4.34.0-rc2` and the Mathlib fork `Deicyde/mathlib4` at `2b73d98`, which is upstream `0383a80e64` plus the commit of [11]. The six statements of the earlier comparator challenge in the same repository pass the comparator `https://github.com/leanprover/comparator` at revision `19e111e`.

Part	Ledger claims	Number
Sections 2–3	2.1–3.3	5
Section 4	4.1–4.6, with both parts of 4.5	7
Section 5	5.1, 5.2	2
Section 6	both parts of 6.1; 6.2–6.5; the bundle statement after 6.4	7
Section 7	7.2, 7.3	2
Section 9	zero dual of universal targets	1
Appendices A–B	A.1–A.3, B.2–B.13, B.15	16
Appendix C	C.1–C.4, C.6–C.8	7
Appendices D–E	D.1–D.9, D.11, E.1, E.2	12
Appendix F	F.1–F.5	5
Appendix G	G.1, G.3–G.5	4
Appendix H	H.3–H.7	5
Appendix I	I.1–I.3	3

Apart from Proposition 6.6, the ledger covers every theorem, proposition, lemma and corollary. It also covers three remarks, the bundle statement after Proposition 6.4, and one claim of this section. It does not cover the definitions or Remarks B.14, C.5 and D.10. Proposition 6.6, the coefficient bound on retracts in Theorem 4.5(1), and the remaining unnumbered arguments of the text are outside the ledger. Apart from the declarations named in Section 8, they have written proofs only.

Some ledger statements differ from the text, and `challenge.md` and the docstrings of the challenge files list the differences. The main ones are these. Theorem 7.2 and Proposition 7.3 are stated on open subsets of a normed space, since Mathlib has no differential forms on manifolds. The Lean statement of Theorem 2.1 groups the variables by variance, uses the canonical

trivializations, and does not state independence of the trivializing cover. Corollary C.7 is stated for the explicit transition and section maps, and Corollary 4.6 for objects and morphisms without the functor laws.

Appendix A. Preliminaries for the obstruction proofs

The constructions below use three elementary facts about valued fields and power series. We record their proofs to make explicit the scope of the coefficient argument, including incomplete base fields.

Lemma A.1 (one-variable uniqueness)**.** *Let K be nontrivially normed and Z a normed K-space, not necessarily complete. Let $z_0, z_1, \ldots \in Z$, $\rho > 0$ and $M \geq 0$ with $\|z_n\|\rho^n \leq M$ for all n. If $\sum_n t^n z_n$ converges to 0 for every $t \in K$ with $|t| < \rho$, then $z_n = 0$ for all n.*

Proof. Otherwise let N be least with $z_N \neq 0$. For $0 < |t| < \rho/2$ the series $\sum_{n>N} t^{n-N} z_n$ converges, being t^{-N} times the convergent tail $\sum_{n>N} t^n z_n$, and $t^N\big(z_N + \sum_{n>N} t^{n-N} z_n\big) = 0$. Hence

$$\|z_N\| = \Big\| \sum_{n>N} t^{n-N} z_n \Big\| \leq \sum_{n>N} M\rho^{-n}|t|^{n-N} = M\rho^{-N}\frac{|t|/\rho}{1-|t|/\rho},$$

where the norm of a convergent series is bounded by the sum of the norms by continuity of the norm on partial sums. Since K is nontrivially normed, there are $t \neq 0$ with $|t|$ arbitrarily small, so $\|z_N\| = 0$, a contradiction. □

Lemma A.2. *A normed field of characteristic $p > 0$ is nonarchimedean.*

Proof. Frobenius and the triangle inequality give, for every $m \geq 0$,

$$|x+y|^{p^m} = |x^{p^m} + y^{p^m}| \leq 2\max(|x|,|y|)^{p^m}.$$

Taking p^m-th roots and letting $m \to \infty$ proves the assertion. □

Lemma A.3. *Let $\mathbb{K}$ be a complete nontrivially normed field and U a finite-dimensional normed $\mathbb{K}$-space. Then every linear functional on U is bounded, and U is complete.*

Proof. We prove both assertions by induction on $n = \dim U$, the case $n = 0$ being immediate. Suppose $n \geq 1$ and let $g \neq 0$ be a linear functional on U. Its kernel H has dimension $n-1$, so the induction hypothesis makes H complete and therefore closed in U. Choose v with $g(v) = 1$. Since $v \notin H$, $\delta := \inf_{h\in H}\|v-h\| > 0$. Write $x = g(x)v + h$ with $h \in H$. If $g(x) \neq 0$, then

$$\|x\| = |g(x)|\,\|v + g(x)^{-1}h\| \geq |g(x)|\delta.$$

Thus $|g(x)| \leq \delta^{-1}\|x\|$ for every x, proving boundedness.

Choose a basis of U. Its coordinate functionals are now bounded, so the coordinate map $U \to \mathbb{K}^n$, with the max norm on $\mathbb{K}^n$, is bounded. The inverse map $(c_i) \mapsto \sum_i c_i e_i$ is bounded by $\sum_i \|e_i\|$. Hence U is bi-Lipschitz equivalent to the complete space $\mathbb{K}^n$, and is itself complete. □

When the scalar field needs to be displayed, $A^{k,L}$ denotes the degree-k precomposition action formed over L. For the precomposition action A, a *bounded lift* means a bounded k-linear map on its operator domain whose diagonal is A, as in Proposition 3.3. That proposition applies at every base point; Lemma A.1 provides the one-variable coefficient comparison used in its proof.

Appendix B. The multiplier theorem over a finite field

This section proves the algebraic obstruction used in the Laurent-series construction: over a finite field in which $k! = 0$, coordinatewise multiplication admits no multilinear lift satisfying a polarized diagonal identity and a uniform bound on exterior support. We give a roadmap of the complete construction, introduce the exterior support estimates, and then distinguish the diagonal identities: over a finite field equality at all scalar points need not be a polynomial identity.

Roadmap of the Banach counterexample. The proof separates into an algebraic obstruction, its realization by a normed space, and passage to the prescribed field. The analytic input is a lift criterion valid in every characteristic: if A is analytic at any point, the degree-k term of its power series has diagonal A. One-variable uniqueness identifies this diagonal without dividing by $k!$ (Proposition 3.3).

The finite-field obstruction. Let κ be finite and put $E_0 = \kappa^{\mathbb{N}}$. Theorem B.9 rules out a multilinear lift of coordinatewise multiplication satisfying the polarized diagonal identity (Pol1), with values in $\Lambda^k_\kappa E_0$ when $k! = 0$ and every value of the lift lies in an exterior power of a subspace of uniformly bounded dimension. Ramsey's theorem first makes the coefficients depend only on the order and equality pattern of their indices. Signed sums over clusters of four indices cancel the terms not tied to the output. A staircase of clusters converts the support bound into invariance under adjacent transpositions; the polarized diagonal identity then forces $k!\,\chi = 1$ for a common cluster value χ, a contradiction.

The Laurent series construction. Over $K_1 = \kappa((X))$, take $E_1 = \ell^\infty(\mathbb{N}, K_1)$ and let B be the completion of $\Lambda^k_{K_1} E_1$ under its projective norm, defined using ordinary sums rather than maxima. The essential estimate is that the constant-coefficient map

$$\eta\colon B \longrightarrow \Lambda^k_\kappa E_0$$

is well defined and has uniformly bounded support dimension on bounded sets (Proposition C.3). Evaluating a hypothetical lift on multiplication operators and on the universal wedge map, then applying η, produces exactly the finite-field lift excluded above. The coefficient map is κ-linear; no K_1-linearity is asserted or needed.

Passage to the prescribed field. Choose $t \in K$ with $0 < |t| < 1$. The closure of $\mathbb{F}_p(t)$ in the completion $\widehat{K}$ is a spherically complete Laurent series field $K_1 \cong \mathbb{F}_p((X))$, with $|X| = |t|$. Ingleton's Hahn–Banach theorem [6] gives a contracting K_1-linear projection $\widehat{K} \to K_1$. It makes the completed projective tensor products

$$E = E_1 \widehat{\otimes}_{\pi,K_1} \widehat{K}, \qquad F = B \widehat{\otimes}_{\pi,K_1} \widehat{K}$$

contain E_1 and B isometrically, and allows a lift over $\widehat{K}$ to descend to one over K_1. Density handles an incomplete K. These are the explicit witnesses in Theorem 6.1(1). Finally, sums of unit wedges on disjoint blocks have norms growing linearly with the number of blocks. This growth persists in F and rules out every equivalent nonarchimedean norm (Proposition E.1).

B.1. **Exterior powers, supports and determinant arrays.** Let L be a field, V an L-vector space and $k \geq 1$. We write $\Lambda^k V = \Lambda^k_L V$ for the algebraic exterior power, in which $x_1 \wedge \cdots \wedge x_k = 0$ whenever two x_i coincide. We use the standard facts: for a subspace $W \subseteq V$ the map $\Lambda^k W \to \Lambda^k V$ is injective, and we regard $\Lambda^k W$ as a subspace of $\Lambda^k V$; if $u_1, \ldots, u_d$ is a basis of a finite-dimensional space U, the wedges $u_I := u_{i_1} \wedge \cdots \wedge u_{i_k}$, $I = \{i_1 < \cdots < i_k\}$, form a basis of $\Lambda^k U$; and every $\omega \in \Lambda^k V$ lies in $\Lambda^k U$ for some finite-dimensional $U \subseteq V$.

Definition B.1. The *support dimension* of $\omega \in \Lambda^k V$ is

$$\operatorname{sdim}(\omega) := \min\{\dim W : \ W \subseteq V \text{ finite-dimensional}, \ \omega \in \Lambda^k W\}.$$

The definition gives $\operatorname{sdim}(0) = 0$, $\operatorname{sdim}(c\omega) = \operatorname{sdim}(\omega)$ for $c \neq 0$, $\operatorname{sdim}(\omega + \omega') \leq \operatorname{sdim}(\omega) + \operatorname{sdim}(\omega')$ (use $W + W'$), and $\operatorname{sdim}(\omega) \leq \dim \operatorname{span} S$ if ω is a sum of wedges of vectors from a finite set S. If $V \subseteq V'$, then sdim computed in V' is at most sdim computed in V.

For spaces of functions, exterior vectors can be detected by determinants of coordinate evaluations. These arrays also turn bounds on support dimension into ordinary matrix-rank bounds. Let S be a set and $V \subseteq L^S$ a space of functions on S. The *determinant array* is the linear map

$$\Omega\colon \Lambda^k V \to L^{S^k}, \qquad \Omega_{y_1 \wedge \cdots \wedge y_k}(c_1, \ldots, c_k) := \det\big(y_b(c_a)\big)_{a,b=1}^{k}.$$

It is well defined because the right side is multilinear and alternating in $(y_1, \ldots, y_k)$ (these are the columns). For each ω, the function Ω_ω is alternating in $(c_1, \ldots, c_k)$ (the rows); in particular $\Omega_\omega(c_{\sigma(1)}, \ldots, c_{\sigma(k)}) = \operatorname{sgn}(\sigma)\, \Omega_\omega(c_1, \ldots, c_k)$.

Lemma B.2. *The determinant array $\Omega\colon \Lambda^k V \to L^{S^k}$ is injective.*

Proof. Let $\Omega_\omega = 0$ and choose a finite-dimensional $U \subseteq V$ with $\omega \in \Lambda^k U$. The evaluations $\operatorname{ev}_s|_U$, $s \in S$, separate the points of U (a function in U vanishing at every s is 0), so they span the dual space U^*. Choose $s_1, \ldots, s_d \in S$ such that the $\operatorname{ev}_{s_i}|_U$ form a basis of U^*, and let $u_1, \ldots, u_d$ be the dual basis of U, so that $u_j(s_i) = \delta_{ij}$. For increasing k-subsets I, J of $\{1, \ldots, d\}$ we get $\Omega_{u_I}(s_{j_1}, \ldots, s_{j_k}) = \det(u_{i_b}(s_{j_a}))_{a,b} = \delta_{IJ}$. Writing $\omega = \sum_I c_I u_I$ therefore gives $0 = \Omega_\omega(s_{j_1}, \ldots, s_{j_k}) = c_J$ for every J, so $\omega = 0$. □

Lemma B.3 (support bounds flattening rank)**.** *Let $V \subseteq L^S$, $\omega \in \Lambda^k V$ and $\alpha \in \{1, \ldots, k\}$. Let $R \subseteq S$ and $R' \subseteq S^{\{1,\ldots,k\}\setminus\{\alpha\}}$ be finite. The $R \times R'$ matrix M with entries $M(c_\alpha, c') := \Omega_\omega(c)$, where c has c_α in slot α and the entries of c' in the other slots, has rank at most* $\operatorname{sdim}(\omega)$.

Proof. Write $\omega = \sum_j y_{j,1} \wedge \cdots \wedge y_{j,k}$ with all $y_{j,b}$ in a subspace W of dimension $\operatorname{sdim}(\omega)$. Expanding each determinant along row α, $\det(y_{j,b}(c_a))_{a,b} = \sum_b \pm y_{j,b}(c_\alpha) \cdot \mu_{j,b}(c')$, where the minor $\mu_{j,b}(c')$ does not involve c_α. So for fixed c' the column $c_\alpha \mapsto M(c_\alpha, c')$ is a linear combination of the restrictions $y_{j,b}|_R$. All columns lie in the restriction of W to R, a space of dimension at most $\operatorname{sdim}(\omega)$. □

A second way to obtain lower bounds on support is to contract all but one exterior factor. The resulting vectors must lie in every subspace supporting the original exterior vector.

Lemma B.4 (contraction)**.** *Let $k \geq 2$ and $\varphi = (\varphi_1, \ldots, \varphi_{k-1}) \in (V^*)^{k-1}$. There is a linear map $c_\varphi\colon \Lambda^k V \to V$ with*

$$c_\varphi(x_1 \wedge \cdots \wedge x_k) = \sum_{i=1}^{k} (-1)^{k+i} \det\big(\varphi_a(x_b)\big)_{1 \leq a \leq k-1,\ b \neq i}\, x_i,$$

and $c_\varphi(\Lambda^k W) \subseteq W$ for every subspace W. Consequently $\operatorname{sdim}(\omega) \geq \dim \operatorname{span}\{c_\varphi(\omega) : \varphi \in (V^*)^{k-1}\}$.

Proof. The right side is multilinear in (x_b). For $\psi \in V^*$, applying ψ to it gives, by Laplace expansion along the last row, the determinant of the $k \times k$ matrix with entries $\varphi_a(x_b)$ in the rows $a < k$ and $\psi(x_b)$ in the last row; it vanishes when two columns coincide. Since V^* separates the points of V, the right side vanishes whenever $x_a = x_b$ for some $a \neq b$, so it factors through $\Lambda^k V$. Each term is a multiple of some x_i, which gives $c_\varphi(\Lambda^k W) \subseteq W$. If $\omega \in \Lambda^k W$ with $\dim W = \operatorname{sdim}(\omega)$, all $c_\varphi(\omega)$ lie in W. □

Proposition B.5 (canonical exterior support)**.** *Let L be any field, V an L-vector space, $k \geq 2$, and $\omega \in \Lambda^k V$. Then*

$$S(\omega) := \operatorname{span}\{c_\varphi(\omega) : \varphi \in (V^*)^{k-1}\}$$

is the unique smallest subspace supporting ω. In particular, $\omega \in \Lambda^k S(\omega)$ and $\operatorname{sdim}(\omega) = \dim S(\omega)$.

Proof. Choose a finite-dimensional supporting subspace $U \subseteq V$. Lemma B.4 gives $S(\omega) \subseteq U$, so $S(\omega)$ is finite-dimensional. Choose a basis $e_1, \dots, e_d$ of U whose first r vectors form a basis of $S(\omega)$, and extend its coordinate functionals algebraically to V. Write $\omega = \sum_{|I|=k} a_I e_I$ in the exterior basis of U. If $j \in I$ and $j > r$, contract against the $k-1$ coordinate functionals indexed by $I \setminus \{j\}$. The e_j-coordinate of this contraction is $\pm a_I$: no other basis wedge contributes to that coordinate. The contraction belongs to $S(\omega)$, so this coordinate is zero. Hence $a_I = 0$ whenever I meets $\{r+1, \dots, d\}$, proving $\omega \in \Lambda^k S(\omega)$. Every supporting subspace contains all the contractions by Lemma B.4, and therefore contains $S(\omega)$. This proves minimality and the dimension formula. Only unit signs occur, so the argument is valid in every characteristic. □

The same conclusion holds in degree one: under $\Lambda^1 V \simeq V$, the empty-tuple contraction is the identity and $S(\omega) = L\omega$. In degree zero, every element of $\Lambda^0 V = L$ is supported by the zero subspace, so its support dimension is zero; no contraction to V is needed.

For matrix calculations, choose any finite-dimensional supporting subspace U and a basis of it. Form the finite matrix whose columns are the coordinates of the contractions against $(k-1)$-tuples of coordinate functionals, extended to V. Arbitrary functionals restrict to linear combinations of these coordinates on U; multilinearity in the covectors therefore shows that the column span is exactly $S(\omega)$. The matrix consequently has rank $\operatorname{sdim}(\omega)$. This uses ordinary matrix rank, and entails no choice of a rank convention for an infinite matrix.

B.2. **Lifts and three diagonal identities.** Fix a field L, an integer $k \geq 1$, L-vector spaces A and V, and a linear map $D\colon A \to \operatorname{End}_L(V)$, $a \mapsto D_a$. A *lift of D in degree k* is a $2k$-linear map $\Psi\colon A^k \times V^k \to \Lambda^k V$ over L that is alternating in the V-slots; its diagonal identity will be specified separately. For every $n \geq 1$ write $[n] := \{1, \dots, n\}$, and for $\alpha \in \mathbb{N}^m$ put $|\alpha| := \sum_i \alpha_i$. Given $m \geq 1$, a map $f\colon [k] \to [m]$, a family $b = (b_1, \dots, b_m) \in A^m$ and $\lambda \in L^m$, put

$$b_f := (b_{f(1)}, \dots, b_{f(k)}), \quad \operatorname{type} f := \big(|f^{-1}(1)|, \dots, |f^{-1}(m)|\big) \in \mathbb{N}^m, \quad \lambda_f := \prod_{j=1}^{k} \lambda_{f(j)} = \lambda^{\operatorname{type} f}.$$

The *pointwise identity* is

$$\Psi(a, \dots, a; x_1, \dots, x_k) = D_a x_1 \wedge \cdots \wedge D_a x_k \qquad \text{for all } a \in A,\ x \in V^k. \tag{Pw}$$

The *polarized identity* is: for all $m \geq 1$, $b \in A^m$, $\alpha \in \mathbb{N}^m$ with $|\alpha| = k$, and $x \in V^k$,

$$\sum_{\operatorname{type} f = \alpha} \Psi(b_f; x) = \sum_{\operatorname{type} f = \alpha} D_{b_{f(1)}} x_1 \wedge \cdots \wedge D_{b_{f(k)}} x_k. \tag{Pol}$$

Its *multilinear instance* is: for all $b_1, \dots, b_k \in A$ and $x \in V^k$,

$$\sum_{\sigma \in S_k} \Psi(b_{\sigma(1)}, \dots, b_{\sigma(k)}; x) = \sum_{\sigma \in S_k} D_{b_{\sigma(1)}} x_1 \wedge \cdots \wedge D_{b_{\sigma(k)}} x_k. \tag{Pol1}$$

Here $\sum_{\operatorname{type} f = \alpha}$ runs over the maps $f\colon [k] \to [m]$ of type α. (Pol1) is the instance $m = k$, $\alpha = (1, \dots, 1)$ of (Pol), since the maps of type $(1, \dots, 1)$ are the permutations. For $b \in A^m$ and

$x \in V^k$ put $a_\lambda := \sum_i \lambda_i b_i$ and

$$\Delta_{b,x}(\lambda) := \Psi(a_\lambda, \dots, a_\lambda; x) - D_{a_\lambda}x_1 \wedge \cdots \wedge D_{a_\lambda}x_k = \sum_{|\alpha|=k} \lambda^\alpha \, \delta_\alpha(b,x),$$

$$\delta_\alpha(b,x) := \sum_{\text{type } f=\alpha} \Big[\Psi(b_f; x) - D_{b_{f(1)}}x_1 \wedge \cdots \wedge D_{b_{f(k)}}x_k\Big].$$

Multilinearity of Ψ in the A-slots, linearity of D and multilinearity of the wedge give the displayed expansion. Thus $\Delta_{b,x}$ is a homogeneous polynomial of degree k with coefficients in $\Lambda^k V$. Identity (Pol) says that every coefficient $\delta_\alpha(b,x)$ is zero. Identity (Pw) says instead that $\Delta_{b,x}$ vanishes at every point of L^m; conversely, this pointwise vanishing gives (Pw) by taking $m = 1$ and $\lambda = 1$. The following two identity principles identify the cases in which pointwise vanishing forces all coefficients to vanish.

Lemma B.6 (identity principle)**.** *Let $\mathbb{K}$ be an infinite field, Y a $\mathbb{K}$-vector space and $g(\lambda) = \sum_\alpha \lambda^\alpha y_\alpha$ a polynomial in $\lambda \in \mathbb{K}^m$ with finitely many nonzero coefficients $y_\alpha \in Y$. If $g(\lambda) = 0$ for all $\lambda \in \mathbb{K}^m$, then every $y_\alpha = 0$.*

Proof. The finitely many coefficients y_α span a finite-dimensional subspace of Y. Applying its coordinate functionals reduces the claim to scalar polynomials. For these, induction on m reduces to the fact that a nonzero polynomial in one variable has only finitely many roots. □

Lemma B.7 (homogeneous forms over a finite field)**.** *Let q be a prime power, $L' \supseteq \mathbb{F}_q$ a field, $1 \le e \le q$, and $g \in L'[\lambda_1, \dots, \lambda_m]$ homogeneous of degree e with $g(\lambda) = 0$ for every $\lambda \in \mathbb{F}_q^m$. Then $g = 0$.*

Proof. Call a monomial λ^β *reduced* if every $\beta_i \le q-1$. The indicator function of a point $a \in \mathbb{F}_q^m$ is $\prod_i \big(1 - (\lambda_i - a_i)^{q-1}\big)$, which expands into reduced monomials. So the q^m reduced monomials span the q^m-dimensional L'-space of functions $\mathbb{F}_q^m \to L'$, and they are linearly independent as functions. A monomial of degree $e \le q$ that is not reduced is λ_i^q (and then $e = q \ge 2$); as a function on $\mathbb{F}_q^m$ it agrees with the reduced monomial λ_i, of degree $1 < e$. If $e = q$, replace each λ_i^q in g by λ_i. These degree-one images are distinct, and homogeneity separates them from all the remaining monomials, which still have degree q. The replacement therefore sends the monomials of g injectively to reduced monomials without changing the associated function. Linear independence of the reduced monomials forces every coefficient to vanish. If $e < q$, no replacement is needed. □

Proposition B.8 (pointwise versus polarized)**.** *Let Ψ be a lift of D in degree k over a field L.*

1. (Pol)⇒(Pw)⇒(Pol1)*, over every field.*
2. *If L is infinite, or if $L = \mathbb{F}_q$ and $k \le q$, then* (Pw)⇒(Pol)*.*
3. *If $L = \mathbb{F}_q$ and $k \ge q+1$, then* (Pw) *does not imply* (Pol) *in general.*
4. *If $k! = 0$ in L, then* (Pol1) *does not imply* (Pw) *in general.*

Proof. (1) Inclusion–exclusion. The implication (Pol)⇒(Pw) is the case $m = 1$ and $\alpha = (k)$. To prove (Pw)⇒(Pol1), fix $b_1, \dots, b_k \in A$ and $x \in V^k$, and put $b_S := \sum_{i \in S} b_i$ for $S \subseteq [k]$. Multilinearity gives

$$\sum_{S \subseteq [k]} (-1)^{k-|S|} \Psi(b_S, \dots, b_S; x) = \sum_S (-1)^{k-|S|} \sum_{f\colon [k] \to S} \Psi(b_f; x)$$

$$= \sum_{f\colon [k]\to[k]} \Psi(b_f; x) \sum_{S \supseteq f([k])} (-1)^{k-|S|} = \sum_{f \text{ bijective}} \Psi(b_f; x),$$

because $\sum_{S\supseteq T}(-1)^{k-|S|}=(1-1)^{k-|T|}$ is 1 if $T=[k]$ and 0 otherwise. The same computation applies to the multilinear map $(a_1,\dots,a_k)\mapsto D_{a_1}x_1\wedge\cdots\wedge D_{a_k}x_k$, whose diagonal is the right side of (Pw). Evaluating (Pw) at the 2^k subset sums b_S therefore gives (Pol1). The computation is valid in every characteristic and involves no division by $k!$.

(2) The identity principles. The coefficients of $\Delta_{b,x}$ span a finite-dimensional subspace of $\Lambda^k V$; in a basis of it each coordinate of $\Delta_{b,x}$ is a scalar form of degree k vanishing on L^m. It is zero by Lemma B.6 if L is infinite, and by Lemma B.7 if $L=\mathbb{F}_q$ and $k\le q$.

(3) Pointwise vanishing without full polarization. Let $\dim A\ge 3$, with $b_1,b_2,b_3\in A$ and functionals g_1,g_2,g_3 on A such that $g_i(b_j)=\delta_{ij}$, and let $y_1,\dots,y_k\in V$ be linearly independent. Put

$$\theta(a_1,\dots,a_k):=\big[g_1(a_1)\cdots g_1(a_q)\,g_2(a_{q+1})-g_1(a_1)\,g_2(a_2)\cdots g_2(a_{q+1})\big]\cdot g_3(a_{q+2})\cdots g_3(a_k)$$

(for $k=q+1$ the g_3-factor is empty and $\dim A\ge 2$ suffices), and $\Theta(a;x):=\theta(a)\,x_1\wedge\cdots\wedge x_k$. Then Θ is a lift of $D=0$. It satisfies (Pw), because $\theta(a,\dots,a)=(g_1(a)^q g_2(a)-g_1(a)g_2(a)^q)\,g_3(a)^{k-q-1}=0$ since $c^q=c$ for $c\in\mathbb{F}_q$. It violates (Pol) for $\alpha=(q,1,k-q-1)$ and $b=(b_1,b_2,b_3)$: the first product contributes to $\sum_{\text{type }f=\alpha}\theta(b_f)$ exactly for $f=(1,\dots,1,2,3,\dots,3)$, with value 1; the second product would need $f(2)=\cdots=f(q+1)=2$, that is, at least $q\ge 2$ values equal to 2, which is impossible for type α. So the coefficient $\delta_\alpha(b,y)$ of Θ is $y_1\wedge\cdots\wedge y_k\neq 0$. When $k=q+1$, the third label and its functional can be omitted throughout, with $\alpha=(q,1)$.

(4) The multilinear identity without the pointwise identity. Take $\bar a\in A$ and a functional g_1 with $g_1(\bar a)=1$, and $y_1,\dots,y_k$ linearly independent. The lift $\Theta'(a;x):=g_1(a_1)\cdots g_1(a_k)\,x_1\wedge\cdots\wedge x_k$ of $D=0$ satisfies (Pol1), since $\sum_\sigma\Theta'(b_\sigma;x)=k!\,g_1(b_1)\cdots g_1(b_k)\,x_1\wedge\cdots\wedge x_k=0$ for all $b_1,\dots,b_k\in A$ and $x\in V^k$, but $\Theta'(\bar a,\dots,\bar a;y)=y_1\wedge\cdots\wedge y_k\neq 0$. □

The finite-field obstruction below requires only (Pol1), which follows from either (Pw) or (Pol). The distinction between the latter two remains important: over $\mathbb{F}_q$ they need not agree when $k>q$. The lift constructed in Appendix C nevertheless satisfies the full identity (Pol) in every degree: the identity is polarized over an infinite Laurent-series field before constant coefficients are taken (Lemma C.4).

B.3. **Statement and proof strategy.** For the proof of the theorem, assume that L is a *finite* field with $k!=0$ in L, that is, $\operatorname{char}L=p\le k$; in particular $k\ge 2$. Put $V:=L^{\mathbb{N}}$ and let $V_{\mathrm{fin}}:=L^{(\mathbb{N})}\subseteq V$ be the subspace of finitely supported sequences. For $u,x\in V$ we write ux for the coordinatewise product; so $D_u x=ux$ is the multiplier family.

Theorem B.9 (multipliers over a finite field)**.** *Let L be a finite field and $k \geq 1$ with $k! = 0$ in L.*

(1) *There is no $2k$-linear map $\Psi\colon V_{\mathrm{fin}}^k \times V_{\mathrm{fin}}^k \to \Lambda_L^k V$ over L such that*
   (a) *Ψ is antisymmetric in the last k slots;*
   (b) *Ψ satisfies (Pol1) for $D_u x = ux$: for all $u_1, \dots, u_k, x_1, \dots, x_k \in V_{\mathrm{fin}}$,*

$$\sum_{\sigma \in S_k} \Psi(u_{\sigma(1)}, \dots, u_{\sigma(k)}; x_1, \dots, x_k) = \sum_{\sigma \in S_k} (u_{\sigma(1)} x_1) \wedge \cdots \wedge (u_{\sigma(k)} x_k);$$

   (c) *for some $d \in \mathbb{N}$, $\operatorname{sdim}(\Psi(u; x)) \leq d$ for all $u, x \in V_{\mathrm{fin}}^k$ with entries in $\{0, \pm 1\}$.*

(2) *Let $E_0 = V$ or $E_0 = V_{\mathrm{fin}}$. There is no $2k$-linear map $\Psi\colon E_0^k \times E_0^k \to \Lambda_L^k E_0$ over L that is antisymmetric in the last k slots, satisfies (Pol1) for all finitely supported u, x, and has $\operatorname{sdim}(\Psi(u; x)) \leq d$ for all $u, x \in E_0^k$, for some d.*

Reduction to (1). Given a map as in (2), restrict its inputs to $V_{\mathrm{fin}}^k \times V_{\mathrm{fin}}^k$ and compose its output with the injective map $\Lambda^k E_0 \to \Lambda^k V$. This preserves antisymmetry and (Pol1). Support dimension can only decrease under the inclusion of the ambient space (§B.1), so the resulting map satisfies (a)–(c).

We therefore prove (1). Suppose that Ψ satisfies its hypotheses, and fix the bound d in (c). Write $e_i \in V_{\mathrm{fin}}$ for the unit vectors, $e_a := (e_{a_1}, \dots, e_{a_k})$ for $a \in \mathbb{N}^k$, and Ω for the determinant array of §B.1 with $S = \mathbb{N}$.

The proof has three steps. Ramsey's theorem first makes the coefficients of $\Omega\Psi$ depend only on the relative order and equalities among their indices. Special weights on four-point clusters then cancel every contribution from an occupied cluster containing no output index. This cancellation produces a two-valued staircase matrix from one exterior vector. Its bounded support forces the two values to agree, and hence makes all cluster order patterns have a common value. Finally, antisymmetry and (Pol1) say that the sum of these $k!$ equal values is 1, contradicting $k! = 0$ in L.

B.4. **Order-homogeneity from Ramsey's theorem.** We use Ramsey's theorem [13] in its infinite form: *for every $n \geq 1$ and every coloring of the n-element subsets of $\mathbb{N}$ with finitely many colors there is an infinite $H \subseteq \mathbb{N}$ all of whose n-element subsets have the same color.*

For a tuple $z \in \mathbb{N}^N$ with distinct values $y_1 < \cdots < y_r$, its *pattern* is the map $i \mapsto j$ determined by $z_i = y_j$. In particular, repeated entries are retained in the pattern. Two tuples z and z' have the same pattern if and only if $z_i < z_j \Leftrightarrow z'_i < z'_j$ and $z_i = z_j \Leftrightarrow z'_i = z'_j$ for all i, j.

Lemma B.10 (pattern homogeneity)**.** *For every $N \geq 1$ and every function $T\colon \mathbb{N}^N \to L$ there is an infinite $H \subseteq \mathbb{N}$ such that $T(z) = T(z')$ whenever $z, z' \in H^N$ have the same pattern.*

Proof. Color an N-element set $s = \{s_1 < \cdots < s_N\} \subseteq \mathbb{N}$ by the vector

$$\big(T(s_{\beta(1)}, \dots, s_{\beta(N)})\big)_{\beta\colon [N] \to [N]} \in L^{N^N}.$$

The coloring records all maps β, including those that repeat indices. It has finitely many values because L is finite; this is the only use of the finiteness of L in the obstruction proof. Ramsey's theorem gives an infinite H on which the color is constant, say equal to $(t_\beta)_\beta$. Let $z \in H^N$ have value set $\{y_1 < \cdots < y_r\}$. Since H is unbounded, add $N - r$ elements of H above y_r to obtain an N-element set $s \subseteq H$. Padding above the largest value leaves the ranks unchanged: $s_j = y_j$ for $j \leq r$, and $z = (s_{\beta(1)}, \dots, s_{\beta(N)})$, where $\beta(i)$ is the rank of z_i in $\{y_1, \dots, y_r\}$. The map β depends only on the pattern of z, and $T(z) = t_\beta$. □

Coefficients. For $(a;y;c) \in \mathbb{N}^k \times \mathbb{N}^k \times \mathbb{N}^k$ put

$$T(a;y;c) := \Omega_{\Psi(e_a;e_y)}(c_1,\dots,c_k) \in L.$$

Lemma B.10 with $N = 3k$ gives an infinite set $H = \{h_0 < h_1 < h_2 < \cdots\}$ on which T depends only on the pattern. For $u_j, v_j \in V_{\mathrm{fin}}$ supported in H, multilinearity of Ψ and linearity of Ω give

$$\Omega_{\Psi(u;v)}(c) = \sum_{a,y\in H^k} \prod_{j=1}^{k} u_j(a_j) \prod_{j=1}^{k} v_j(y_j) \cdot T(a;y;c). \tag{B.1}$$

Although the sum is indexed by $H^k \times H^k$, only finitely many summands are nonzero, because all the input vectors have finite support. Every output tuple c used below also lies in H^k, so pattern homogeneity applies to all $3k$ indices in (B.1).

B.5. **Clusters and the cluster value.** A *cluster* is a set $C = \{h_n, h_{n+1}, h_{n+2}, h_{n+3}\}$ of four *consecutive* elements of H; we write $C(i) := h_{n+i-1}$ for $i = 1,\dots,4$. Since a cluster is an interval of H, two disjoint clusters are *comparable*: every element of one is smaller than every element of the other; we write $C < C'$. Distinct clusters need not be adjacent in H.

We choose two weight vectors, one for each kind of input slot:

$$s = (s_1,s_2,s_3,s_4) := (1,0,-1,0) \qquad \text{on the operator slots,}$$
$$w = (w_1,w_2,w_3,w_4) := (1,-1,1,-1) \qquad \text{on the vector slots.}$$

Over $\mathbb{Z}$, hence over every field,

$$\sum_i s_i = \sum_i w_i = \sum_i s_i w_i = \sum_{i<i'} s_i w_{i'} = \sum_{i>i'} s_i w_{i'} = 0, \qquad s_1 w_1 = 1. \tag{W}$$

Indeed $\sum_i s_i = 1-1$, $\sum_i w_i = 1-1+1-1$, $s_1 w_1 = 1$, and

$$\sum_i s_i w_i = s_1 w_1 + s_3 w_3 = 1 - 1 = 0,$$
$$\sum_{i<i'} s_i w_{i'} = s_1(w_2+w_3+w_4) + s_3 w_4 = -1 + 1 = 0,$$
$$\sum_{i>i'} s_i w_{i'} = s_3(w_1+w_2) = -(1-1) = 0.$$

The first two zero sums will cancel a cluster occupied by only one input index. The other three will cancel a cluster occupied by an operator index and a vector index, according to their relative order. For a cluster C put

$$s_C := \sum_{i=1}^{4} s_i e_{C(i)}, \qquad w_C := \sum_{i=1}^{4} w_i e_{C(i)}.$$

These are vectors of V_{fin} with entries in $\{0,\pm 1\}$, and $s_C w_C = e_{C(1)} - e_{C(3)}$.

The cluster value. For pairwise disjoint clusters $C_1,\dots,C_k$, one for each slot, define

$$\chi(C_1,\dots,C_k) := \Omega_{\Psi(s_{C_1},\dots,s_{C_k};\, w_{C_1},\dots,w_{C_k})}\big(C_1(1),\dots,C_k(1)\big)$$
$$= \sum_{i,i'\in[4]^k} \prod_{j=1}^{k} s_{i_j} w_{i'_j} \cdot T\big(C_1(i_1),\dots,C_k(i_k);\, C_1(i'_1),\dots,C_k(i'_k);\, C_1(1),\dots,C_k(1)\big).$$

Let $\tau \in S_k$ be the *order pattern* of $(C_1,\dots,C_k)$: $\tau(j)$ is the position of C_j among $C_1,\dots,C_k$. The pattern of every $3k$-tuple in the sum is determined by (i,i') and τ: entries in different clusters compare as the clusters do, and the entries $C_j(i_j)$, $C_j(i'_j)$, $C_j(1)$ in the same cluster

compare as their indices do, including equalities. Pattern homogeneity therefore shows that the displayed sum depends only on τ. Every permutation occurs as the order pattern of some disjoint clusters in H; we denote the resulting scalar by $\chi(\tau)$.

Lemma B.11 (cancellation of output-free clusters)**.** *Let $\mathcal{C}_1, \dots, \mathcal{C}_k$ be finite families of clusters that are pairwise disjoint as sets of clusters, such that all clusters in $\bigcup_j \mathcal{C}_j$ are pairwise disjoint. Put $u_j := \sum_{C \in \mathcal{C}_j} s_C$ and $v_j := \sum_{C \in \mathcal{C}_j} w_C$. Let $C_l \in \mathcal{C}_l$ for $l = 1, \dots, k$ and $c := (C_1(1), \dots, C_k(1))$. Then $\Omega_{\Psi(u;v)}(c) = \chi(C_1, \dots, C_k)$.*

Proof. We show that every cluster assignment except the one matching the output clusters has total contribution zero. Expand by (B.1). Since the clusters are disjoint, a nonzero term has, for each slot j, an operator index $a_j = C_j^{\mathrm{op}}(i_j)$ and a vector index $y_j = C_j^{\mathrm{vec}}(i'_j)$ with $C_j^{\mathrm{op}}, C_j^{\mathrm{vec}} \in \mathcal{C}_j$, and weight $\prod_j s_{i_j} w_{i'_j}$. Group the terms according to the *cluster assignment* $(C_j^{\mathrm{op}}, C_j^{\mathrm{vec}})_j$.

Call an occupied cluster *output-free* if it is none of $C_1, \dots, C_k$. Suppose a group has such a cluster Γ. Disjointness of the families assigns Γ to exactly one slot j, so the only tuple entries it can contain are a_j and y_j. Disjointness of the clusters excludes every output index from Γ.

Fix the positions of all indices outside Γ and sum over the positions inside it. Since Γ consists of consecutive elements of H, every outside entry lies either below the whole cluster or above it. Moving an index within Γ cannot change any comparison with an outside entry. By pattern homogeneity, the coefficient T therefore depends on the internal positions only as follows:

(i) if only $a_j \in \Gamma$, not at all, and the partial sum is $(\sum_i s_i) \cdot T = 0$;
(ii) if only $y_j \in \Gamma$, not at all, and the partial sum is $(\sum_i w_i) \cdot T = 0$;
(iii) if both lie in Γ, only through whether $i_j < i'_j$, $i_j = i'_j$ or $i_j > i'_j$; with the corresponding values $T_<, T_=, T_>$ of T the partial sum is

$$T_< \sum_{i<i'} s_i w_{i'} + T_= \sum_i s_i w_i + T_> \sum_{i>i'} s_i w_{i'} = 0.$$

In each case the weights of the other slots are a common factor, and the displayed partial sum vanishes by (W). Thus every assignment with an output-free cluster contributes zero. In any surviving assignment, every occupied cluster must be an output cluster. The only output cluster belonging to $\mathcal{C}_j$ is C_j, which forces $C_j^{\mathrm{op}} = C_j^{\mathrm{vec}} = C_j$ for every j. The contribution of this one assignment is exactly $\chi(C_1, \dots, C_k)$. □

B.6. **The staircase and the diagonal.** We now use the support bound to compare two order patterns that differ by an adjacent exchange. The key is to realize arbitrarily large staircase matrices as flattenings of a single value of Ψ.

Lemma B.12 (staircase)**.** *Let $\tau \in S_k$, let $\alpha \neq \beta$ be slots with $\tau(\alpha) = \ell$ and $\tau(\beta) = \ell + 1$, and let τ' be τ with these two positions exchanged, so $\tau'(\alpha) = \ell + 1$, $\tau'(\beta) = \ell$ and $\tau'(\gamma) = \tau(\gamma)$ otherwise. Then $\chi(\tau) = \chi(\tau')$.*

Proof. Let $m \geq 1$. Choose pairwise disjoint clusters, in increasing order in H: first a cluster C_γ for each slot γ with $\tau(\gamma) < \ell$, ordered as τ prescribes; then $2m$ clusters $A_1 < B_1 < A_2 < B_2 < \cdots < A_m < B_m$; then a cluster C_γ for each slot γ with $\tau(\gamma) > \ell + 1$, ordered as τ prescribes. Put $\mathcal{C}_\alpha := \{A_1, \dots, A_m\}$, $\mathcal{C}_\beta := \{B_1, \dots, B_m\}$ and $\mathcal{C}_\gamma := \{C_\gamma\}$ for the other slots, and let u, v be as in Lemma B.11 and $\omega := \Psi(u; v)$. Disjointness of the clusters ensures that the entries of u and v still lie in $\{0, \pm 1\}$, so $\mathrm{sdim}(\omega) \leq d$ by hypothesis (c) of Theorem B.9(1). For this fixed m, keep u, v and hence ω fixed while varying only the output indices.

Consider the $m \times m$ matrix M with rows indexed by $c_\alpha = A_n(1)$ and columns by $c_\beta = B_{n'}(1)$ $(n, n' = 1, \dots, m)$, all other output indices being fixed at $c_\gamma = C_\gamma(1)$, and entries $M_{nn'} := \Omega_\omega(c)$.

By Lemma B.11, applied with the output clusters $A_n \in \mathcal{C}_\alpha$, $B_{n'} \in \mathcal{C}_\beta$ and $C_\gamma \in \mathcal{C}_\gamma$, $M_{nn'}$ is the cluster value of the tuple with A_n in slot α, $B_{n'}$ in slot β, and C_γ in every other slot. The interlacing order gives $A_n < B_{n'}$ exactly when $n \le n'$. The tuple therefore has pattern τ on and above the matrix diagonal, and pattern τ' below it.

Put $\delta := \chi(\tau) - \chi(\tau')$, let $\mathbf{1} \in L^m$ be the all-ones column, and let U be the upper unitriangular matrix with $U_{nn'} = 1$ for $n \le n'$ and $U_{nn'} = 0$ otherwise. Then

$$M = \begin{pmatrix} \chi(\tau) & \chi(\tau) & \cdots & \chi(\tau) \\ \chi(\tau') & \chi(\tau) & \cdots & \chi(\tau) \\ \vdots & \ddots & \ddots & \vdots \\ \chi(\tau') & \cdots & \chi(\tau') & \chi(\tau) \end{pmatrix} = \chi(\tau')\,\mathbf{1}\mathbf{1}^{\mathsf{T}} + \delta\, U,$$

The rows vary only slot α; each column fixes all the other slots, including the chosen value in slot β. Thus M is a submatrix of the flattening of this one array Ω_ω along slot α. Lemma B.3 gives $\operatorname{rank} M \le \operatorname{sdim}(\omega) \le d$.

If $\delta \neq 0$, the matrix δU is invertible, whereas the constant matrix $\chi(\tau')\mathbf{1}\mathbf{1}^{\mathsf{T}}$ has rank at most one. Applying rank subadditivity to $\delta U = M - \chi(\tau')\mathbf{1}\mathbf{1}^{\mathsf{T}}$ gives

$$m = \operatorname{rank}(\delta U) \le \operatorname{rank} M + 1 \le d + 1.$$

The construction is available for every $m \ge 1$; choosing $m = d + 2$ is a contradiction. Hence $\delta = 0$, as required. □

The support bound has now forced equality between adjacent order patterns. The remaining identity comes from (Pol1); it does not use the support bound.

Lemma B.13 (the diagonal)**.** $\sum_{\tau \in S_k} \chi(\tau) = 1$.

Proof. Take clusters $C_1 < \cdots < C_k$, and put $u_j := s_{C_j}$, $v_j := w_{C_j}$ and $c := (C_1(1), \ldots, C_k(1))$. Apply $\omega \mapsto \Omega_\omega(c)$ to both sides of (Pol1), with the operator vectors $u_1, \ldots, u_k$ and the vector arguments $v_1, \ldots, v_k$.

Right side. If $\sigma \neq \mathrm{id}$, some slot j has $\sigma(j) \neq j$. Disjoint supports then give $u_{\sigma(j)} v_j = 0$, so the corresponding wedge vanishes. Only $\sigma = \mathrm{id}$ survives. It gives $\Omega_{(u_1 v_1) \wedge \cdots \wedge (u_k v_k)}(c) = \det\big((u_b v_b)(C_a(1))\big)_{a,b}$. Since $u_b v_b = e_{C_b(1)} - e_{C_b(3)}$ and $C_a(1) \neq C_b(3)$ for all a, b, this is the determinant of the identity matrix, so the right side is 1.

Left side. Fix $\sigma \in S_k$ and write $u_\sigma := (u_{\sigma(1)}, \ldots, u_{\sigma(k)})$ and $v_\sigma := (v_{\sigma(1)}, \ldots, v_{\sigma(k)})$. To identify the summand with a cluster value, we must permute both the vector arguments and the output coordinates to match the operator arguments. Each permutation contributes the same sign. More precisely, antisymmetry of Ψ in the vector slots gives $\Psi(u_\sigma; v) = \operatorname{sgn}(\sigma)\Psi(u_\sigma; v_\sigma)$. For every $\omega \in \Lambda^k V$ the function Ω_ω is alternating in its arguments, so $\Omega_\omega(c) = \operatorname{sgn}(\sigma)\,\Omega_\omega(c_{\sigma(1)}, \ldots, c_{\sigma(k)})$; we apply this with $\omega = \Psi(u_\sigma; v_\sigma)$. Since $\operatorname{sgn}(\sigma)^2 = 1$,

$$\begin{aligned} \Omega_{\Psi(u_\sigma; v)}(c) &= \operatorname{sgn}(\sigma)\,\Omega_{\Psi(u_\sigma; v_\sigma)}(c) = \operatorname{sgn}(\sigma)^2\,\Omega_{\Psi(u_\sigma; v_\sigma)}(c_{\sigma(1)}, \ldots, c_{\sigma(k)}) \\ &= \Omega_{\Psi(u_\sigma; v_\sigma)}(c_{\sigma(1)}, \ldots, c_{\sigma(k)}). \end{aligned}$$

In the last expression slot j has operator $s_{C_{\sigma(j)}}$, vector $w_{C_{\sigma(j)}}$ and output $C_{\sigma(j)}(1)$, so it equals $\chi(C_{\sigma(1)}, \ldots, C_{\sigma(k)})$ by definition. Its order pattern is σ, since $C_{\sigma(j)}$ is the $\sigma(j)$-th cluster in increasing order. Thus the convention for patterns gives σ, not σ^{-1}. No cluster cancellation is needed in this calculation.

As σ runs over S_k, each order pattern occurs exactly once. Comparing the two sides gives

$$\sum_{\sigma \in S_k} \Omega_{\Psi(u_\sigma; v)}(c) = \sum_{\tau \in S_k} \chi(\tau) = 1.$$ □

Proof of Theorem B.9. Exchanging adjacent ranks acts on an order pattern by $\tau \mapsto (\ell\ \ell{+}1) \circ \tau$, $1 \le \ell < k$. These transpositions generate S_k, and Lemma B.12 makes χ invariant under each exchange. Hence χ is constant; write its value as χ_0. Lemma B.13 now gives $k!\,\chi_0 = 1$ in L, contradicting $k! = 0$. This proves (1), and the preceding reduction gives (2). □

Remark B.14 (where the hypotheses enter). Ramsey homogeneity is the only step in the obstruction proof that uses the finiteness of L. Multilinearity then gives the cluster expansion and cancellation. The support bound is used in Lemma B.12 only at the finitely many test inputs obtained by taking $m = d + 2$; all have entries in $\{0, \pm 1\}$. Antisymmetry and (Pol1) enter in Lemma B.13, which does not use that bound. Finally, $k! = 0$ is used only to turn the diagonal identity into a contradiction.

The inclusion–exclusion argument in Proposition B.8(1) uses only multilinearity in the operator slots. The theorem therefore remains true with the pointwise identity (Pw) at L-points in place of (Pol1), for this family of multipliers. In characteristic 2 antisymmetry is the same as symmetry, and the two signs in Lemma B.13 are trivial.

Remark B.15 (when $k!$ is invertible). The vanishing of $k!$ is essential. If instead $k! \neq 0$ in L, the *normalized alternatization*

$$\Psi_{\mathrm{alt}}(u; x) := \frac{1}{k!} \sum_{\sigma \in S_k} (u_{\sigma(1)} x_1) \wedge \cdots \wedge (u_{\sigma(k)} x_k)$$

is alternating in x and satisfies (Pol1). Every summand is supported in the span of the k^2 vectors $u_i x_j$, so $\operatorname{sdim}(\Psi_{\mathrm{alt}}(u; x)) \le k^2$. The preceding homogeneity, cancellation, staircase and diagonal arguments apply, but now yield $\chi \equiv 1/k!$: for pairwise disjoint clusters only $\sigma = \mathrm{id}$ survives in $\Psi_{\mathrm{alt}}(s_{C_1}, \dots, s_{C_k}; w_{C_1}, \dots, w_{C_k})$, and the determinant at $(C_1(1), \dots, C_k(1))$ is that of the identity. The final equation $k! \cdot (1/k!) = 1$ is then consistent. So the obstruction is exactly the vanishing of $k!$, not positive characteristic as such. For example, at $(p, k) = (5, 3)$ every step is consistent, since $3! = 6 = 1$ in $\mathbb{F}_5$ and $\chi_0 = 1$.

Appendix C. The Banach exterior target over a Laurent series field

We now realize the finite-field obstruction in a Banach-space setting. The target is the completion of an exterior power with its ordinary projective norm. Its determinant arrays admit a constant-coefficient map into the *algebraic* exterior power over the coefficient field, with support dimension controlled by the Banach norm. This estimate turns a hypothetical bounded lift into the uniformly bounded-support map ruled out by Theorem B.9. The passage to the completion is the key point: constant coefficients stabilize exactly once the approximation error is less than one.

Theorem C.1. *Let κ be a finite field of characteristic p, let $k \ge p$ and $r \in (0, 1)$, and let $K_1 = \kappa((X))$ with $|X| = r$. Let $E_1 = \ell^\infty(\mathbb{N}, K_1)$, and let B be the completion of $\Lambda^k_{K_1} E_1$ under the projective exterior norm. Then $A^{k,K_1}_{E_1,E_1;B}$ admits no bounded k-linear lift over K_1. Consequently it is analytic at no point of $\mathcal{L}(E_1, E_1)$.*

C.1. **Setting.** Throughout this section $k \ge 1$, κ is a field, $r \in (0, 1)$, and $K_1 := \kappa((X))$ is the Laurent series field with absolute value $|a| := r^{\operatorname{ord} a}$ ($|0| = 0$). This field is complete, nontrivially normed and nonarchimedean, and $|c| = 1$ for every $c \in \kappa^\times$. Write $\operatorname{coeff}_n(a) \in \kappa$ for the X^n-coefficient of $a \in K_1$. For a set S and $y \in \ell^\infty(S, K_1)$ we take the constant coefficient coordinatewise:

$$\operatorname{coeff}_0(y) := \big(\operatorname{coeff}_0(y(s))\big)_{s \in S} \in \kappa^S.$$

The resulting *constant-coefficient map* coeff_0 is κ-linear but not K_1-linear. A nonzero Laurent series of absolute value less than one has order at least one. Consequently,

$$\text{(C.1)}\quad \|y\|_\infty < 1 \implies \mathrm{coeff}_0(y) = 0, \qquad \text{hence} \qquad \|y - y'\|_\infty < 1 \implies \mathrm{coeff}_0(y) = \mathrm{coeff}_0(y').$$

Put $E_1 := \ell^\infty(\mathbb{N}, K_1)$ with the sup norm, a nonarchimedean K_1-Banach space, and let $E_0 := \kappa^{\mathbb{N}} \subseteq E_1$ be the κ-subspace of sequences with entries in κ. Every nonzero $z \in E_0$ has $\|z\|_\infty = 1$. For $x \in E_1$ and $n \in \mathbb{Z}$ let $x_{[n]} := (\mathrm{coeff}_n(x(i)))_{i\in\mathbb{N}} \in E_0$ be the n-th Laurent component. For $x \neq 0$ the set $\{\mathrm{ord}\, x(i) : x(i) \neq 0\}$ is bounded below, because $r^{\mathrm{ord}\, x(i)} \leq \|x\|_\infty$, so it has a minimum $\nu(x)$; then $\|x\|_\infty = r^{\nu(x)}$ and $x_{[n]} = 0$ for $n < \nu(x)$. Finally put

$$M_r := \max_{l\in\mathbb{N}} (l+1)r^l.$$

The maximum exists because $(l+1)r^l \to 0$. This constant will control the number of Laurent components that can contribute to a constant coefficient. For $r \leq \frac{1}{2}$ one has $(l+1)r^l \leq (l+1)2^{-l} \leq 1$, so $M_r = 1$.

C.2. **The projective exterior norm.** On $\Lambda := \Lambda^k_{K_1} E_1$ put

$$\|\omega\|_\pi := \inf\Big\{\sum_j \prod_{i=1}^k \|x_{j,i}\|_\infty \;:\; \omega = \sum_j x_{j,1} \wedge \cdots \wedge x_{j,k}\Big\},$$

where the infimum is over finite decompositions and the cost is an ordinary sum. Let $\Omega^{K_1}\colon \Lambda \to \ell^\infty(\mathbb{N}^k, K_1)$ be the determinant array of §B.1 with $L = K_1$, $S = \mathbb{N}$ and $V = E_1$, so that $\Omega^{K_1}(x_1 \wedge \cdots \wedge x_k)(i_1, \ldots, i_k) = \det(x_b(i_a))_{a,b}$; its values are bounded by the next lemma.

Lemma C.2. *$\|\cdot\|_\pi$ is a K_1-norm on Λ with $\|x_1 \wedge \cdots \wedge x_k\|_\pi \leq \prod_i \|x_i\|_\infty$, and Ω^{K_1} is K_1-linear and injective with $\|\Omega^{K_1}(\omega)\|_\infty \leq \|\omega\|_\pi$. Let B be the completion of $(\Lambda, \|\cdot\|_\pi)$, a K_1-Banach space, $J\colon B \to \ell^\infty(\mathbb{N}^k, K_1)$ the continuous extension of Ω^{K_1} (so $\|J\| \leq 1$), and $W_B\colon E_1^k \to B$, $W_B(x) := x_1 \wedge \cdots \wedge x_k$. Then $W_B \in \mathrm{Alt}^k_{K_1}(E_1; B)$, $\|W_B\| \leq 1$, and $J(W_B(x)) = \Omega^{K_1}(x_1 \wedge \cdots \wedge x_k)$.*

Proof. For a pure wedge, each entry of the determinant array is a signed sum of products $x_1(i_{\sigma(1)}) \cdots x_k(i_{\sigma(k)})$. Each product has absolute value at most $\prod_b \|x_b\|_\infty$, so the nonarchimedean inequality in K_1 gives the same bound for the determinant. For any finite decomposition of ω, it follows that

$$\|\Omega^{K_1}(\omega)\|_\infty \leq \max_j \prod_i \|x_{j,i}\|_\infty \leq \sum_j \prod_i \|x_{j,i}\|_\infty.$$

Thus the array is bounded, and taking the infimum gives $\|\Omega^{K_1}(\omega)\|_\infty \leq \|\omega\|_\pi$. The map Ω^{K_1} is linear and injective by Lemma B.2, with $L = K_1$, $S = \mathbb{N}$ and $V = E_1$. In particular, $\|\omega\|_\pi > 0$ whenever $\omega \neq 0$.

The remaining norm laws follow directly from decompositions. Concatenation gives the triangle inequality. Since $k \geq 1$, a scalar can be absorbed into the first factor of each wedge; for $\lambda \neq 0$ this gives $\|\lambda\omega\|_\pi \leq |\lambda|\|\omega\|_\pi$, and applying the same inequality to λ^{-1} gives equality. The pure-wedge bound follows by using a single summand.

The array space $\ell^\infty(\mathbb{N}^k, K_1)$ is complete, so Ω^{K_1} extends to the contraction J on B. The wedge map into B is K_1-multilinear and vanishes whenever two arguments coincide. The pure-wedge bound makes it continuous with norm at most one, and the asserted identity with J holds on its values by construction. □

For $k \geq 2$, B is not nonarchimedean (Proposition E.1).

C.3. **The support estimate.** We next show that the constant coefficient of every completed determinant array is represented by a finite algebraic exterior vector. The quantitative bound comes from counting the Laurent components needed as wedge *factors*; the number of wedge summands may be much larger.

Let $\Omega^\kappa\colon \Lambda^k_\kappa E_0 \to \kappa^{\mathbb{N}^k}$ be the determinant array over κ; it is injective by Lemma B.2 with $L=\kappa$ and $V=E_0$. For $y_1,\dots,y_k\in E_0$ we have $\Omega^{K_1}(y_1\wedge\cdots\wedge y_k)=\Omega^\kappa(y_1\wedge\cdots\wedge y_k)$: the determinants are the same, with entries in $\kappa\subseteq K_1$. Support dimensions of elements of $\Lambda^k_\kappa E_0$ are computed in E_0 over κ.

Proposition C.3 (support estimate)**.** *For every $b\in B$ there is a unique $\eta(b)\in\Lambda^k_\kappa E_0$ with $\Omega^\kappa(\eta(b))=\operatorname{coeff}_0(Jb)$. The map η is κ-linear, and*

$$\operatorname{sdim}\big(\eta(b)\big)\le kM_r\,\|b\|_B \qquad \text{for all } b\in B. \tag{C.2}$$

Proof. Step 1: one decomposable. Let $x_1,\dots,x_k\in E_1$. If some $x_b=0$, the wedge is zero and we take $\eta=0$. Otherwise put $\nu_b:=\nu(x_b)$ and $l:=-\sum_b\nu_b$, so that $\prod_b\|x_b\|_\infty=r^{-l}$. The possible exponent tuples contributing to a constant coefficient form the set

$$\mathcal{N}:=\Big\{n\in\mathbb{Z}^k:\ n_b\ge\nu_b \text{ for all } b,\ \sum_b n_b=0\Big\}.$$

For such a tuple, the nonnegative integers $n_b-\nu_b$ sum to l. Hence $\mathcal{N}$ is empty if $l<0$; if $l\ge0$, every coordinate satisfies $\nu_b\le n_b\le\nu_b+l$, so $\mathcal{N}$ is finite.

Fix $(i_1,\dots,i_k)\in\mathbb{N}^k$ and expand each entry as $x_b(i_a)=\sum_{n\ge\nu_b}\operatorname{coeff}_n(x_b(i_a))X^n$. In the determinant expansion

$$\det(x_b(i_a))_{a,b}=\sum_\sigma\operatorname{sgn}(\sigma)\prod_b x_b(i_{\sigma(b)}),$$

the constant coefficient of each product is its finite coefficient convolution over $\mathcal{N}$. Exchanging this finite sum with the permutation sum gives

$$\operatorname{coeff}_0\det\big(x_b(i_a)\big)_{a,b}=\sum_{n\in\mathcal{N}}\det\big(x_{b,[n_b]}(i_a)\big)_{a,b}.$$

The same finite set $\mathcal{N}$ works for every array coordinate. Thus $\operatorname{coeff}_0\big(\Omega^{K_1}(x_1\wedge\cdots\wedge x_k)\big)=\Omega^\kappa(\eta)$, where

$$\eta:=\sum_{n\in\mathcal{N}}x_{1,[n_1]}\wedge\cdots\wedge x_{k,[n_k]}\in\Lambda^k_\kappa E_0.$$

This is a finite sum in the algebraic exterior power. When $l<0$ it is zero. When $l\ge0$, all its wedge factors belong to the set $\{x_{b,[n]}:1\le b\le k,\ \nu_b\le n\le\nu_b+l\}$, which has at most $k(l+1)$ elements. Taking their span gives

$$\operatorname{sdim}(\eta)\le k(l+1)=k\,(l+1)r^l\cdot r^{-l}\le kM_r\prod_b\|x_b\|_\infty.$$

The final bound therefore holds in both cases.

Step 2: an element of Λ. Write $\omega=\sum_j w_j$, where $w_j=x_{j,1}\wedge\cdots\wedge x_{j,k}$, and take η_j from Step 1. Additivity gives

$$\operatorname{coeff}_0(\Omega^{K_1}(\omega))=\sum_j\operatorname{coeff}_0(\Omega^{K_1}(w_j))=\Omega^\kappa\Big(\sum_j\eta_j\Big).$$

Injectivity of Ω^κ shows that $\eta(\omega):=\sum_j\eta_j$ is independent of the chosen decomposition. Subadditivity of support dimension now yields

$$\operatorname{sdim}(\eta(\omega))\le\sum_j\operatorname{sdim}(\eta_j)\le kM_r\sum_j\prod_i\|x_{j,i}\|_\infty.$$

The left side is fixed while the decomposition varies. Taking the infimum therefore gives $\operatorname{sdim}(\eta(\omega)) \le kM_r\|\omega\|_\pi$, without requiring a minimizing decomposition.

Step 3: completion. Choose $\omega_n \in \Lambda$ with $\|b-\omega_n\|_B \to 0$. Since J is a contraction,

$$\|Jb - \Omega^{K_1}(\omega_n)\|_\infty \le \|b-\omega_n\|_B < 1 \qquad (n \ge n_0).$$

Equation (C.1) therefore gives the *exact* identities

$$\operatorname{coeff}_0(Jb) = \operatorname{coeff}_0(\Omega^{K_1}(\omega_n)) = \Omega^\kappa(\eta(\omega_n)) \qquad (n \ge n_0).$$

By injectivity of Ω^κ, the algebraic vectors $\eta(\omega_n)$ are eventually equal. Define $\eta(b)$ to be this common value. Its array is $\operatorname{coeff}_0(Jb)$, so the same injectivity also proves independence of the approximating sequence and uniqueness. In particular, no limit in $\Lambda^k_\kappa E_0$ is involved.

For every $n \ge n_0$ the fixed vector $\eta(b)$ satisfies $\operatorname{sdim}(\eta(b)) \le kM_r\|\omega_n\|_\pi$. Since $\|\omega_n\|_\pi \to \|b\|_B$, this gives (C.2). Finally, the array characterization proves κ-linearity: J is K_1-linear, coeff_0 is κ-linear, and Ω^κ is linear and injective. □

C.4. **The coefficient lift.** The support estimate converts norm control into an algebraic obstruction. We apply it to a hypothetical lift, evaluated on coordinatewise multiplication operators and on the universal wedge map W_B.

Suppose that P is a bounded k-linear lift of $A^{k,K_1}_{E_1,E_1;B}$ over K_1. For $a \in E_1$ let $D_a \in \mathcal{L}(E_1,E_1)$ be coordinatewise multiplication, $D_a x = ax$. The map $a \mapsto D_a$ is K_1-linear and $\|D_a\| \le \|a\|_\infty$. Define

$$\Phi(a_1,\dots,a_k;x_1,\dots,x_k) := P(D_{a_1},\dots,D_{a_k})(W_B)(x_1,\dots,x_k) \in B,$$
$$\Psi := \eta\circ\Phi|_{E_0^{2k}} \colon E_0^k \times E_0^k \to \Lambda^k_\kappa E_0.$$

Since $\Omega^\kappa \circ \eta = \operatorname{coeff}_0 \circ J$, the map Ψ is $\operatorname{coeff}_0 \circ J \circ \Phi|_{E_0^{2k}}$, read in $\Lambda^k_\kappa E_0$ through the injective map Ω^κ.

Lemma C.4. *The map Ψ has the following properties.*

- (Ψ1) *Ψ is $2k$-linear over κ.*
- (Ψ2) *Ψ is alternating in the last k slots.*
- (Ψ3) $\operatorname{sdim}(\Psi(z)) \le d := \lfloor kM_r\|P\|\rfloor$ *for* all $z \in E_0^{2k}$.
- (Ψ4) *Ψ satisfies* (Pol) *for the multiplier family $D^\kappa\colon E_0 \to \operatorname{End}_\kappa(E_0)$, $D^\kappa_a x = ax$: for all $m \ge 1$, $b_1,\dots,b_m \in E_0$, $\alpha \in \mathbb{N}^m$ with $|\alpha| = k$, and $x \in E_0^k$,*

$$\sum_{\text{type } f=\alpha} \Psi(b_f;x) = \sum_{\text{type } f=\alpha} (b_{f(1)}x_1)\wedge\cdots\wedge(b_{f(k)}x_k) \qquad \text{in } \Lambda^k_\kappa E_0.$$

In particular Ψ satisfies (Pol1).

Proof. The map Φ is K_1-multilinear, so its restriction to E_0^{2k} is κ-multilinear. Composing with the κ-linear map η proves (Ψ1). Since $P(D_{a_1},\dots,D_{a_k})(W_B)$ belongs to $\operatorname{Alt}^k_{K_1}(E_1;B)$, it vanishes when two of the last k arguments coincide; the same holds after applying η, proving (Ψ2).

For (Ψ3), all inputs from E_0 have norm at most one. Thus

$$\|\Phi(a;x)\| \le \|P\| \prod_i \|D_{a_i}\| \, \|W_B\| \prod_i \|x_i\|_\infty \le \|P\|.$$

Apply (C.2) and use that support dimension is an integer to obtain the stated bound by d.

To prove (Ψ4), we first compare coefficients over the infinite field K_1, while the values still lie in B. Fix $b_1, \dots, b_m \in E_0$ and $x \in E_0^k$. For $\lambda \in K_1^m$ put $a_\lambda := \sum_i \lambda_i b_i \in E_1$. The diagonal identity for the lift gives

$$\Phi(a_\lambda, \dots, a_\lambda; x) = P(D_{a_\lambda}, \dots, D_{a_\lambda})(W_B)(x) = A(D_{a_\lambda})(W_B)(x) = W_B(a_\lambda x_1, \dots, a_\lambda x_k).$$

Expanding both sides by multilinearity gives $\sum_\alpha \lambda^\alpha (L_\alpha - R_\alpha) = 0$ for all $\lambda \in K_1^m$, where

$$L_\alpha := \sum_{\text{type } f = \alpha} \Phi(b_f; x) \in B, \qquad R_\alpha := \sum_{\text{type } f = \alpha} W_B(b_{f(1)} x_1, \dots, b_{f(k)} x_k) \in B.$$

The field K_1 is infinite even when κ is finite. Lemma B.6, applied in the K_1-vector space B, therefore yields $L_\alpha = R_\alpha$ for every α.

We can now apply η. Additivity gives $\eta(L_\alpha) = \sum_{\text{type } f=\alpha} \Psi(b_f; x)$. To identify the other side, take $y_1, \dots, y_k \in E_0$. By Lemma C.2 and the determinant-array agreement preceding Proposition C.3,

$$J(W_B(y)) = \Omega^{K_1}(y_1 \wedge \cdots \wedge y_k) = \Omega^{\kappa}(y_1 \wedge \cdots \wedge y_k).$$

This array has entries in κ, so taking its constant coefficient leaves it unchanged. Uniqueness in Proposition C.3 then gives $\eta(W_B(y)) = y_1 \wedge \cdots \wedge y_k \in \Lambda_\kappa^k E_0$. Using $y_j = b_{f(j)} x_j$ in each summand of R_α proves

$$\eta(R_\alpha) = \sum_{\text{type } f = \alpha} (b_{f(1)} x_1) \wedge \cdots \wedge (b_{f(k)} x_k),$$

which is the required identity. □

The order of these operations is essential for the full identity (Ψ4). Evaluating the diagonal only at κ-points would give (Pw) there, which for $\kappa = \mathbb{F}_q$ and $k > q$ is strictly weaker than (Pol) (Proposition B.8(3)). The proof below uses only the consequence (Pol1).

C.5. **No bounded lift over a Laurent series field.** We now combine the support estimate (Proposition C.3) and the coefficient lift (Lemma C.4) with the finite-field theorem of Appendix B (Theorem B.9) to prove Theorem C.1.

Proof of Theorem C.1. Suppose P exists, and form Ψ as in §C.4. By Lemma C.4, $\Psi\colon E_0^k \times E_0^k \to \Lambda_\kappa^k E_0$ with $E_0 = \kappa^{\mathbb{N}}$ is $2k$-linear over κ, alternating and hence antisymmetric in the last k slots, satisfies (Pol1), and has $\operatorname{sdim}(\Psi(z)) \le d$ for all inputs z. Moreover $k! = 0$ in κ, since $k \ge p = \operatorname{char}\kappa$. This contradicts Theorem B.9(2) with $L = \kappa$ and $E_0 = \kappa^{\mathbb{N}}$. The last sentence follows from Proposition 3.3. □

Remark C.5. Lemmas C.2 and C.4 and Proposition C.3 hold for every field κ, and nothing in them uses a normalization of r: the constant M_r makes the argument uniform in $r \in (0,1)$. Only the final appeal to Theorem B.9 needs κ finite. We do not need, and do not discuss, the analogue of Theorem C.1 for infinite κ with this particular pair (E_1, B). For an infinite κ of characteristic p the field $\kappa((X))$ is covered by Theorem 6.1(1), with the pair obtained by ascent from its subfield $\mathbb{F}_p((X))$.

C.6. **Multiplier families and c_0 bases.** The proof of Theorem C.1 uses the hypothetical lift only on coordinatewise multiplication operators and only at the single form W_B. Moreover, Theorem B.9(1) needs only finitely supported test vectors with entries in $\{0, \pm 1\}$. Both observations give a much smaller bad parameter space. Throughout this subsection κ, k, r, K_1, E_1, B and W_B are as in Theorem C.1, $c_0(\mathbb{N}, K_1) \subseteq E_1$ carries the supremum norm, and $D_a \in \mathcal{L}(E_1, E_1)$ denotes multiplication by a.

Proposition C.6 (multipliers on c_0)**.** *The homogeneous map*

$$\sigma\colon c_0(\mathbb{N},K_1)\to \mathrm{Alt}^k_{K_1}(E_1;B),\qquad \sigma(a)(x_1,\dots,x_k)=W_B(ax_1,\dots,ax_k)=(ax_1)\wedge\cdots\wedge(ax_k),$$

has no bounded k-linear lift with values in $\mathrm{Alt}^k_{K_1}(E_1;B)$. Consequently, for every $u_0\in\mathcal{L}(E_1,E_1)$, the map $a\mapsto A^{k,K_1}_{E_1,E_1;B}(u_0+D_a)(W_B)$ is analytic at no point of $c_0(\mathbb{N},K_1)$. In particular $a\mapsto A^{k,K_1}_{E_1,E_1;B}(u_0+D_a)$ is analytic at no point.

Proof of Proposition C.6. Suppose $p\colon c_0(\mathbb{N},K_1)^k\to\mathrm{Alt}^k_{K_1}(E_1;B)$ is a bounded k-linear lift of σ. Let $V_{\mathrm{fin}}=\kappa^{(\mathbb{N})}$ be the finitely supported sequences with entries in κ; it lies in both $c_0(\mathbb{N},K_1)$ and E_0. Define $\Psi(a;x)=\eta\big(p(a_1,\dots,a_k)(x_1,\dots,x_k)\big)$ for $a,x\in V_{\mathrm{fin}}^k$. The proof of Lemma C.4 applies verbatim. The map Ψ is $2k$-linear over κ and alternating in the last k slots. Its inputs have supremum norm at most one, so (C.2) gives $\operatorname{sdim}\Psi(a;x)\le\lfloor kM_r\|p\|\rfloor$. The polarization points $a_\lambda=\sum_i\lambda_i b_i$, with $b_i\in V_{\mathrm{fin}}$, are finitely supported and hence lie in $c_0(\mathbb{N},K_1)$, so the identity (Pol), and in particular (Pol1), follows from the diagonal identity as before. This contradicts Theorem B.9(1), since $k!=0$ in κ.

For the second assertion, suppose that the map is analytic at a_0, with expansion $\sum_n p_n$ there. For $h\in c_0(\mathbb{N},K_1)$ and $t\in K_1$, multilinearity gives, in $\mathrm{Mult}^k_{K_1}(E_1;B)$,

$$W_B\circ(u_0+D_{a_0}+tD_h,\dots,u_0+D_{a_0}+tD_h)=\sum_{r=0}^{k}t^rC_r(h),\qquad C_k(h)=\sigma(h).$$

Comparing this with the given expansion on scalar lines, Lemma A.1 shows, exactly as in Proposition 3.3, that p_k is a bounded lift of σ. This is impossible. The operator-valued statement follows by evaluation at W_B. □

Corollary C.7 (diagonal transitions over a c_0 base)**.** *Let U be the open unit ball of $c_0(\mathbb{N},K_1)$, an analytic manifold with one chart. Let $\mathcal{V}\to U$ be the trivial bundle with fiber E_1 and the two global trivializations $\tau_0(a,z)=(a,z)$ and $\tau_1(a,z)=(a,D_{1+a}z)$, and let $\mathcal{B}=U\times B$. Then $\mathcal{V}$ is an analytic normed vector bundle whose transitions are isometric diagonal automorphisms of E_1. The naturally induced atlas on $\mathrm{Alt}^k(\mathcal{V};\mathcal{B})$ is not analytic: the section that is constantly W_B in the chart induced by τ_1 is, in the chart induced by τ_0, the map $a\mapsto W_B\circ(D_{1+a},\dots,D_{1+a})$, which is analytic at no point of U.*

Proof. For $a\in U$ every coordinate satisfies $|1+a_i|=1$, so D_{1+a} is an isometry of E_1. Its inverse $D_{(1+a)^{-1}}=\sum_{n\ge0}(-D_a)^n$ is a power series of radius one in a, since $D_a^n=D_{a^n}$ has norm at most $\|a\|^n$. Hence both transitions are analytic. If $\omega_a\in\mathrm{Alt}^k(\mathcal{V}_a;B)$ has coordinates $m_0(a)$ and $m_1(a)$ in the two induced charts, then $m_0(a)=m_1(a)\circ(D_{1+a},\dots,D_{1+a})$. With $m_1\equiv W_B$, this is the map of Proposition C.6 with $u_0=\mathrm{id}$, which is analytic at no point. □

The obstruction concerns the atlas obtained from the two analytic frames. The underlying pointwise bundle is trivial and has an analytic product structure; the failure is compatibility of that structure with the natural construction from all analytic frames. The complete target B admits no equivalent nonarchimedean norm by Proposition E.1, so this example does not contradict the c_0 parameter theorem with nonarchimedean targets.

Corollary C.8 (finite dimension is sharp for nonarchimedean bases)**.** *Let $K=\mathbb{F}_q((u))$, where q is a power of p, with a u-adic absolute value, and let $k\ge p$. Let P be a K-Banach space admitting an equivalent nonarchimedean norm. The alternating construction preserves analytic bundles and their operator-valued analytic morphisms over every analytic manifold modeled on P, with arbitrary normed fibers, if and only if P is finite-dimensional.*

Proof. If P is finite-dimensional, Lemma A.3 supplies continuous coordinates and Corollary 4.6 applies. Conversely, suppose that P is infinite-dimensional. Round an equivalent nonarchimedean norm upwards to the discrete value group $|K^\times|$; the result is still an equivalent nonarchimedean K-norm, and its nonzero values lie in $|K^\times|$. Serre's orthonormal-basis theorem [17, §1, Proposition 1] then identifies P topologically with $c_0(I, K)$ for an infinite set I. Choosing a countable subset of I gives bounded linear maps

$$\iota\colon c_0(\mathbb{N}, K) \longrightarrow P, \qquad \pi\colon P \longrightarrow c_0(\mathbb{N}, K), \qquad \pi\iota = \mathrm{id}.$$

Apply the Laurent construction with $\kappa = \mathbb{F}_q$ and $K_1 = K$. On the open set $V = \{x \in P : \|\pi x\| < 1\}$, take the trivial bundle with fiber E_1 and two analytic frames whose transition is $D_{1+\pi(x)}$, as in Corollary C.7, and the trivial target bundle with fiber B. If the induced alternating transition were analytic at $x_0 \in V$, restriction along $b \mapsto x_0 + \iota b$ would make

$$b \longmapsto A^k_{E_1,E_1;B}\big(D_{1+\pi(x_0)} + D_b\big)$$

analytic at zero. This contradicts Proposition C.6 with $u_0 = D_{1+\pi(x_0)}$. □

The dichotomy permits arbitrary normed target fibers. In the nonarchimedean Banach target category, Corollary 4.3 gives analyticity over every base for this discretely valued field.

Appendix D. Ascent to an arbitrary base field

To pass from a Laurent series field to a prescribed field, we work first inside its completion. A suitable Laurent subfield is spherically complete, so Ingleton's theorem supplies a contractive scalar retraction. The ordinary projective tensor norm then gives isometric scalar extensions of the source and target, together with a retraction onto the original Banach target. Operators and alternating forms extend compatibly with these embeddings. As a result, any lift after scalar extension descends to the Laurent subfield, and nonexistence of a lift ascends. The final density argument removes the completeness assumption on the prescribed field.

D.1. **The Laurent subfield.** We use spherical completeness in the sense used in Theorem 4.2: every nonempty family of pairwise-meeting closed balls has a common point. For the field case of the following lemma, see Schneider [16, Lemmas 1.3 and 1.6].

Lemma D.1 (discrete distances)**.** *Let Y be a complete nonarchimedean space whose distances lie in $r^{\mathbb{Z}} \cup \{0\}$ for some $r \in (0, 1)$. Then Y is spherically complete.*

Proof. Write $\mathcal{D}(a, \rho) = \{y \in Y : \mathrm{d}(y, a) \le \rho\}$. In a nonarchimedean space, every point of a ball is a center. Consequently, whenever two closed balls meet, the one with the smaller radius lies in the other. The discrete distance assumption also gives

$$\mathcal{D}(a, \rho) = \mathcal{D}(a, \rho^\flat), \qquad \rho^\flat := \max\{s \in r^{\mathbb{Z}} \cup \{0\} : s \le \rho\}.$$

We may therefore replace every radius by a value in $r^{\mathbb{Z}} \cup \{0\}$.

Let $(\mathcal{D}(a_i, \rho_i))_{i\in I}$ be a nonempty family of pairwise-meeting closed balls. If its radii have a least element ρ_{i_0}, then $\mathcal{D}(a_{i_0}, \rho_{i_0})$ lies in every ball of the family, so a_{i_0} is a common point. Otherwise, discreteness implies that all the radii are positive and that their infimum is zero. Choose $i_1, i_2, \dots$ so that ρ_{i_n} decreases strictly to zero. The corresponding balls are nested, and

$$\mathrm{d}(a_{i_{n'}}, a_{i_n}) \le \rho_{i_n} \qquad (n' \ge n).$$

Their centers form a Cauchy sequence. By completeness it has a limit a, and closedness puts a in every ball of the sequence. For any $i \in I$, choose n with $\rho_{i_n} \le \rho_i$. The n-th ball then lies in $\mathcal{D}(a_i, \rho_i)$, so a belongs to every ball of the original family. □

Lemma D.2 (Laurent subfield)**.** *Let K' be a complete nontrivially normed field of characteristic p, and let $t \in K'$ with $0 < |t| = r < 1$.*

(1) *K' is nonarchimedean, and $|c| = 1$ for every $c \in \mathbb{F}_p^\times$.*
(2) *For $a_m, \dots, a_M \in \mathbb{F}_p$, not all zero, $\left|\sum_{i=m}^{M} a_i t^i\right| = r^{\min\{i : a_i \neq 0\}}$.*
(3) *Give $\mathbb{F}_p((X))$ the absolute value with $|X| = r$. There is a continuous isometric ring homomorphism $\mathrm{ev}_t \colon \mathbb{F}_p((X)) \to K'$ such that $\mathrm{ev}_t\big(\sum_{i \geq m} a_i X^i\big) = \sum_{i \geq m} a_i t^i$. Its image K_1 is a closed subfield of K'; it is the closure of $\mathbb{F}_p(t)$, and it is isometrically isomorphic to $\mathbb{F}_p((X))$.*
(4) *K_1 is spherically complete.*

Proof. (1) Lemma A.2 makes K' nonarchimedean. For $c \in \mathbb{F}_p^\times$, the identity $c^{p-1} = 1$ gives $|c| = 1$.

(2) The nonzero terms $a_i t^i$ have pairwise distinct absolute values r^i. In a nonarchimedean field, $|x + y| = |x|$ whenever $|y| < |x|$. Repeatedly applying this identity shows that the sum has the absolute value of its largest term, namely $r^{\min\{i : a_i \neq 0\}}$.

(3) Evaluation on Laurent polynomials defines a ring homomorphism $\mathrm{ev}_t \colon \mathbb{F}_p[X, X^{-1}] \to K'$, and (2) shows that $|\mathrm{ev}_t(g)| = r^{\operatorname{ord} g} = |g|$. Laurent polynomials are dense in $\mathbb{F}_p((X))$, since truncations converge. Completeness of K' therefore gives a unique continuous extension to $\mathbb{F}_p((X))$. Passing to limits preserves the isometry and the ring operations. It also gives the stated evaluation formula, because the partial sums of $\sum_{i \geq m} a_i X^i$ map to the partial sums of $\sum_{i \geq m} a_i t^i$.

The image is complete as the isometric image of a complete space, hence closed in K'. It is a subfield: for $g \neq 0$, the identity $\mathrm{ev}_t(g)\mathrm{ev}_t(g^{-1}) = 1$ provides the inverse. This subfield contains $\mathbb{F}_p(t)$, while the Laurent polynomials $\mathbb{F}_p[t, t^{-1}] \subseteq \mathbb{F}_p(t)$ are dense in it. The image is consequently exactly the closure of $\mathbb{F}_p(t)$.

(4) By (3), K_1 is complete and has value group $|K_1^\times| = r^{\mathbb{Z}}$. It is nonarchimedean by (1), so its distances lie in $r^{\mathbb{Z}} \cup \{0\}$. Lemma D.1 gives spherical completeness. □

D.2. **Ingleton's theorem and its consequences.** Throughout the rest of this section we fix fields $K_1 \subseteq K'$ with the following properties.

(H1) K_1, with the restricted absolute value, is nontrivially normed, complete and spherically complete.
(H2) K' is nonarchimedean.

By multiplication, K' is a nonarchimedean normed K_1-space. In our application $K' = \widehat{K}$ and K_1 is the field of Lemma D.2.

Theorem D.3 (Ingleton [6]; see also [14])**.** *Let K_1 be a spherically complete nonarchimedean normed field, V a K_1-vector space with a nonarchimedean norm, $V_0 \subseteq V$ a subspace, $C \geq 0$, and $T_0 \colon V_0 \to K_1$ linear with $|T_0 v| \leq C\|v\|$ for $v \in V_0$. Then T_0 extends to a linear map $T \colon V \to K_1$ with $|Tx| \leq C\|x\|$ for all $x \in V$.*

Schneider [16, Prop. 9.2] gives the more general formulation with a dominating nonarchimedean seminorm.

We will use Theorem D.3 on the nonarchimedean K_1-space K' in two ways: to obtain the scalar retraction below, and to extend finite-dimensional coordinate functionals in the proof of positivity of the projective tensor norm. Both applications have codomain K_1. The spaces E_1 and F_1 introduced later need not be nonarchimedean.

Corollary D.4 (Ingleton projection)**.** *There is a K_1-linear map $\varpi \colon K' \to K_1$ with $\varpi|_{K_1} = \mathrm{id}$ and $|\varpi(\lambda)| \leq |\lambda|$ for all $\lambda \in K'$.*

Proof. Theorem D.3 with $V = K'$, $V_0 = K_1$, $T_0 = \mathrm{id}$ and $C = 1$; its hypotheses hold by (H1) and (H2). □

Lemma D.5 (extension)**.** *Let $\mathbb{K} \in \{K_1, K'\}$ and let $C \geq 0$.*

(1) *Let V_0 be a dense $\mathbb{K}$-subspace of a normed $\mathbb{K}$-space V, let Y be a $\mathbb{K}$-Banach space and $T_0 \colon V_0 \to Y$ a $\mathbb{K}$-linear map with $\|T_0 x\| \leq C\|x\|$. Then T_0 extends uniquely to a continuous map $T \colon V \to Y$, which is $\mathbb{K}$-linear with $\|Tx\| \leq C\|x\|$.*

(2) *Let $V_1, \dots, V_k$ be normed K'-spaces with dense subspaces V_i^0, Y a K'-Banach space, and $M \colon \prod_i V_i^0 \to Y$ a K'-multilinear map with $\|M(d)\| \leq C \prod_i \|d_i\|$. Then M extends uniquely to a continuous K'-multilinear map $\overline{M} \colon \prod_i V_i \to Y$ with $\|\overline{M}\| \leq C$.*

Proof. (1) The bound makes T_0 Lipschitz, hence uniformly continuous. Density of V_0 and completeness of Y give a unique continuous extension. Linearity and the norm bound pass to limits by continuity of addition, scalar multiplication and the norm.

(2) The case $k = 0$ is immediate. For $k \geq 1$, multilinearity gives the telescoping estimate

$$\|M(d) - M(d')\| \leq C\rho^{k-1} \sum_i \|d_i - d_i'\|$$

whenever $\|d_i\|, \|d_i'\| \leq \rho$ for every i, with $\rho > 0$. For any $v \in \prod_i V_i$, choose tuples $d^{(n)} \in \prod_i V_i^0$ converging to v. Their coordinates are uniformly bounded, so the estimate makes $M(d^{(n)})$ Cauchy in Y. Define $\overline{M}(v)$ to be its limit. The same estimate, applied to two approximating sequences, proves that this definition is independent of the choices and is continuous on bounded sets. Multilinearity and the product norm bound pass to limits. Finally, density of $\prod_i V_i^0$ proves uniqueness. □

D.3. **Projective base change.** The next construction extends scalars while retaining the original norm and a contractive retraction whenever the original space is complete. The scalar retraction from Corollary D.4 is the essential ingredient; the construction applies to arbitrary normed spaces.

For a normed K_1-space V let $V_{K'} := V \otimes_{K_1} K'$, a K'-vector space through the second factor, and put

$$\|u\|_\pi := \inf\Big\{ \sum_j \|x_j\| \, |\lambda_j| \ : \ u = \sum_j x_j \otimes \lambda_j \Big\},$$

the infimum over finite decompositions, with an ordinary sum. For a bounded K_1-linear $\psi \colon K' \to K_1$ let $T_\psi \colon V_{K'} \to V$ be the K_1-linear map with $T_\psi(x \otimes \lambda) = \psi(\lambda)x$; it exists because $(x, \lambda) \mapsto \psi(\lambda)x$ is K_1-bilinear. Let $\iota_V \colon V \to V_{K'}$, $x \mapsto x \otimes 1$; it is K_1-linear, since $cx \otimes 1 = x \otimes c = c \cdot (x \otimes 1)$ for $c \in K_1$.

Lemma D.6 (projective base change)**.** *Let V be a normed K_1-space; no nonarchimedean hypothesis is made on V.*

(1) *$\|T_\psi u\| \leq \|\psi\| \, \|u\|_\pi$ for every bounded K_1-linear $\psi \colon K' \to K_1$.*

(2) *$\|\cdot\|_\pi$ is a K'-norm on $V_{K'}$.*

(3) *ι_V is isometric.*

(4) *Let $V \widehat{\otimes}_\pi K'$ be the completion of $(V_{K'}, \|\cdot\|_\pi)$. If V is complete, then T_ϖ, with ϖ as in Corollary D.4, extends to a K_1-linear map $\Pi_V \colon V \widehat{\otimes}_\pi K' \to V$ with $\|\Pi_V\| \leq 1$ and $\Pi_V \circ \iota_V = \mathrm{id}_V$.*

Proof. (1) For every decomposition $u = \sum_j x_j \otimes \lambda_j$,

$$\|T_\psi u\| = \Big\| \sum_j \psi(\lambda_j) x_j \Big\| \leq \sum_j \|\psi\| |\lambda_j| \|x_j\|.$$

Taking the infimum over decompositions proves the contraction estimate.

(2) Concatenating decompositions gives the triangle inequality. Multiplying their second factors by $\mu \in K'$ gives $\|\mu u\|_\pi \le |\mu|\|u\|_\pi$; for $\mu \neq 0$, applying the same inequality with μ^{-1} gives equality. The zero-scalar case is immediate.

It remains to prove positivity. Let $u \neq 0$. Choose a finite tensor representation and rewrite its second factors in a basis of their K_1-span. This gives

$$u = \sum_{j=1}^{n} x_j \otimes \lambda_j$$

with $\lambda_1, \dots, \lambda_n$ linearly independent over K_1. Some x_{j_0} is nonzero, since otherwise $u = 0$. On the finite-dimensional subspace $U := \operatorname{span}_{K_1}\{\lambda_1, \dots, \lambda_n\} \subseteq K'$, take the coordinate functional $\psi_0(\lambda_j) = \delta_{jj_0}$. It is bounded by some $C \ge 0$ by Lemma A.3, using completeness of K_1 in (H1). Apply Theorem D.3 to the nonarchimedean K_1-space K' to extend it to $\psi\colon K' \to K_1$ with $\|\psi\| \le C$. Then $T_\psi u = x_{j_0}$, so (1) gives

$$0 < \|x_{j_0}\| \le C\|u\|_\pi.$$

Thus $\|u\|_\pi > 0$. This argument detects a nonzero vector coefficient by a functional on the *scalar* factor K'; it requires no separating continuous dual on V.

(3) The elementary tensor representation gives $\|x \otimes 1\|_\pi \le \|x\|$. For the reverse inequality, use $\varpi(1) = 1$ and $\|\varpi\| \le 1$ in (1):

$$\|x\| = \|T_\varpi(x \otimes 1)\| \le \|x \otimes 1\|_\pi.$$

(4) When V is complete, the contraction T_ϖ on the dense tensor subspace extends by Lemma D.5(1), with $\mathbb{K} = K_1$, to a K_1-linear contraction Π_V with values in V. On the embedded copy of V it satisfies $\Pi_V \iota_V x = T_\varpi(x \otimes 1) = \varpi(1)x = x$. □

We now fix the spaces to which the descent argument will be applied. Let E_1 be a normed K_1-space and F_1 a K_1-Banach space, and put

$$E := E_1 \widehat{\otimes}_\pi K', \qquad F := F_1 \widehat{\otimes}_\pi K',$$

the completions of $((E_1)_{K'}, \|\cdot\|_\pi)$ and $((F_1)_{K'}, \|\cdot\|_\pi)$. They are K'-Banach spaces; $\iota_E\colon E_1 \to E$ and $\iota_F\colon F_1 \to F$ are K_1-linear isometries, and $\Pi_F\colon F \to F_1$ is K_1-linear with $\|\Pi_F\| \le 1$ and $\Pi_F \circ \iota_F = \mathrm{id}_{F_1}$ (Lemma D.6).

Lemma D.7 (operators)**.** *For $f \in \mathcal{L}_{K_1}(E_1, E_1)$ the map $f \otimes \mathrm{id}$ on $(E_1)_{K'}$ is K'-linear with $\|(f \otimes \mathrm{id})u\|_\pi \le \|f\|\|u\|_\pi$. Its extension $f_{K'} \in \mathcal{L}_{K'}(E, E)$ satisfies $\|f_{K'}\| \le \|f\|$ and $f_{K'} \circ \iota_E = \iota_E \circ f$, and $f \mapsto f_{K'}$ is K_1-linear.*

Proof. For any decomposition $u = \sum_j x_j \otimes \lambda_j$, the image under $f \otimes \mathrm{id}$ has the decomposition $\sum_j f(x_j) \otimes \lambda_j$, whose cost is at most $\|f\| \sum_j \|x_j\||\lambda_j|$. Taking the infimum proves the norm bound on the algebraic tensor product. Since E is complete, Lemma D.5(1), with $\mathbb{K} = K'$, gives the continuous extension $f_{K'}$ with the same bound. Its action on $x \otimes 1$ proves the embedding identity. Additivity and K_1-homogeneity in f hold on elementary tensors, hence on their span, and then on E by continuity. □

Lemma D.8 (forms)**.** *For $m \in \mathrm{Alt}^k_{K_1}(E_1; F_1)$ there is $m_{K'} \in \mathrm{Alt}^k_{K'}(E; F)$ with $\|m_{K'}\| \le \|m\|$ and*

$$m_{K'}(\iota_E x_1, \dots, \iota_E x_k) = \iota_F\big(m(x_1, \dots, x_k)\big) \qquad (x_i \in E_1),$$

and $m \mapsto m_{K'}$ is K_1-linear.

Proof. We first define the form on algebraic tensors, then establish a bound independent of their decompositions, and finally pass to the completions. Fix a K_1-basis $(\lambda_\beta)_{\beta\in\mathfrak{B}}$ of K'. Every $u \in (E_1)_{K'}$ has a unique finite expansion $u = \sum_\beta u^\beta \otimes \lambda_\beta$, with $u^\beta \in E_1$, and each coordinate map $u \mapsto u^\beta$ is K_1-linear. Define

$$m^{\mathrm{alg}}(u_1,\dots,u_k) := \sum_{\beta_1,\dots,\beta_k} m\big(u_1^{\beta_1},\dots,u_k^{\beta_k}\big) \otimes \lambda_{\beta_1}\cdots\lambda_{\beta_k} \in (F_1)_{K'}.$$

The sum is finite, since each input has finitely many nonzero coordinates.

(i) Linearity of the coordinate maps and multilinearity of m make m^{alg} K_1-multilinear.

(ii) On elementary tensors, the definition gives

$$m^{\mathrm{alg}}(x_1\otimes\mu_1,\dots,x_k\otimes\mu_k) = m(x_1,\dots,x_k)\otimes\mu_1\cdots\mu_k \qquad (x_i\in E_1,\ \mu_i\in K').$$

Indeed, writing $\mu_i = \sum_\beta c_{i\beta}\lambda_\beta$ with $c_{i\beta}\in K_1$ gives $x_i\otimes\mu_i = \sum_\beta (c_{i\beta}x_i)\otimes\lambda_\beta$. Substitution in the defining sum yields

$$\sum_{\beta_1,\dots,\beta_k}\Big(\prod_i c_{i\beta_i}\Big)m(x)\otimes\prod_i\lambda_{\beta_i} = m(x)\otimes\prod_i\Big(\sum_\beta c_{i\beta}\lambda_\beta\Big),$$

which is the asserted formula.

(iii) The map m^{alg} is K'-multilinear. By additivity in each slot, it suffices to check K'-homogeneity when every argument is an elementary tensor; there it follows from (ii). The same formula determines the map on all tuples of algebraic tensors, so the initial basis choice does not affect it.

(iv) For arbitrary decompositions $u_i = \sum_j x_{ij}\otimes\mu_{ij}$, multilinearity and (ii) give

$$m^{\mathrm{alg}}(u) = \sum_{j_1,\dots,j_k} m(x_{1j_1},\dots,x_{kj_k})\otimes\prod_i\mu_{ij_i}.$$

The cost of this tensor decomposition is at most $\|m\|\prod_i\big(\sum_j\|x_{ij}\|\|\mu_{ij}\|\big)$. Taking the infimum separately over the decompositions in each slot proves

$$\|m^{\mathrm{alg}}(u_1,\dots,u_k)\|_\pi \le \|m\|\prod_i\|u_i\|_\pi.$$

(v) To prove strong alternation, suppose two distinct slots a,b both contain $u=\sum_j x_j\otimes\mu_j$. Use this *same* decomposition in both slots and expand all arguments into elementary tensors. Terms using the same index j in slots a and b vanish because m has a repeated argument. For $j\neq j'$, the term using indices (j,j') in these slots pairs with the term using (j',j). Their scalar factors agree, while the values of m are negatives of one another, so the pair sums to zero. Thus m^{alg} vanishes whenever any two arguments coincide. Pair cancellation remains valid in characteristic 2; merely checking $m^{\mathrm{alg}}(u,\dots,u)=0$ would not establish strong alternation for $k\ge 3$.

(vi) Regard m^{alg} as taking values in the completion F. By (iii), (iv) and Lemma D.5(2), it extends to a continuous K'-multilinear map $m_{K'}\colon E^k\to F$ with $\|m_{K'}\|\le\|m\|$. Strong alternation also passes to the limit: if $v_a=v_b$, approximate this common vector by one sequence in $(E_1)_{K'}$, and approximate the other entries separately. Every approximating tuple has equal entries in slots a,b, so (v) and continuity give $m_{K'}(v)=0$.

(vii) Taking every scalar factor equal to one in (ii) gives

$$m_{K'}(\iota_E x_1,\dots,\iota_E x_k) = m^{\mathrm{alg}}(x_1\otimes 1,\dots,x_k\otimes 1) = m(x)\otimes 1 = \iota_F(m(x)).$$

The defining formula is K_1-linear in m. These linear identities pass to the continuous extensions by uniqueness, proving K_1-linearity of $m\mapsto m_{K'}$. □

Proposition D.9 (descent of a lift)**.** *Assume* (H1) *and* (H2)*. Let E_1 be a normed K_1-space, F_1 a K_1-Banach space, and E, F as above. If $A^{k,K'}_{E,E;F}$ admits a bounded k-linear lift P over K', then $A^{k,K_1}_{E_1,E_1;F_1}$ admits a bounded k-linear lift P_1 over K_1 with $\|P_1\| \le \|P\|$. Equivalently, by Proposition 3.3 over K_1 and over K': if $A^{k,K_1}_{E_1,E_1;F_1}$ is analytic at no point, then $A^{k,K'}_{E,E;F}$ is analytic at no point either.*

Proof. Extend the operator and form inputs to K', apply the given lift, and retract the value to F_1. Explicitly, for $f_1, \dots, f_k \in \mathcal{L}_{K_1}(E_1, E_1)$, $m \in \mathrm{Alt}^k_{K_1}(E_1; F_1)$ and $x \in E_1^k$, set

$$P_1(f_1, \dots, f_k)(m)(x_1, \dots, x_k) := \Pi_F\Big(P\big((f_1)_{K'}, \dots, (f_k)_{K'}\big)(m_{K'})(\iota_E x_1, \dots, \iota_E x_k)\Big).$$

We check the norm bound and then the diagonal identity.

For fixed f and m, the map $P((f_i)_{K'})(m_{K'})$ is a continuous alternating K'-multilinear form on E. Precomposing each slot with the K_1-linear isometry ι_E and postcomposing with Π_F gives a K_1-multilinear map on E_1, still vanishing whenever two arguments coincide. Lemmas D.6–D.8 give

$$\|P_1(f)(m)\| \le \|\Pi_F\|\|P\| \prod_i \|(f_i)_{K'}\| \|m_{K'}\| \le \|P\| \prod_i \|f_i\| \|m\|.$$

In particular, $P_1(f)(m) \in \mathrm{Alt}^k_{K_1}(E_1; F_1)$. The form extension is K_1-linear in m, the operator extension is K_1-linear in each f_i, and P is K'-multilinear with values in K'-linear operators. Consequently P_1 is K_1-linear in m and K_1-multilinear in $(f_1, \dots, f_k)$. The displayed estimate proves boundedness at both levels and yields $\|P_1\| \le \|P\|$.

On the diagonal, use $P(g, \dots, g) = A^{k,K'}(g)$ with $g = f_{K'}$, followed by the embedding identities in Lemmas D.7 and D.8 and the retraction identity $\Pi_F \iota_F = \mathrm{id}$:

$$\begin{aligned} P_1(f, \dots, f)(m)(x) &= \Pi_F\big(m_{K'}(f_{K'}\iota_E x_1, \dots, f_{K'}\iota_E x_k)\big) \\ &= \Pi_F\big(m_{K'}(\iota_E f x_1, \dots, \iota_E f x_k)\big) \\ &= \Pi_F\big(\iota_F(m(f x_1, \dots, f x_k))\big) \\ &= m(f x_1, \dots, f x_k) = A^{k,K_1}(f)(m)(x). \end{aligned}$$

□

Remark D.10. Proposition D.9 is characteristic-free and uses no hypothesis on k. It transports negative results upward only, from K_1 to K'. Ingleton's theorem is used only on K', with no nonarchimedean norm hypothesis on F_1. The completeness of F_1 is used essentially: it is what makes the retraction Π_F land in F_1.

D.4. **Incomplete base fields.** The preceding construction takes place over the completion $\widehat{K}$ of the prescribed field. To return to K, we restrict scalars on the resulting Banach spaces. Density of K in $\widehat{K}$ forces every continuous K-multilinear map between these spaces to be $\widehat{K}$-multilinear, so a lift cannot appear under this restriction.

Lemma D.11. *Let K be a nontrivially normed field with completion $\widehat{K}$, and let E, E', F be $\widehat{K}$-Banach spaces, regarded as K-Banach spaces by restriction of scalars (with the same norms). Then $\mathcal{L}_K(E, E') = \mathcal{L}_{\widehat{K}}(E, E')$, $\mathrm{Alt}^k_K(E; F) = \mathrm{Alt}^k_{\widehat{K}}(E; F)$ and $\mathrm{Alt}^k_K(E'; F) = \mathrm{Alt}^k_{\widehat{K}}(E'; F)$ with the same norms, $A^{k,K}_{E,E';F} = A^{k,\widehat{K}}_{E,E';F}$, and every bounded k-linear lift of $A^{k,K}_{E,E';F}$ over K is a bounded k-linear lift of $A^{k,\widehat{K}}_{E,E';F}$ over $\widehat{K}$.*

Proof. Let g be a continuous K-linear map between normed spaces that already carry compatible $\widehat{K}$-scalar actions. For $\lambda \in \widehat{K}$, choose $\lambda_n \in K$ with $\lambda_n \to \lambda$. Since $\|\lambda_n x - \lambda x\| = |\lambda_n - \lambda|\|x\|$,

continuity gives

$$g(\lambda x) = \lim_n g(\lambda_n x) = \lim_n \lambda_n g(x) = \lambda g(x).$$

Thus g is $\widehat{K}$-linear. Applying the same argument with all but one slot fixed shows that every continuous K-multilinear map is $\widehat{K}$-multilinear. Alternation is the same equal-input vanishing condition over either field.

The spaces of bounded maps therefore coincide as sets. Their operator norms agree as well, since the norms are defined by the same pointwise bounds on the same vectors. They are $\widehat{K}$-Banach spaces, and $A^{k,K}_{E,E';F}$ and $A^{k,\widehat{K}}_{E,E';F}$ agree by their defining precomposition formula. Finally, apply the slotwise density argument once more to a bounded k-linear lift P between these map spaces. It becomes $\widehat{K}$-multilinear with the same norm and the same diagonal identity, and hence is a lift over $\widehat{K}$. □

Appendix E. Proof of the counterexample theorem

Appendices B–D give the obstruction to a bounded lift over a Laurent series field and its transfer to a prescribed nontrivially normed field of characteristic p. We first establish the remaining property of the target, that it admits no equivalent nonarchimedean norm (Proposition E.1), and then assemble the proof of Theorem 6.1(1). The classification in Corollary 6.2 follows from this obstruction and the invertible-factorial construction.

E.1. **The target has no equivalent nonarchimedean norm.**

Proposition E.1. *Let κ be any field, $k \geq 2$, $r \in (0,1)$, and let $K_1 = \kappa((X))$, E_1, E_0 and B be as in Appendix C. Let $e_i \in E_0$ be the unit vectors and $\omega_j := e_{kj+1} \wedge \cdots \wedge e_{kj+k} \in B$ $(j \geq 0)$, wedges on pairwise disjoint blocks of indices. Then $\|\omega_j\|_B = 1$ and*

$$\frac{N}{M_r} \;\leq\; \Big\|\sum_{j<N} \omega_j\Big\|_B \;\leq\; N \qquad (N \geq 1).$$

Hence the norm of B is not nonarchimedean, and B admits no equivalent nonarchimedean norm. The same holds for $F = B\widehat{\otimes}_\pi K'$ as in Appendix D, for any $K' \supseteq K_1$ satisfying (H1) *and* (H2)*, regarded as a normed space over K' or over any subfield of K'.*

Proof. Each ω_j is a wedge of unit vectors, so $\|\omega_j\|_B \leq 1$ by the defining projective norm. Evaluation of its determinant array at the block indices gives $\det I = 1$, hence $\|\omega_j\|_B \geq \|J\omega_j\|_\infty \geq 1$. The triangle inequality now gives the upper bound N.

For the lower bound, write ξ_N for the sum of the same block wedges in $\Lambda^k_\kappa E_0$, and put $\omega := \sum_{j<N} \omega_j \in B$. Its array $J\omega$ is the constant Laurent array $\Omega^\kappa(\xi_N)$. Thus $\mathrm{coeff}_0(J\omega) = \Omega^\kappa(\xi_N)$, and uniqueness in Proposition C.3 gives $\eta(\omega) = \xi_N$.

We apply Lemma B.4 to ξ_N. Fix one of the first N blocks and an index i_0 in it. Take for φ the $k-1$ coordinate functionals corresponding to the other indices of that block. On every other block all these functionals vanish, so its contraction is zero; here $k \geq 2$ ensures that the minors have positive size. On the chosen block, the coefficient of e_{i_0} is ± 1, while every other cofactor contains the zero column obtained by evaluating the functionals at e_{i_0}. Consequently $c_\varphi(\xi_N) = \pm e_{i_0}$. These contractions span all kN unit vectors in the first N blocks, so

$$kN \leq \mathrm{sdim}(\xi_N) = \mathrm{sdim}(\eta(\omega)) \leq kM_r\|\omega\|_B$$

by (C.2). This proves the lower bound.

If $\|\cdot\|_u$ were an equivalent nonarchimedean norm, choose positive constants c, c' with $\|\cdot\|_B \le c\|\cdot\|_u \le c'\|\cdot\|_B$. Then

$$\frac{N}{M_r} \le \Big\|\sum_{j<N} \omega_j\Big\|_B \le c \max_{j<N} \|\omega_j\|_u \le c'$$

for every $N \ge 1$, a contradiction. In particular, the original norm is not nonarchimedean. This argument uses only the nonarchimedean inequality for $\|\cdot\|_u$, not its scalar homogeneity.

Finally, $\iota_F \colon B \to F$ is a K_1-linear isometry by Lemma D.6(3). The images $\iota_F \omega_j$ satisfy the same unit-norm and sum-growth estimates, so the argument applies to F as well. Restriction to a subfield leaves these estimates unchanged. □

If $r \le \frac{1}{2}$, then $M_r = 1$ and $\|\sum_{j<N} \omega_j\|_B = N$ exactly.

E.2. **Proof of Theorem 6.1(1).**

Proof. Let K be a nontrivially normed field of characteristic $p > 0$, and let $k \ge p$.

(1) *Choose a Laurent subfield.* By Lemma A.2, K is nonarchimedean. Its completion $\widehat{K}$ is a complete, nontrivially normed, nonarchimedean field of characteristic p. Choose $t \in K$ with $0 < |t| = r < 1$, and let K_1 be the closure of $\mathbb{F}_p(t)$ in $\widehat{K}$. Lemma D.2 identifies K_1 isometrically with $\mathbb{F}_p((X))$, with $|X| = r$, by ev_t. In particular, K_1 is complete and spherically complete.

(2) *Obtain the obstruction over K_1.* Put $E_1 := \ell^\infty(\mathbb{N}, K_1)$, and let B be the completion of $\Lambda^k_{K_1} E_1$ under the projective exterior norm. Transport of scalars along ev_t preserves the norms, exterior powers and completions in this construction. It also identifies the operator and alternating-map spaces, their precomposition maps, and bounded lifts. Theorem C.1, with $\kappa = \mathbb{F}_p$, therefore shows that $A^{k,K_1}_{E_1,E_1;B}$ has no bounded k-linear lift.

(3) *Ascend to $\widehat{K}$.* The inclusion $K_1 \subseteq \widehat{K}$ satisfies (H1) and (H2), and B is complete. Proposition D.9, with $K' = \widehat{K}$, gives Banach spaces

$$E := E_1 \widehat{\otimes}_\pi \widehat{K}, \qquad F := B \widehat{\otimes}_\pi \widehat{K}$$

for which $A^{k,\widehat{K}}_{E,E;F}$ has no bounded k-linear lift.

(4) *Descend to K.* Regard the same E and F as K-Banach spaces. By Lemma D.11, a bounded lift of $A^{k,K}_{E,E;F}$ would extend to a bounded lift of $A^{k,\widehat{K}}_{E,E;F}$, which is impossible. Proposition 3.3 applies over the possibly incomplete field K: $A^{k,K}_{E,E;F}$ is analytic at no point. Reindexing the arguments by a bijection gives the same conclusion, with the same spaces E, F, for every finite index set with k elements.

(5) *Exclude an equivalent nonarchimedean target norm.* Since $k \ge p \ge 2$, Proposition E.1 applies to B and F, after the scalar identification in step (2). Its conclusion persists when F is regarded over K. □

Remark E.2. If $K_1 = \widehat{K}$, as when $K = \mathbb{F}_p((u))$ and $t = u$, the completed tensor products identify isometrically with E_1 and B: use Lemma D.6(3) and $x \otimes \lambda = \lambda x \otimes 1$. Thus E is nonarchimedean in this case. For $K = \mathbb{F}_q((u))$, $q = p^n$, with its u-adic absolute value, one can instead apply Theorem C.1 directly with $\kappa = \mathbb{F}_q$ and $X = u$, obtaining $E = \ell^\infty(\mathbb{N}, K)$ and $F = B$. The proof over a general base field does not require, or assert, that E is nonarchimedean.

Appendix F. A counterexample with scalar values

We now prove Theorem 6.1(2), in the more explicit form below. The obstruction persists when the value space is the ground field. The construction combines a finite-dimensional incompatibility with a family of limit conditions on a countable tree. The finite lemma applies directly in every degree whose factorial vanishes; no passage from degree p to a higher degree is needed.

Theorem F.1 (scalar counterexample)**.** *Let K be a complete, nontrivially normed field that is not spherically complete, and let $k \geq 1$ satisfy $k! = 0$ in K. There are nonarchimedean Banach spaces E, E' for which*

$$A^k_{E,E';K} \colon \mathcal{L}(E, E') \longrightarrow \mathcal{L}\big(\mathrm{Alt}^k(E'; K), \mathrm{Alt}^k(E; K)\big)$$

is analytic at no point. The spaces may be taken to be closed subspaces of $\ell^\infty(\Lambda; K^{k+1})$ and $\ell^\infty(\Lambda; K^k)$, respectively, containing the corresponding c_0 spaces, where Λ is countable. Both spaces are nonseparable.

The factorial hypothesis says precisely that K has characteristic $p > 0$ and $k \geq p$. In particular $k \geq 2$, and K is nonarchimedean by Lemma A.2. We first establish the three ingredients of the construction.

F.1. **The countable multilinear tail property.** For a set I, a family $(a_i)_{i \in I}$ is *null* if, for every $\varepsilon > 0$, only finitely many indices satisfy $|a_i| \geq \varepsilon$. Over a complete nonarchimedean field, every null scalar family is unconditionally summable: its finite partial sums form a Cauchy net, since the norm of a difference is at most the largest omitted term. Its sum has norm at most $\sup_i |a_i|$, and such sums may be regrouped.

We derive the multilinear tail property from van der Put's description of the dual of countable ℓ^∞. The induction is independent of the characteristic.

Lemma F.2 (multilinear tails)**.** *Let K be complete, nonarchimedean and not spherically complete, let I be countably infinite, and let $d \geq 1$. Every bounded d-linear map $\lambda \colon \ell^\infty(I, K)^d \to K$ has an expansion*

$$\lambda(x^1, \dots, x^d) = \sum_{(i_1, \dots, i_d) \in I^d} a_{i_1, \dots, i_d} x^1_{i_1} \cdots x^d_{i_d}, \qquad a_{i_1, \dots, i_d} = \lambda(e_{i_1}, \dots, e_{i_d}), \tag{F.1}$$

where $a \in c_0(I^d, K)$. Consequently, for every $\varepsilon > 0$, there is a finite $S \subseteq I$ such that $|\lambda(x^1, \dots, x^d)| \leq \varepsilon$ whenever all $\|x^r\| \leq 1$ and at least one x^r vanishes on S. In particular $\lambda(e_i, \dots, e_i) \to 0$ outside finite subsets of I.

Proof. After reindexing, take $I = \mathbb{N}$. Van der Put's theorem [12, Théorème 4.1], applied to countably many copies of K (a countable index set satisfies its nonmeasurability hypothesis), gives

$$\varphi(x) = \sum_i x_i \varphi(e_i), \qquad (\varphi(e_i))_i \in c_0(\mathbb{N}, K) \quad \big(\varphi \in \mathcal{L}(\ell^\infty, K)\big).$$

This is the case $d = 1$. Suppose the assertion about the array and expansion holds in smaller degrees. Fixing a coordinate vector in any slot of λ gives a form of degree $d - 1$. Hence, for every $\varepsilon > 0$, each slice of the set

$$W = \{(i_1, \dots, i_d) : |a_{i_1, \dots, i_d}| \geq \varepsilon\}$$

obtained by fixing one coordinate is finite.

If W were infinite, we could choose $w^{(0)}, w^{(1)}, \dots \in W$ with the coordinates $w^{(n)}_r$ distinct as n varies, for each r: a finite list of forbidden coordinates excludes only finitely many elements

of W. Define an isometric embedding $T_r\colon \ell^\infty(\mathbb{N},K)\to\ell^\infty(\mathbb{N},K)$ by putting $(T_rx)_{w_r^{(n)}}=x_n$ and zero elsewhere. The pulled-back form $\mu=\lambda\circ(T_1,\dots,T_d)$ has array b with $|b_{n,\dots,n}|\ge\varepsilon$ for all n. Its slices are null by induction. Thus

$$F_n=\{v\in\mathbb{N}^d:|b_v|\ge\varepsilon,\ v_r=n\text{ for some }r\}$$

is finite. Choose $n_1<n_2<\cdots$ so that n_s exceeds every coordinate occurring in $F_{n_1}\cup\cdots\cup F_{n_{s-1}}$. Every nonconstant tuple in $\{n_s:s\ge1\}^d$ then has coefficient of norm $<\varepsilon$: its least coordinate puts a large tuple in the corresponding F_{n_r}, whereas its greatest coordinate violates the choice of the subsequence.

Let y be the indicator of $\{n_s:s\ge1\}$ and set $\psi(x)=\mu(x,y,\dots,y)$. The expansion of each fixed-coordinate slice gives

$$\psi(e_{n_s})=b_{n_s,\dots,n_s}+\sum_{v'\in\{n_r:r\ge1\}^{d-1},\ v'\neq(n_s,\dots,n_s)}b_{n_s,v'}.$$

The summands in the last sum form a null family and all have norm $<\varepsilon$. Their supremum is also $<\varepsilon$: outside a finite set they have norm $<\varepsilon/2$, and the remaining finite maximum is $<\varepsilon$. The nonarchimedean inequality therefore gives $|\psi(e_{n_s})|\ge\varepsilon$ for every s, contradicting the displayed dual formula. Thus a is null. Applying that formula in the first slot and the inductive expansion in the remaining slots gives (F.1); the null full array permits regrouping the iterated sums.

Finally choose a finite S containing all coordinates occurring in tuples with $|a_{i_1,\dots,i_d}|>\varepsilon$. If one input vanishes on S, all these terms vanish. Every remaining term has norm at most ε on the product of unit balls, so the same holds for the sum. Apply this conclusion to coordinate vectors outside S to obtain the last assertion. □

F.2. **Chains and a finite-dimensional incompatibility.** Let L be a nonempty finite set and $\Lambda=L^{<\omega}$ the set of finite words in L, including the empty word. Write $s\preceq t$ when t extends s, and write sj for the word obtained by appending $j\in L$. A *j-chain* is a set $T=\{s_0,s_1,\dots\}$ with $s_rj\preceq s_{r+1}$ for every r. Let Γ_j denote all such chains.

Lemma F.3 (chain gap)**.** *Chains with different labels have at most one common word. If sets $a_j\subseteq\Lambda$, $j\in L$, contain all but finitely many words of every chain in Γ_j, then $\bigcap_{j\in L}a_j\neq\varnothing$.*

Proof. If distinct words s,t belong to both a j-chain and an h-chain, take s shorter than t. Then t extends both sj and sh, forcing $j=h$.

For each word r and label j, there is an extension $s\succeq r$ such that every extension of sj belongs to a_j. Otherwise start with $s_0=r$ and choose successively an extension $s_{m+1}\succeq s_mj$ outside a_j. This is a j-chain with infinitely many words outside a_j, a contradiction. Enumerate $L=\{j_1,\dots,j_q\}$ and start at the empty word r_0. Choose $s_m\succeq r_{m-1}$ with the asserted property for j_m, and put $r_m=s_mj_m$. The word r_q extends every s_mj_m and hence belongs to every a_j. □

The next lemma is purely algebraic; the alternating maps in its statement need no topology. Over a field κ, put $V=\kappa^{k+1}$ and $V'=\kappa^k$. Write $e_1,\dots,e_{k+1}$ and $e'_1,\dots,e'_k$ for their bases, ε^a for the coordinate forms on V, and δ' for the determinant on V'. Set

$$N=\{\varepsilon^a:1\le a\le k+1\}\cup\{\varepsilon^a+\varepsilon^b:1\le a<b\le k+1\},$$
$$\Sigma_\nu=\ker\nu,\qquad C'_s=\{z:z_s=0\},\qquad H'_{cd}=\{z:z_c=z_d\},$$

and consider the finite family of pairs

$$\mathcal{F}=\{(V,H'_{cd}):c<d\}\cup\{(\Sigma_\nu,C'_s):\nu\in N,\ 1\le s\le k\}. \tag{F.2}$$

Let $g_0\colon V \to V'$ be the coordinate projection onto the first k coordinates.

Lemma F.4 (finite fiber obstruction)**.** *Suppose $k! = 0$ in κ. There is no k-linear map*

$$\tau\colon \operatorname{Hom}_\kappa(V,V')^k \longrightarrow \operatorname{Alt}^k_\kappa(V;\kappa)$$

satisfying both

1. *$\tau(g_0,\dots,g_0) = \delta' \circ (g_0,\dots,g_0)$;*
2. *for every $(\Sigma,\Sigma') \in \mathcal{F}$, the restriction $\tau(g^1,\dots,g^k)|_{\Sigma^k}$ is zero whenever $g^r(\Sigma) \subseteq \Sigma'$ for every r.*

Proof. Let D be the determinant on V. Every alternating k-form α has a unique expression $\alpha = D(w_\alpha,\cdot,\dots,\cdot)$: its evaluations on the basis tuples omitting one vector give

$$(w_\alpha)_a = (-1)^{a-1}\alpha(e_1,\dots,\widehat{e}_a,\dots,e_{k+1}).$$

For a nonzero ν, this description also gives

$$\alpha|_{(\ker\nu)^k} = 0 \quad\Longleftrightarrow\quad \nu(w_\alpha) = 0. \tag{F.3}$$

Indeed choose ζ with $\nu(\zeta) = 1$ and a basis $y_1,\dots,y_k$ of $\ker\nu$. Writing $w_\alpha = \nu(w_\alpha)\zeta + z$ with $z \in \ker\nu$ gives $D(w_\alpha,y_1,\dots,y_k) = \nu(w_\alpha)D(\zeta,y_1,\dots,y_k)$, whose last factor is nonzero.

Suppose τ exists. For a function $a\colon \{1,\dots,k\} \to \{1,\dots,k\}$ define

$$B_a(\varphi_1,\dots,\varphi_k) = \tau(\varphi_1 \otimes e'_{a(1)},\dots,\varphi_k \otimes e'_{a(k)}), \qquad (\varphi \otimes e'_c)(x) = \varphi(x)e'_c.$$

If a omits s, every input map takes V into C'_s. Applying condition (2) to $(\ker\varepsilon^b, C'_s)$ for each b and using (F.3) makes every coordinate of $w_{B_a(\varphi)}$ zero. Hence $B_a = 0$ unless a is a permutation.

For a permutation a and distinct c,d, replace the target vector in slots $a^{-1}(c)$ and $a^{-1}(d)$ by $e'_c + e'_d$. All input maps then take V into H'_{cd}, so their image under τ is zero. Expansion in these two slots gives $B_a + B_{(c\ d)\circ a}$ and two terms with a missing target coordinate. The latter vanish by the preceding paragraph. Thus, with $B = B_{\mathrm{id}}$,

$$B_a = \operatorname{sgn}(a)B \qquad (a \in S_k). \tag{F.4}$$

Define the $(k+1)$-linear form

$$\beta(\varphi_1,\dots,\varphi_k;\chi) = \chi\big(w_{B(\varphi_1,\dots,\varphi_k)}\big).$$

If the sth argument is $\nu \in N$, its rank-one input vanishes on Σ_ν, and every other rank-one input takes values in C'_s. Conditions (2) and (F.3) give $\beta(\varphi_1,\dots,\nu,\dots,\varphi_k;\nu) = 0$. Use this identity successively for ε^a, ε^b and $\varepsilon^a + \varepsilon^b$ to obtain

$$\beta(\dots,\varepsilon^a,\dots;\varepsilon^b) = -\beta(\dots,\varepsilon^b,\dots;\varepsilon^a).$$

Consequently $t(\sigma) = \beta(\varepsilon^{\sigma(1)},\dots,\varepsilon^{\sigma(k)};\varepsilon^{\sigma(k+1)})$ changes sign under every transposition of a slot with the last slot. These transpositions generate S_{k+1}, so $t(\sigma) = \operatorname{sgn}(\sigma)t(\mathrm{id})$.

Expand $g_0 = \sum_{c=1}^k \varepsilon^c \otimes e'_c$ in condition (1). The missing-coordinate terms vanish, and (F.4) gives

$$\varepsilon^1 \wedge \cdots \wedge \varepsilon^k = \sum_{a\in S_k} \operatorname{sgn}(a)B(\varepsilon^{a(1)},\dots,\varepsilon^{a(k)}).$$

Apply $\alpha \mapsto \varepsilon^{k+1}(w_\alpha)$. Its value on the left is $(-1)^k$; on the right it is $\sum_{a\in S_k}\operatorname{sgn}(a)^2 t(\mathrm{id}) = k!t(\mathrm{id}) = 0$. This is a contradiction. All identities remain valid in characteristic 2: repeated arguments vanish by alternation, and no step divides by 2 or by a factorial. □

We will need a quantitative consequence of this finite incompatibility. Give K^{k+1} and K^k their maximum norms and use the same family (F.2) over K.

Lemma F.5 (finite test certificate)**.** *Assume $k! = 0$ in K, as in Theorem F.1. There are finite sets of tests $\mathcal{T}_j$, one for each $j = (\Sigma^j, \Sigma'^j) \in \mathcal{F}$, consisting of tuples $(g^1, \dots, g^k; \xi_1, \dots, \xi_k)$ with $g^r \Sigma^j \subseteq \Sigma'^j$ and $\xi_r \in \Sigma^j$, such that every K-multilinear τ satisfying condition (1) of Lemma F.4 obeys*

$$\max_{j \in \mathcal{F}} \max_{(g;\xi) \in \mathcal{T}_j} |\tau(g^1, \dots, g^k)(\xi_1, \dots, \xi_k)| \geq 1. \tag{F.5}$$

Proof. Write $\operatorname{char} K = p$. All spaces in (F.2) are defined over $\mathbb{F}_p$. For each j, choose an $\mathbb{F}_p$-basis of the subspace Σ^j defined over $\mathbb{F}_p$, and one of

$$\{g \in \operatorname{Hom}_{\mathbb{F}_p}(\mathbb{F}_p^{k+1}, \mathbb{F}_p^k) : g\Sigma^j \subseteq \Sigma'^j\}.$$

Their scalar extensions are bases of the corresponding K-spaces: row reduction of the defining prime-field matrices has the same pivot positions over K. Take all k-tuples from each basis to form $\mathcal{T}_j$. Multilinearity in the maps and the vectors makes these tests equivalent to condition (2).

Choose coordinates θ for the finite-dimensional space of possible τ. Condition (1) is a system $A\theta = b$, and the tests for condition (2) form a system $C\theta = 0$, with all matrices and constants over $\mathbb{F}_p$. Lemma F.4 over $\mathbb{F}_p$ makes the combined system inconsistent. Row reduction of its augmented matrix therefore gives row vectors λ, μ over $\mathbb{F}_p$ with

$$\lambda A + \mu C = 0, \qquad \lambda b = 1.$$

For every K-valued solution of $A\theta = b$ it follows that $\mu C\theta = -1$. Every nonzero coefficient of μ has absolute value 1, since $\mathbb{F}_p^\times$ is finite. The nonarchimedean inequality now gives (F.5). □

F.3. **The Banach spaces defined by chain limits.** Set $L = \mathcal{F}$ and $\Lambda = L^{<\omega}$. Write $\Sigma^j \subseteq V = K^{k+1}$ and $\Sigma'^j \subseteq V' = K^k$ for the pair indexed by $j \in L$. Define

$$\begin{aligned} E &= \{x \in \ell^\infty(\Lambda; V) : \operatorname{dist}(x_i, \Sigma^j) \longrightarrow 0 \text{ along every } T \in \Gamma_j,\ j \in L\}, \\ E' &= \{z \in \ell^\infty(\Lambda; V') : \operatorname{dist}(z_i, \Sigma'^j) \longrightarrow 0 \text{ along every } T \in \Gamma_j,\ j \in L\}. \end{aligned} \tag{F.6}$$

Limits along a chain are taken outside its finite subsets. We verify the properties of these full solution spaces that will enter the proof.

Each subspace in the definition admits a projection R with $\|R\| \leq 1$ and $\|1 - R\| \leq 1$. For $\ker \varepsilon^a$, delete coordinate a; for $\ker(\varepsilon^a + \varepsilon^b)$, replace coordinate a by the negative of coordinate b; for V, use the identity. On V', delete coordinate s for C'_s, and replace coordinate c by coordinate d for H'_{cd}. These formulas are projections onto the stated spaces. Their norm bounds follow directly from the maximum norm and the nonarchimedean inequality. Moreover

$$\operatorname{dist}(v, \Sigma) = \|(1 - R)v\|. \tag{F.7}$$

Indeed $Rv \in \Sigma$ gives one inequality, and $(1 - R)(v - y) = (1 - R)v$ for every $y \in \Sigma$ gives the other.

For fixed j and $T \in \Gamma_j$, the corresponding condition in (F.6) is thus the inverse image of $c_0(T; V)$ under the bounded linear map $x \mapsto ((1 - R_j)x_i)_{i \in T}$, and similarly for V'. The space c_0 is closed in the sup norm: a uniform approximation within $\varepsilon/2$ by a null family leaves only finitely many coordinates of norm at least ε. Each condition is therefore closed and linear. Their intersection is a closed subspace of the Banach space $\ell^\infty(\Lambda; V)$, respectively $\ell^\infty(\Lambda; V')$, so E and E' are nonarchimedean Banach spaces.

Both contain their corresponding c_0 spaces. They are also modules over $\ell^\infty(\Lambda, K)$ under coordinatewise multiplication, with $\|tx\| \leq \|t\|\|x\|$: each residual family in (F.7) is multiplied by the same bounded scalar family and remains null. Finally, if $T \in \Gamma_j$, every bounded Σ^j-valued family supported on T belongs to E. For chains with label j its residual is zero; for a chain

with another label, its support meets that chain in at most one word by Lemma F.3. The identical assertion holds for E' and Σ'^{j}.

Since $k \geq 2$, the label $j = (V, H'_{12})$ occurs and both its spaces contain a vector of norm 1. Fix a j-chain T. Multiplying either such vector by arbitrary bounded scalar families on T gives an isometric embedding of $\ell^\infty(\mathbb{N}, K)$ into E, respectively E'. The binary-valued sequences form an uncountable set at pairwise distance 1, so both spaces are nonseparable.

F.4. **The global contradiction.**

Proof of Theorem F.1. Use the spaces (F.6). Suppose that $A = A^k_{E,E';K}$ has a bounded k-linear lift P. For $i \in \Lambda$, let $\xi_{[i]} \in E$ denote the family equal to $\xi \in V$ at i and zero elsewhere. Single-coordinate families belong to $c_0 \subseteq E$, and likewise in E'. For $g \in \mathrm{Hom}_K(V, V')$ let $g_{[i]} \in \mathcal{L}(E, E')$ send x to the family equal to $g(x_i)$ at i and zero elsewhere. Its norm is at most $\|g\|$. Set

$$\delta'_i(z_1, \dots, z_k) = \delta'(z_{1,i}, \dots, z_{k,i}) \in \mathrm{Alt}^k(E'; K).$$

The determinant estimate for the maximum norm gives $\|\delta'_i\| \leq 1$. Define the fiber map

$$\pi_i(g^1, \dots, g^k)(\xi_1, \dots, \xi_k) = P(g^1_{[i]}, \dots, g^k_{[i]})(\delta'_i)(\xi_{1[i]}, \dots, \xi_{k[i]}). \tag{F.8}$$

It is k-linear in the maps and alternating in the vectors, and the lift identity gives

$$\pi_i(g, \dots, g) = \delta' \circ (g, \dots, g) \qquad \text{for every } g. \tag{F.9}$$

In particular every π_i satisfies condition (1) of Lemma F.4.

Fix $j = (\Sigma, \Sigma') \in L$, a chain $T \in \Gamma_j$, maps $g^1, \dots, g^k$ with $g^r\Sigma \subseteq \Sigma'$, and vectors $\xi_1, \dots, \xi_k \in \Sigma$. We claim that

$$\pi_i(g^1, \dots, g^k)(\xi_1, \dots, \xi_k) \longrightarrow 0 \qquad (i \in T). \tag{F.10}$$

For $t \in \ell^\infty(\Lambda, K)$ define u^r_t by

$$(u^r_t x)_i = \begin{cases} t_i g^r(x_i), & i \in T, \\ 0, & i \notin T. \end{cases}$$

This is a bounded map from E to E'. To see that its values lie in E', let R be the projection onto Σ above and decompose, on T,

$$t_i g^r(x_i) = t_i g^r(Rx_i) + t_i g^r((1 - R)x_i).$$

The first term is a bounded Σ'-valued family supported on T, hence belongs to E'. The second is null on T by (F.7), hence its zero extension belongs to $c_0 \subseteq E'$. The coordinate formula gives $\|u^r_t\| \leq \|t\|\|g^r\|$, so $t \mapsto u^r_t$ is bounded linear with values in $\mathcal{L}(E, E')$.

For $c \in \ell^\infty(\Lambda, K)$ put

$$m_c(z_1, \dots, z_k) = \sum_{i \in T} c_i \delta'(z_{1,i}, \dots, z_{k,i}). \tag{F.11}$$

This sum converges. In fact, if R' projects onto Σ', write $z_{r,i} = a_{r,i} + b_{r,i}$ with $a_{r,i} = R'z_{r,i}$ and $b_{r,i} = (1 - R')z_{r,i}$. Both families are bounded by $\|z_r\|$, and $b_{r,i} \to 0$ along T. Expanding the determinant, the term with only a's is zero because $\dim \Sigma' = k - 1$. Every remaining term contains an error factor, giving

$$|\delta'(z_{1,i}, \dots, z_{k,i})| \leq \max_{1 \leq r \leq k} \left(\|b_{r,i}\| \prod_{q \neq r} \|z_q\| \right) \longrightarrow 0.$$

Multiplication by bounded c preserves this limit. Completeness of K now gives the unconditional scalar sum in (F.11). Linearity in each argument and in c follows by taking limits of finite partial sums; repeated arguments make every summand zero. Moreover the determinant estimate gives

$$|m_c(z_1,\dots,z_k)| \le \|c\| \prod_{r=1}^{k} \|z_r\|.$$

Thus $c \mapsto m_c$ is a bounded linear map into $\mathrm{Alt}^k(E';K)$, of norm at most 1.

Let x^r be the family equal to ξ_r on T and zero elsewhere; it belongs to E by the chain-generator inclusion already proved. Coordinatewise multiplication defines a bounded linear map $\rho \mapsto \rho x^r$ from $\ell^\infty(\Lambda,K)$ to E. Consequently

$$\Phi(t^1,\dots,t^k,c,r^1,\dots,r^k) = P(u^1_{t^1},\dots,u^k_{t^k})(m_c)(r^1x^1,\dots,r^kx^k)$$

is a bounded $(2k+1)$-linear scalar form on $\ell^\infty(\Lambda,K)$, with bound

$$\|P\| \prod_{s=1}^{k} \big(\|g^s\|\|\xi_s\|\big) \prod_{s=1}^{k} \|t^s\| \, \|c\| \prod_{s=1}^{k} \|r^s\|.$$

The set Λ is countably infinite, so Lemma F.2 gives $\Phi(e_i,\dots,e_i) \to 0$ outside finite sets. For $i \in T$ we have $u^s_{e_i} = g^s_{[i]}$, $m_{e_i} = \delta'_i$ and $e_i x^s = \xi_{s[i]}$. Equation (F.8) therefore identifies this diagonal value with the expression in (F.10), proving the claim.

Choose the finite tests of Lemma F.5, and define

$$G_j(i) = \max_{(g;\xi)\in\mathcal{T}_j} |\pi_i(g^1,\dots,g^k)(\xi_1,\dots,\xi_k)|, \qquad a_j = \{i \in \Lambda : G_j(i) < 1\}.$$

For fixed j and $T \in \Gamma_j$, equation (F.10) applies to every test in the finite set $\mathcal{T}_j$. Hence $G_j(i) \to 0$ along T, and $T \setminus a_j$ is finite. Lemma F.3 supplies an i belonging to every a_j. But (F.9) and Lemma F.5 force $G_j(i) \ge 1$ for some j, a contradiction.

There is therefore no bounded k-linear lift of A. Proposition 3.3 applies in this degree without dividing by $k!$: analyticity at any point would make the degree-k Taylor coefficient such a lift. Thus A is analytic at no point, as claimed. □

Appendix G. Universal analytic reflection and projection norms

The parameter theorems for alternating bundles exploit the fixed degree of the outer operator polynomial. Reflection of every Banach-valued analytic map through every closed linear inclusion is a stronger property. We characterize that property and give an explicit example separating the two.

Throughout this appendix, K is complete and nontrivially normed, and P is Banach. For $n \ge 1$, let $T_n(P)$ be the separated completed projective tensor power representing bounded n-linear maps from P^n to Banach spaces. We use the ordinary projective seminorm, defined by infima of sums of products of norms, and take its separated completion before invoking the universal property. Its canonical multilinear map has norm at most one. Define the closed subspace

$$\Delta_n(P) = \overline{\mathrm{span}\{x^{\otimes n} : x \in P\}} \subseteq T_n(P).$$

A projection onto $\Delta_n(P)$ will mean a bounded linear map $R_n\colon T_n(P) \to \Delta_n(P)$ whose restriction to $\Delta_n(P)$ is the identity.

Proposition G.1 (homogeneous reflection)**.** *For a fixed $n \ge 1$, the following conditions are equivalent.*

(1) *For every Banach space Z, closed subspace $W \subseteq Z$, and bounded n-linear map $B\colon P^n \to Z$ with diagonal values in W, there is a bounded W-valued n-linear map with the same diagonal.*
(2) *The subspace $\Delta_n(P)$ is the range of a bounded projection in $T_n(P)$.*

Proof. Suppose first that a projection R_n exists. Linearize B to a bounded map $\widetilde{B}\colon T_n(P) \to Z$. The diagonal hypothesis and closedness of W imply $\widetilde{B}(\Delta_n(P)) \subseteq W$. Composing $\widetilde{B}|_{\Delta_n(P)}$ with R_n and the canonical tensor map gives the required multilinear representative, of norm at most $\|B\|\|R_n\|$. Conversely, take $Z = T_n(P)$, $W = \Delta_n(P)$, and B equal to the canonical tensor map. Linearizing the representative supplied by the first condition gives a bounded map into $\Delta_n(P)$ fixing every pure power. It fixes their linear span and, by continuity, their closed span, so it is a projection. □

Definition G.2. The Banach parameter space P has *universal analytic reflection* if, for every Banach space Z, closed subspace $W \subseteq Z$, every open $U \subseteq P$, and every map $f\colon U \to W$, analyticity of the composite $U \to W \hookrightarrow Z$ implies analyticity of f.

Theorem G.3 (analytic tensor criterion)**.** *A Banach space P has universal analytic reflection if and only if there are projections $R_n\colon T_n(P) \to \Delta_n(P)$ for all $n \geq 1$ such that*

$$\limsup_{n\to\infty} \|R_n\|^{1/n} < \infty. \tag{G.1}$$

Proof. Assume first that the projections exist. Expand an ambient analytic map with values in W at a point. The scalar-line uniqueness argument in Theorem 3.1 puts every homogeneous diagonal in W. The construction in Proposition G.1 replaces its coefficient p_n by a W-valued coefficient of norm at most $\|p_n\|\|R_n\|$ and with the same diagonal. The exponential bound preserves a positive radius of convergence. The degree-zero coefficient is already in W, and the resulting expansion represents the given map.

Conversely, form the Banach spaces with supremum norms

$$Z = c_0\big(\{T_n(P)\}_{n\geq 1}\big), \qquad W = c_0\big(\{\Delta_n(P)\}_{n\geq 1}\big) \subseteq Z.$$

Here c_0 denotes families tending to zero in their respective Banach coordinate spaces; W is closed. On the open unit ball of P, define

$$f(x) = (x^{\otimes n})_{n\geq 1}.$$

Since $\|x^{\otimes n}\| \leq \|x\|^n$, this vector lies in W. As a Z-valued map it is analytic: its degree-n coefficient is the canonical tensor map placed in coordinate n, and has norm at most one. This gives an expansion at zero. To re-expand at a with $\|a\| < 1$, choose $r > 0$ with $\|a\| + r < 1$. Expanding each tensor power in $a + h$ gives translated coefficients, and the sum of all term bounds is at most $\sum_{n\geq 1}(\|a\| + r)^n < \infty$. Completeness of Z supplies the coefficients and the rearranged sum, proving analyticity throughout the unit ball. Universal reflection gives a W-valued expansion $\sum_m q_m(x, \dots, x)$ near zero with $\|q_m\| \leq C\rho^{-m}$ for some $C, \rho > 0$. Compose this expansion with the projection $\pi_n\colon W \to \Delta_n(P)$ and compare on scalar lines with $x \mapsto x^{\otimes n}$. Uniqueness of scalar power-series coefficients gives

$$\pi_n q_n(x, \dots, x) = x^{\otimes n} \qquad (x \in P).$$

Linearizing $\pi_n q_n$ therefore gives a projection R_n onto $\Delta_n(P)$ with $\|R_n\| \leq \|q_n\| \leq C\rho^{-n}$, proving (G.1). □

G.1. **Projection bounds and the special role of fixed degree.** For $P = K^d$ with the maximum norm, the coordinate construction in Theorem 4.4, applied to the canonical tensor map, gives $\|R_n\| \le d^n$ for $d \ge 1$. A bounded linear change of coordinates introduces only a fixed distortion factor to the n-th power, so it does not change the exponential-growth condition.

For the ordinary sum-norm space $P = \ell^1(I, K)$, the projective tensor power is isometrically $\ell^1(I^n, K)$. Indeed, the coordinate tensor map has norm at most one, while expansion in coordinate tensors gives the reverse projective-norm estimate; completion then gives the identification. In these coordinates, $\Delta_n(P)$ consists exactly of arrays constant on each permutation orbit in I^n. One inclusion follows because every pure power is constant on such orbits. For the other, finite-coordinate polynomial interpolation puts each finite orbit sum in the span of pure powers, and finite sums of orbit sums are dense in the space of orbit-constant ℓ^1 arrays.

Choose one representative of each orbit, send its basis vector to the sum of all basis vectors in that orbit, and send the other basis vectors in the orbit to zero. This gives a projection with norm at most $n!$. Over a nonarchimedean field and for infinite I, this estimate is optimal even among projections that do not preserve the individual orbit blocks.

Proposition G.4. *If K is nonarchimedean and complete, and I is infinite, then*

$$\inf\{\|R\| : R\colon T_n(\ell^1(I,K)) \to \Delta_n(\ell^1(I,K)) \text{ is a bounded projection}\} = n!.$$

Consequently $\ell^1(I, K)$ does not have universal analytic reflection.

Proof. The preceding construction gives the upper bound. For the lower bound, take an orbit $\mathcal{O}$ of a tuple with n distinct indices; it has $n!$ elements. For each $w \in \mathcal{O}$, let λ_w be the common coefficient on $\mathcal{O}$ of $R(e_w)$. The orbit sum $\sum_{w\in\mathcal{O}} e_w$ belongs to $\Delta_n(P)$ and is fixed by R, hence

$$\sum_{w\in\mathcal{O}} \lambda_w = 1.$$

The nonarchimedean inequality implies that some $|\lambda_w| \ge 1$. For this w, the ordinary sum norm satisfies $\|R(e_w)\|_1 \ge n!|\lambda_w| \ge n!$. Thus $\|R\| \ge n!$. Finally, $(n!)^{1/n}$ is unbounded, so Theorem G.3 excludes universal analytic reflection. □

Corollary G.5 (entire polynomial approximation of a nonanalytic smooth map)**.** *Let K be a complete nonarchimedean nontrivially normed field and put $P = \ell^1(\mathbb{N}, K)$, with the ordinary sum norm. There are Banach spaces $W \subseteq Z$, with W closed, and a map f on the open unit ball of P such that f is analytic into Z and C^∞ into W, but is not analytic into W at zero. Moreover, f is a uniform limit on every smaller ball of entire W-valued polynomials.*

Proof. Take

$$Z = c_0\big(\{T_n(P)\}_{n\ge1}\big), \qquad W = c_0\big(\{\Delta_n(P)\}_{n\ge1}\big), \qquad f(x) = (x^{\otimes n})_{n\ge1}.$$

The spaces are Banach and W is closed in Z. As in the proof of Theorem G.3, the ambient expansion has coefficient norms at most one and defines an analytic map on the open unit ball. Smooth reflection through the closed isometric inclusion makes f C^∞ as a map into W.

Suppose a W-valued expansion at zero has coefficients q_n. Uniqueness on scalar lines gives

$$\pi_n q_n(x, \dots, x) = x^{\otimes n},$$

where $\pi_n\colon W \to \Delta_n(P)$ is the coordinate projection. Linearizing $\pi_n q_n$ gives a projection $T_n(P) \to \Delta_n(P)$ of norm at most $\|q_n\|$. Proposition G.4 therefore gives $\|q_n\| \ge n!$. Since $(n!)^{1/n}$ is unbounded, these coefficients give a formal multilinear series of radius zero, contradicting the assumed analytic expansion.

For each n, choose the sorted-orbit projection $R_n\colon T_n(P) \to \Delta_n(P)$. Its composition with the canonical tensor map is a bounded multilinear representative of $x \mapsto x^{\otimes n}$ with values in $\Delta_n(P)$. Consequently the truncation

$$f_N(x) = (x, x^{\otimes 2}, \dots, x^{\otimes N}, 0, \dots)$$

is an entire W-valued polynomial. For $0 < r < 1$ and $\|x\| \le r$,

$$\|f(x) - f_N(x)\|_W \le r^{N+1},$$

which proves the asserted uniform convergence. □

This example is characteristic-free within the nonarchimedean setting. Every individual homogeneous term has an analytic lift, but their best multilinear bounds grow factorially. The fixed-degree argument for alternating families avoids this obstruction by using one outer degree throughout.

There is no conflict with Theorem 4.5(2): every alternating-map family on these parameter spaces is still analytic as an operator family when its final output fiber is complete. The ambient alternating action has a fixed polynomial degree. Factoring a general analytic parameter map through a free ℓ^1 space therefore incurs a single factorial bound, rather than a sequence of factorial bounds across Taylor degrees. Over a complete nonarchimedean field, the alternating-family property is consequently strictly weaker than universal analytic reflection. Over $\mathbb{R}$ and $\mathbb{C}$ every Banach space has universal analytic reflection, by Theorem G.3: the symmetrization $\frac{1}{n!}\sum_{\sigma\in S_n}\sigma$ has norm at most one on $T_n(P)$ and, by polarization, is a projection onto $\Delta_n(P)$.

For comparison, in the category of nonarchimedean normed spaces with the *nonarchimedean* projective tensor norm, the corresponding tensor power of $c_0(I, K)$ is $c_0(I^n, K)$ and the same sorted-orbit projection has norm at most one. This uses the infimum of maxima in the tensor seminorm, rather than the ordinary projective seminorm used in Theorem G.3. It is the tensor counterpart of the norm-preserving coefficient construction in Theorem 4.5(1); the two tensor norms must not be interchanged.

Theorem G.3 thus gives a complete answer for universal Banach-valued analytic reflection. It supplies a sufficient parameter criterion for alternating families. Indeed, if the final fiber is complete, the ambient multilinear operator space is Banach, and universal reflection through the closed inclusion of alternating operators makes every alternating-map family analytic as an operator family. Over a complete nonarchimedean field, Proposition G.4 shows that the criterion is not a necessary one. An intrinsic characterization of the parameter spaces for which every alternating-map family is analytic as an operator family remains a separate question.

Appendix H. Why a largest full analytic domain need not exist

The positive bundle theorems restrict either the fibers or the parameter spaces. Another possible restriction keeps arbitrary analytic bases but chooses a full subcategory of fiber pairs on whose hom spaces the alternating construction is analytic. There need not be a largest such subcategory: endomorphism actions at two objects can each be analytic while their cross-action is not. As in Proposition 3.3, a bounded d-linear *lift* of a homogeneous map f means a bounded multilinear map B satisfying $B(x, \dots, x) = f(x)$.

Definition H.1. A full subcategory $\mathcal{D} \subseteq \mathsf{Vec}_K^{\mathrm{op}} \times \mathsf{Vec}_K$ is an *analytic domain* for Alt^k if the restriction of Alt^k to $\mathcal{D}$ is analytic on every hom space.

Definition H.2. A pair (E, F) is *split in degree k* if the isometric inclusion

$$j_{E,F}\colon \operatorname{Alt}^k(E;F) \longrightarrow \operatorname{Mult}^k(E;F)$$

has a bounded linear retraction.

Lemma H.3. *Every morphism-space action whose destination is a split pair is analytic. In particular, the full subcategory of split pairs is an analytic domain.*

Proof. Let (E, F) be split, with retraction R, and let the other object be (D, G). For $u\colon E \to D$ and $v\colon G \to F$, the action is

$$m \longmapsto R\big(v \circ m \circ (u, \dots, u)\big).$$

Before applying R, replace the k copies of u by independent variables. This gives a bounded $(k+1)$-linear lift of the joint action, with values in $\mathcal{L}(\operatorname{Alt}^k(D;G), \operatorname{Mult}^k(E;F))$. Postcomposition with R gives a lift with the required alternating codomain. No compatibility between the retractions chosen for different objects is needed. □

Theorem H.4 (no largest full analytic domain)**.** *Let p be prime, let $k \geq p$, and equip $K = \mathbb{F}_p(t)$ with its t-adic absolute value. There is no largest full analytic domain for*

$$\operatorname{Alt}^k\colon \mathsf{Vec}_K^{\mathrm{op}} \times \mathsf{Vec}_K \longrightarrow \mathsf{Vec}_K.$$

The assertion remains true if the domains must be closed under isomorphisms, and if their objects are restricted to pairs of finite-dimensional, nonarchimedean normed K-spaces. In the latter case, isomorphism closure is understood inside that restricted ambient category.

More precisely, there are objects $X_k = (E_k, G)$ and $Y_k = (D_k, G)$ whose singleton full subcategories are analytic, such that Y_k is split and the action on morphisms from Y_k to X_k is not analytic.

Here finite-dimensional means finite *algebraic* dimension over the incomplete field K. These spaces need not admit continuous coordinates over K. The theorem does not give a counterexample over a complete field, or a counterexample with Banach objects.

We give the entire construction. Put $L = \mathbb{F}_p((t))$, the completion of K. We first work in degree p, then extend to every $k \geq p$ by adjoining coordinate summands.

H.1. **The rigid source and the split pair.** We use the following elementary completion principle. Let K be dense in a complete valued extension L, let (E_i) be a finite family of dense K-subspaces of Banach L-spaces H_i, and let Z be Banach over L. A bounded K-multilinear map $\prod_i E_i \to Z$ extends uniquely to a bounded L-multilinear map $\prod_i H_i \to Z$, with the same norm. Indeed, approximate each input by vectors in E_i. A telescoping expansion and the multilinear bound make the resulting values Cauchy, independently of the approximations. The bound persists in the limit, and density of K in L gives L-linearity in each slot. In the applications below the extended codomain is the complete space L or A, while membership of values on the original inputs in the smaller spaces is retained.

Put $L = \mathbb{F}_p((t))$, the completion of K, and consider the normed L-algebra

$$A = L[\epsilon]/(\epsilon^p), \qquad e_i = \epsilon^i \quad (0 \leq i < p),$$

with the maximum norm in this basis. Multiplication satisfies $\|xy\| \leq \|x\|\,\|y\|$. Choose elements $a_0, \dots, a_{p-1} \in L$, together with the finitely many auxiliary scalars τ_w introduced below, algebraically independent over K. Such a choice is possible because K is countable and

L is uncountable: the elements of L algebraic over any finitely generated extension of K form a countable set. Set

$$a = \sum_{i=0}^{p-1} a_i e_i, \qquad E = K^p + Ka \subseteq A.$$

Every subspace used below carries the norm induced by A or L. In particular, E has K-dimension $p+1$ and completion A.

Lemma H.5. *Every bounded endomorphism of E is multiplication by a scalar in K, and every bounded linear map $E \to K$ is zero.*

Proof. A bounded K-linear map extends to completions, and the extension is L-linear by density of K in L. If it preserves E, its columns, the images of the e_i, give a matrix $M + ac^{\mathsf{T}}$, with $M \in M_p(K)$ and $c \in K^p$. Its value at a lies in $K^p + Ka$. Comparison of homogeneous degrees in the algebraically independent a_i shows first that $a(c^{\mathsf{T}}a) = 0$, hence $c = 0$, and then that $M = sI$ for some $s \in K$.

Similarly, the extension of a bounded map $E \to K$ is an L-linear functional on A. Its coefficients belong to K, since its values on the e_i do. Its value on a also lies in K, so algebraic independence makes all its coefficients zero. □

Multiplication by a has lower triangular Toeplitz matrix

$$(M_a)_{ij} = \begin{cases} a_{i-j}, & i \geq j, \\ 0, & i < j. \end{cases}$$

Define

$$D_0 = \operatorname{span}_K\{e_i, \epsilon^i a : 0 \leq i < p\} + Ka^2, \qquad \mathcal{W} = \{e_i : 0 \leq i < p\} \cup \{e_i + e_j : 0 \leq i < j < p\},$$

and put

$$D = D_0 + \sum_{w \in \mathcal{W}} K\tau_w w, \qquad G = \operatorname{span}_K\{\det(d_1, \ldots, d_p) : d_i \in D\} \subseteq L.$$

The determinant is taken in the basis $(e_0, \ldots, e_{p-1})$. These are finite-dimensional K-spaces, with completions A and L: D contains K^p, and G contains 1. By multilinearity, G is spanned by determinants of the displayed finite generating set for D. We may therefore identify G with a finite-dimensional subspace of $K[a_0, \ldots, a_{p-1}, (\tau_w)_w]$. Every element has degree at most one in each individual τ_w, since a nonzero determinant cannot repeat the generator $\tau_w w$. Algebraic independence gives

$$\tau_w^2 G \cap G = \{0\} \qquad (w \in \mathcal{W}). \tag{H.1}$$

Lemma H.6. *Every bounded p-linear map $D^p \to G$ is alternating. Thus (D, G) is split in degree p.*

Proof. Compose such a map m with $G \hookrightarrow L$ and extend to a bounded L-multilinear map on A^p. Fix two argument positions. Put the same $w \in \mathcal{W}$ in those positions and arbitrary standard basis vectors in the others, and call the value $c \in G$. Replacing both copies of w by $\tau_w w \in D$ shows that $\tau_w^2 c \in G$, so (H.1) gives $c = 0$. Taking $w = e_i$ gives vanishing on repeated basis vectors. Taking $w = e_i + e_j$ then gives the signed interchange identity in the two positions. Expanding two equal arbitrary arguments in the basis proves that the completed map, and hence m, is alternating, including when $p = 2$. The inclusion of alternating maps into multilinear maps is therefore an isometric bijection, whose inverse is the desired retraction. □

H.2. **A missing coefficient.** The coefficient space of alternating maps on E is

$$C = \{\lambda \in G : a_i\lambda \in G \text{ for every } i\}.$$

Indeed, an alternating map extends to $\lambda \det$ on the p-dimensional completion A. Determinants of tuples from E span $\operatorname{span}_K\{1, a_0, \dots, a_{p-1}\}$: a determinant can contain at most one copy of a, and replacing one basis vector by a realizes each a_i up to sign. Consequently

$$\operatorname{Alt}^p(E; G) = C \det . \tag{H.2}$$

This identification is isometric, since the determinant bound gives one inequality and evaluation on the standard basis gives the reverse inequality.

Algebraically set every τ_w to zero. The image of G is

$$G_0 = \operatorname{span}_K\{\det(d_1, \dots, d_p) : d_i \in D_0\}.$$

Put $C_0 = \{\lambda \in G_0 : a_i\lambda \in G_0 \text{ for every } i\}$. Specialization sends C into C_0; no continuity of specialization is asserted. The displayed generators of D_0 are homogeneous polynomial vectors of degrees zero, one, and two. Their determinants are homogeneous, so G_0 is graded, and the defining conditions show that C_0 is graded as well.

Lemma H.7. *The homogeneous degree-one part of C_0 is zero.*

Proof. The degree-two part of G_0 is spanned by the 2×2 minors of M_a and the coordinates of a^2. These correspond respectively to choosing two degree-one columns, or one degree-two column, in a determinant of generators. On homogeneous quadratic polynomials define the K-linear map $\Phi\colon K[a_0, \dots, a_{p-1}]_2 \to K[T]$ by

$$\Phi(a_ia_j) = \begin{cases} T^{i+j}, & i + j \geq p - 1, \\ 0, & i + j < p - 1. \end{cases}$$

Every 2×2 minor of M_a is killed by Φ. When both product terms are nonzero, their index sums coincide. If only one is nonzero, write the rows as $r < s$ and columns as $i < j$. The surviving term has index sum $r + s - i - j$, and the missing term implies $r < j$. Hence $r + s - i - j \leq s - i - 1 \leq p - 2$, so its image is again zero.

The coefficient of ϵ^r in a^2 is $\sum_{i+j=r} a_ia_j$. Its image is zero for $r < p - 1$; for $r = p - 1$ there are exactly p summands and its image is $pT^{p-1} = 0$. Thus Φ kills the entire degree-two part of G_0. If $\ell = \sum_j c_ja_j$ lies in the degree-one part of C_0, then $a_{p-1}\ell \in G_0$, whereas

$$\Phi(a_{p-1}\ell) = \sum_j c_jT^{p-1+j}.$$

The powers are distinct, so all c_j vanish. □

H.3. **The nonanalytic cross-action.** Set $X = (E, G)$ and $Y = (D, G)$. Lemma H.5 identifies $\mathcal{L}(E, E)$ isometrically with K, and the self-action of X is

$$(s, v) \longmapsto s^pv_*.$$

This is a bounded polynomial, so the singleton full subcategory on X is analytic. The singleton on Y is analytic by Lemmas H.6 and H.3.

For $x \in E$, multiplication by x in A restricts to a bounded map $M_x\colon E \to D$. Indeed, writing $x = b + ha$ with $b \in K^p$ and $h \in K$, its values on the generators e_j, a lie in

$$K^p + \sum_i K\epsilon^i a + Ka^2 \subseteq D.$$

Submultiplicativity gives a bounded linear family $x \mapsto M_x$ from E to $\mathcal{L}(E, D)$. Its matrix is lower triangular with diagonal entry x_0, the e_0-coordinate of x in A, so $\det M_x = x_0^p$. In the cross-action from Y to X, fix the target morphism to id_G, evaluate at $\det \in \mathrm{Alt}^p(D; G)$, and use (H.2). The resulting map is

$$f\colon E \longrightarrow C, \qquad f(x) = x_0^p. \tag{H.3}$$

Its values lie in C because it was obtained by pulling back an alternating G-valued map.

Suppose that f were analytic at zero. Since $f(sx) = s^p f(x)$, uniqueness of power-series coefficients on scalar lines gives a bounded p-linear map $B\colon E^p \to C$ with diagonal f. This is the homogeneous coefficient argument used in Proposition 3.3; it requires no factorial division or completeness. Compose B with $C \hookrightarrow L$ and extend to an L-multilinear map on A^p. Put

$$b_i = B(e_0, \dots, e_0, e_i) \in C.$$

There are $p-1$ initial copies of e_0, and $b_0 = f(e_0) = 1$. Evaluation at a in the last position gives $\sum_i a_i b_i \in C$. Specializing all τ_w to zero yields elements $d_i \in C_0$ with

$$d_0 = 1, \qquad \sum_i a_i d_i \in C_0.$$

The degree-one part of this last expression is $\sum_i a_i (d_i)_0$, where $(d_i)_0 \in K$ is the constant term. Its a_0-coefficient equals one. This contradicts Lemma H.7. Thus (H.3), and therefore the cross-action from Y to X, is not analytic.

H.4. **Higher degrees and the categorical conclusion.** For $k = p + r$, let

$$E_k = E \oplus K^r, \qquad D_k = D \oplus K^r$$

with maximum norms. Their completions have dimension k over L. Determinants of tuples from E_k still span $\mathrm{span}_K\{1, a_0, \dots, a_{p-1}\}$, so extraction of the top-degree coefficient identifies

$$\mathrm{Alt}^k(E_k; G) = C \det_k \tag{H.4}$$

isometrically. In particular, the coefficient takes values in C, a fact needed to retain the preceding obstruction after increasing the degree.

By Lemma H.5, the extension of every bounded endomorphism of E_k to its L-completion $A \oplus L^r$ has block form

$$\begin{pmatrix} sI_p & B \\ 0 & T \end{pmatrix}, \qquad s \in K, \quad T \in M_r(K).$$

The scalar s and entries of T are bounded linear functions of the endomorphism, as is seen by evaluation on the standard basis. Its determinant is $s^p \det T$, so its action on (H.4) is multiplication by this bounded polynomial. Including postcomposition shows that the singleton full subcategory on $X_k = (E_k, G)$ is analytic.

The pair $Y_k = (D_k, G)$ is split. Given $m \in \mathrm{Mult}^k(D_k; G)$, restrict the first p arguments to D and fix the last r arguments at the standard auxiliary basis $e'_1, \dots, e'_r$ of K^r. The resulting p-linear map α_m is alternating by Lemma H.6. Write $\pi_D\colon D_k \to D$ and $\pi_r\colon D_k \to K^r$ for the two projections. Explicitly, set

$$(\mathcal{R}m)(v_1, \dots, v_k) = \sum_{\substack{S \subseteq \{1,\dots,k\} \\ |S| = p}} \varepsilon(S)\, \alpha_m(\pi_D v_{s_1}, \dots, \pi_D v_{s_p}) \det(\pi_r v_{t_1}, \dots, \pi_r v_{t_r}),$$

where $s_1 < \cdots < s_p$ list S, $t_1 < \cdots < t_r$ list its complement, and $\varepsilon(S)$ is the sign of the permutation $(s_1, \dots, s_p, t_1, \dots, t_r)$. The determinant and the sign are scalars multiplying the G-valued factor. In degree $r = 0$ the determinant is 1. This shuffle expression is alternating by

the same repeated-argument cancellation as the wedge product, involves no factorial division, and gives a bounded linear map $\mathcal{R}\colon \mathrm{Mult}^k(D_k;G) \to \mathrm{Alt}^k(D_k;G)$ with norm at most $\binom{k}{p}$. It fixes every alternating map: after extension to the k-dimensional completion, both maps are determined by their value on the standard basis, where the shuffle formula has exactly one nonzero term. This is the required retraction. For $r > 0$, we do not need or assert that every k-linear map on D_k is alternating.

Finally, restrict the cross-action along the bounded affine family

$$E \longrightarrow \mathcal{L}(E_k, D_k), \qquad x \longmapsto M_x \oplus I_r.$$

Evaluation at $\det_k$, which is G-valued on D_k by Laplace expansion along the first p rows, and coefficient extraction through (H.4) give exactly the nonanalytic map (H.3). Thus the cross-action from Y_k to X_k is not analytic.

A largest analytic full subcategory would have to contain both individually analytic singleton subcategories, and fullness would then require this cross-action to be analytic. This is impossible. If closure under isomorphisms is required, replace each singleton by its full isomorphism closure. Every action between two copies is transported from the self-action by bounded linear changes of coordinates, so both closures remain analytic. The same contradiction proves Theorem H.4.

The scope of this obstruction is precise. Finite algebraic dimension over an incomplete field does not ensure continuous coordinates. The same construction does not answer the largest-domain question over complete fields or for Banach pairs: completion replaces the exotic finite-dimensional K-spaces by ordinary finite-coordinate L-spaces and removes the missing-coefficient obstruction.

Finite-dimensional bases. The same spaces show that Theorem 4.4 needs continuous coordinates on the base, not only finite algebraic dimension. Let $k = p + r$, and regard E, of dimension $p + 1$, as an analytic manifold with one chart. It has no continuous coordinates, since every bounded linear map $E \to K$ is zero by Lemma H.5. Over E take the trivial bundle with fiber $D_k \oplus E_k$ and the two frames id and

$$S_x(d, e) = \big(d + (M_x \oplus I_r)e, e\big), \qquad S_x^{-1}(d, e) = \big(d - (M_x \oplus I_r)e, e\big),$$

and the trivial bundle with fiber G. Both S_x and S_x^{-1} are affine in x, so these are analytic bundles with finite-dimensional fibers. By (6.1), analyticity of the induced alternating transition at $x = 0$ would make $x \mapsto A^k_{E_k,D_k;G}(M_x \oplus I_r)$ analytic at zero. Evaluation at $\det_k$ and (H.4) would then make (H.3) analytic at zero, which was excluded above. So the induced alternating atlas is not analytic.

Appendix I. Further coordinate methods

I.1. **Countable-type scalar sources.** The obstruction requires large source spaces in a stronger sense than infinite dimension. A normed space is *of countable type* if a countable subset has dense linear span.

Proposition I.1 (countable-type scalar sources)**.** *Let K be complete and nonarchimedean. If either E or D is of countable type, then $A^k_{E,D;K}$ is analytic in every degree. Neither completeness nor a nonarchimedean norm is required of E, D. In particular, the two spaces in Theorem 6.1(2) must both fail to be of countable type.*

Proof. The case $k = 0$ is constant. For a normed space V, set

$$q_V(x) = \inf_{x=\sum_i x_i} \max_i \|x_i\|,$$

where the decompositions use nonempty finite families, allowing a zero term. Scaling and concatenating them show that q_V is a nonarchimedean seminorm. Let V^u be the completion of its separated quotient; the canonical map $j_V\colon V \to V^u$ is contractive and has dense range. Expanding a bounded scalar multilinear form m along decompositions of each argument gives

$$|m(x_1,\dots,x_k)| \le \|m\| \prod_a q_V(x_a).$$

Thus scalar multilinear forms extend uniquely and isometrically to V^u, as do alternating forms. Extension preserves alternation by density of tuples with repeated entries, and commutes with permutations. Every bounded $u\colon V \to W$ induces $u^u\colon V^u \to W^u$ with $\|u^u\| \le \|u\|$. Uniqueness on the dense image makes $u \mapsto u^u$ bounded linear, and extension of forms intertwines precomposition by u and u^u.

If V is of countable type, so is V^u. By van der Put's basis theorem [12, Proposition 1.4], it has a t-orthogonal basis (b_i) for any fixed $0 < t < 1$. Writing f_i for its coordinate functionals, define

$$S(m)(x_1,\dots,x_k) = \sum_{i_1<\cdots<i_k} m(b_{i_1},\dots,b_{i_k}) \prod_{a=1}^{k} f_{i_a}(x_a).$$

The weighted coordinates $|f_i(x)|\|b_i\|$ form a null family bounded by $t^{-1}\|x\|$. Products of finitely many bounded null families are null, so the displayed sum converges unconditionally in K and $\|S(m)\| \le t^{-k}\|m\|$. Let $\mathcal{A}_V$ be unnormalized antisymmetrization, the signed sum over permutations of the arguments. It is bounded and alternating-valued in every characteristic. On finite-coordinate vectors, sorting distinct indices shows $\mathcal{A}_V S(m) = m$ for alternating m; repeated indices contribute zero. Density proves the same identity everywhere. Hence $R = \mathcal{A}_V S$ retracts all multilinear forms onto alternating forms, while $S|_{\mathrm{Alt}}$ is a bounded right inverse of $\mathcal{A}_V$. Transport through j_V gives both operators on the original V.

If E is of countable type, a lift of A is $R_E\big(m \circ (u_1,\dots,u_k)\big)$. If D is of countable type, use instead $\mathcal{A}_E\big(S_D(m) \circ (u_1,\dots,u_k)\big)$. Both are bounded multilinear in the u_i, linear in m, and have diagonal $A(u)(m)$; the second identity uses that antisymmetrization commutes with simultaneous precomposition. Proposition 3.3 concludes the proof. □

I.2. **Continuous algebraic polynomials on c_0.** The reflection theorem starts with bounded multilinear coefficients in an ambient space. For one homogeneous polynomial, interpolation supplies a lift even when only the diagonal is known to be continuous. The resulting bound is degreewise; it gives no assertion about a common radius across an infinite sequence of degrees.

Proposition I.2 (continuous algebraic polynomials on c_0)**.** *Let K be nontrivially normed and nonarchimedean, let Z be complete and nonarchimedean, and let $n \ge 1$. Suppose $p\colon c_0(I,K) \to Z$ is continuous and is an algebraic homogeneous polynomial: there exists a K-multilinear map $b\colon c_0(I,K)^n \to Z$, not assumed continuous, with $p(x) = b(x,\dots,x)$. Then p has a bounded n-linear lift q. More precisely, there is $C_{K,n} < \infty$, independent of I, Z, p, such that*

$$\|q\| \le C_{K,n}\|p\|_{\mathrm{diag}}, \qquad \|p\|_{\mathrm{diag}} := \inf\{D \ge 0 : \|p(x)\| \le D\|x\|^n \text{ for all } x\}.$$

No completeness assumption on K is needed.

Proof. Continuity at zero and $p(ax) = a^n p(x)$ imply $\|p\|_{\mathrm{diag}} < \infty$. Explicitly, choose $t \in K$ with $0 < |t| < 1$ and $\delta > 0$ such that $\|p(x)\| \le 1$ when $\|x\| < \delta$. For $x \ne 0$, choose an integer m with $|t|\delta \le |t^m|\|x\| < \delta$. Homogeneity gives $\|p(x)\| \le (|t|\delta)^{-n}\|x\|^n$.

Fix a total order on I. On each finite coordinate subspace, the algebraic formula for p is an ordinary homogeneous polynomial of degree n. Polynomial uniqueness over the infinite field K makes its coefficients compatible as the finite coordinate set varies. Denote the coefficient

of x^α by c_α, where $\alpha\colon I \to \mathbb{N}$ is finitely supported and $\sum_i \alpha_i = n$. Its support has at most n elements.

Choose distinct $a_0, \dots, a_n$ in the valuation ring of K, and let $L_n \geq 1$ bound the absolute values of all entries of the inverse Vandermonde matrix $(a_i^j)_{0\leq i,j\leq n}$. If the support of α has $d \leq n$ elements, interpolate the restriction of p to those coordinates on the grid $\{a_0, \dots, a_n\}^d$. Each grid vector has norm at most one. Applying the inverse Vandermonde matrix in each coordinate and using the nonarchimedean inequality in Z yields

$$\|c_\alpha\| \leq L_n^d \|p\|_{\mathrm{diag}} \leq L_n^n \|p\|_{\mathrm{diag}}.$$

For a nondecreasing word $i_1 \leq \cdots \leq i_n$, write $c_{i_1,\dots,i_n}$ for the coefficient with those multiplicities. Define

$$q(x_1, \dots, x_n) = \sum_{i_1 \leq \cdots \leq i_n} c_{i_1,\dots,i_n} (x_1)_{i_1} \cdots (x_n)_{i_n}.$$

The coefficients are uniformly bounded and each input is a null family, so the summands tend to zero outside finite sets of words. Completeness and the nonarchimedean inequality give convergence, multilinearity, and the bound $\|q\| \leq L_n^n \|p\|_{\mathrm{diag}}$. On finite supports its diagonal is p; density and continuity extend the identity to all of $c_0(I, K)$. Thus $C_{K,n} = L_n^n$ suffices. □

This bound is uniform in the number of coordinates, but the interpolation nodes and L_n depend on the degree. No exponential growth bound for $C_{K,n}$ is asserted. Consequently this proposition does not justify replacing lifting norms by diagonal norms in an arbitrary infinite analytic expansion. Theorem 4.5(1) obtains its stronger, radius-preserving conclusion from the already bounded ambient coefficients.

I.3. **A nonspherical target norm with analytic precomposition.**

Remark I.3. Spherical completeness of the given target norm is not necessary for analyticity, even in degrees at least the characteristic. Let $K = \mathbb{F}_p((t))$ with $|t| = 1/2$ and let $F = c_0(\mathbb{N}, K)$, the space of sequences tending to zero. Define

$$w_j = 1 + \frac{1}{j+2}, \qquad \|x\|_w = \sup_{j\geq 0} w_j |x_j|.$$

This is a nonarchimedean norm satisfying

$$\|x\|_\infty \leq \|x\|_w \leq \frac{3}{2} \|x\|_\infty.$$

Thus F is Banach for either norm. The usual supremum norm has nonzero values in $2^{\mathbb{Z}}$, since every nonzero null sequence attains its maximum absolute value. The discrete-distance argument in the proof of Corollary 4.3 therefore makes the supremum norm spherically complete. Theorem 4.2 gives analytic precomposition in every finite degree for that norm, for all normed E, E'. Equivalent norms preserve the bounded maps and their analytic power-series expansions: the norms of coefficients change by a fixed factor and convergence is unchanged. Thus precomposition remains analytic for $\|\cdot\|_w$, also directly by Corollary 4.3.

Nevertheless, the weighted norm is not spherically complete. Write e_j for the coordinate vectors, set $a_n = \sum_{j<n} e_j$, and consider the closed balls

$$B_n = \overline{B}_w(a_n, w_n).$$

They are nested because $w_{n+1} < w_n$ and $\|a_{n+1} - a_n\|_w = w_n$; the nonarchimedean inequality gives $B_{n+1} \subseteq B_n$. If x belonged to all of them, then, for each j and any $n > j$,

$$|x_j - 1| \leq \frac{w_n}{w_j} < 1.$$

Hence $|x_j| = 1$ for every j, contradicting $x_j \to 0$. Their intersection is empty. This example separates spherical completeness of a particular norm from analyticity, which is invariant under equivalent norms. It does not exclude a characterization using an equivalent spherically complete norm: this target has exactly such a norm.

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